

PARIKSHA365 — STUDY NOTES

Arithmetic — for SSC & RRB

Tier-1/2 focus: arithmetic core + geometry + mensuration + DI; less DI-caselet

SSC

RRB

v 2026-04

The promise. Read this book end-to-end (with the usual two re-reads + active recall on the embedded prompts) and you will be able to attempt every quiz question in the Pariksha365 pool for this subject with a strong fundamental grasp.

How to use this book

This book is different from the static-GK books. Here **every topic is broken into a focused set of TYPES**. Most topics settle at 8-15 types; some (Geometry, Input-Output, Puzzles) go higher — **there is no artificial cap**. What matters is that each type unlocks a *distinct* pattern or method. Almost every quantitative question you will ever see in SSC / Banking / RRB / PSU / State-PSC exams is a **tweak of one of these types**. If you can recognise the type in 5 seconds, you are already halfway to the answer.

For every TYPE you will see:

1. **What it looks like** — the recognition cue. Read the stem and ask: "which type is this?"
2. **Core insight / formula** — the single equation or idea that drives it.
3. **Smart / shortcut method FIRST** — the way a trained student solves it in their head in under 20 seconds, **with a full derivation** so you know *why* it works (so a twist will not break you).
4. **Traditional method SECOND** — the longer, safe fallback when the shortcut's precondition is not met.
5. **Variations** — 2-3 important tweaks that examiners love.
6. **Self-check** — one mini question to prove to yourself you have the type.

Before each topic, you will find a "Opener" box with:

- **Why the topic matters** (where you meet it in exams).
- **Mental-math arsenal** — the tables / constants you **MUST** know cold (no pen-paper).
- **Memorised formulas** — the 3-6 equations that run the topic.
- **Generic attack plan** — what to identify first, what to compute mentally, when to switch methods.

Golden rule of arithmetic prep: *the student who knows 10 types cold beats the student who has solved 1,000 questions without classifying them.*

PART 0 — THE MENTAL-MATH ARSENAL

You cannot be fast at arithmetic without *cached* values in your head. Before any topic, master this core.

0.1 Fraction ↔ Percentage ↔ Decimal table (MUST memorise)

Fraction	Percentage	Decimal
1/1	100 %	1.000
1/2	50 %	0.500
1/3	33.33 %	0.333...
1/4	25 %	0.250
1/5	20 %	0.200
1/6	16.67 %	0.166...
1/7	14.28 %	0.142857...
1/8	12.5 %	0.125
1/9	11.11 %	0.111...
1/10	10 %	0.100
1/11	9.09 %	0.0909...
1/12	8.33 %	0.0833...
1/13	7.69 %	0.0769...
1/14	7.14 %	0.0714...
1/15	6.66 %	0.0667...
1/16	6.25 %	0.0625
1/20	5 %	0.050
1/25	4 %	0.040
1/40	2.5 %	0.025
1/50	2 %	0.020

Also the multiples: $3/8 = 37.5\%$, $5/8 = 62.5\%$, $2/3 = 66.67\%$, $3/7 = 42.85\%$, $4/7 = 57.14\%$, $5/11 = 45.45\%$, etc.

0.2 Squares, cubes, roots

- Squares of 1-30 (NON-NEGOTIABLE), bonus 31-50.
- Cubes of 1-15 (NON-NEGOTIABLE), bonus 16-20.
- $\sqrt{2} \approx 1.414$, $\sqrt{3} \approx 1.732$, $\sqrt{5} \approx 2.236$, $\sqrt{7} \approx 2.645$, $\sqrt{10} \approx 3.162$.
- $\pi \approx 3.1416$, $\pi^2 \approx 9.87$, $e \approx 2.718$.

0.3 Tables 11-20 (yes, all of them)

If you still compute 17×14 by column multiplication, you are losing 15 seconds per question. Drill these.

0.4 Universal shortcuts (used everywhere)

- **10% trick** — move decimal one place left. 10% of 742 = 74.2. Everything else is built on this.
- **1% trick** — move decimal two places left. 1% of 742 = 7.42.
- **5% = half of 10%**. 5% of 742 = 37.1.

- **Break a nasty percentage into a sum of easy ones.** $17\% = 10\% + 5\% + 2\% = 10\% + 5\% + 2 \times 1\%$.
- **Fraction conversion:** "x is what % of y" is faster as the fraction x/y matched to the table above.
- **Percent-change symmetry:** $a\% \text{ of } b = b\% \text{ of } a$. $32\% \text{ of } 50 = 50\% \text{ of } 32 = 16$. Use whichever side is easier.

0.5 Generic problem-solving flow

1. **Read once — label the type.** Don't start solving yet.
 2. **Identify the one equation / ratio / insight** the type uses.
 3. **Check if values suit the shortcut** (round numbers, known fractions, symmetric setups). If yes, shortcut.
 4. **Else traditional method**, but compute mentally where possible (step 0 onwards).
 5. **Sanity-check the sign / order of magnitude.** A 120% increase cannot land below the original.
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PART 1 — PERCENTAGE

1.0 Opener

Why: Percentage is the single highest-frequency topic. It appears *inside* profit-loss, simple/compound interest, data interpretation, DI caselets, ratio problems, population, discount, tax, exam-marks, mixtures, error measurement. If percentage is slow, everything downstream is slow.

Mental-math arsenal: - The fraction-percent table (§ 0.1) — the single most powerful table in arithmetic. - Squares 1-30. - 10%, 1%, 5%, 25% as reflexes.

Memorised formulas: 1. $x\% \text{ of } y = (x/100) \cdot y = (y/100) \cdot x$ (*interchange*). 2. Percentage change = $(\text{New} - \text{Old}) / \text{Old} \times 100$. 3. Successive % changes $a, b \rightarrow \text{net} = a + b + (a \cdot b)/100$. 4. If A is $r\%$ more than B, then B is $(r / (100 + r)) \times 100\%$ **less** than A. Symmetric for "less than". 5. Population after n years at $r\%$ p.a. growth = $P \cdot (1 + r/100)^n$. (Same as CI.)

Generic attack plan: ask "what is the base (100)?" before anything. Then match to a type.

Type 1 — Convert fraction ↔ percentage ↔ decimal

Cue: "Express $5/8$ as a percentage" / " $37.5\% = ?$ fraction".

Insight: The table in § 0.1. Done.

Shortcut: Table lookup. $5/8 = 62.5\%$.

Traditional: $5/8 = 5 \times 100 / 8 = 62.5$.

Variations: Recurring decimals $\rightarrow 0.333... = 1/3$; $0.142857... = 1/7$ (period 6 — a known pattern).

Self-check: $7/8 = ?\%$. $\rightarrow 87.5\%$ ($50 + 25 + 12.5$).

Type 2 — "x % of y" direct calculation

Cue: "Find 23 % of 850".

Insight: Decompose x into multiples of 10, 5, 1.

Shortcut: - 23 % of 850 = 20 % + 3 %. - 20 % of 850 = 170. 1 % of 850 = 8.5. 3 % = 25.5. Total = 195.5. - Took < 10 sec mentally.

Traditional: $23/100 \times 850 = 19,550/100 = 195.5$.

Variations: - Use the symmetry trick: 36 % of 25 = 25 % of 36 = 9. - Ugly percentages (17.5 %) $\rightarrow 17.5 = 10 + 5 + 2.5$.

Self-check: 36 % of 250 = ? $\rightarrow$ 25 % of 360 = **90**.

Type 3 — "A is what percentage of B?"

Cue: "What percent of 64 is 24?"

Insight: Compute A/B , then map to the table.

Shortcut: $24/64 = 3/8 = 37.5\%$.

Traditional: $(24 \times 100) / 64 = 2400/64 = 37.5$.

Variations: - "By what percentage is A greater than B?" $\rightarrow (A-B)/B \times 100$ (base is B). - "By what percentage is A less than B?" $\rightarrow (B-A)/B \times 100$ (base is B).

Self-check: 45 is what % of 75? $\rightarrow 45/75 = 3/5 = 60\%$.

Type 4 — Percentage increase / decrease (basic)

Cue: "Price rose from 240 to 300. % increase?"

Insight: Change / Original $\times 100$, original (the *old* value) is the base.

Shortcut: $300 - 240 = 60$. $60/240 = 1/4 = 25\%$.

Traditional: $60 \times 100 / 240 = 25$.

Variations: - Watch the base direction: "decreased from 300 to 240" $\rightarrow 60/300 = 1/5 = 20\%$. - Two different bases give two different answers — the exam exploits this.

Self-check: Salary up from ₹20k to ₹28k $\rightarrow 8/20 = 40\%$.

Type 5 — Successive percentage change (the single most tested type)

Cue: "A salary is increased by 20 %, then decreased by 10 %. Net change?"

Insight: Net = $a + b + (a \cdot b)/100$ where a, b are **signed** (+ for rise, – for fall).

Shortcut (derivation shown once): - After a: amount = $100 + a$. - After b: $(100 + a)(1 + b/100) = 100 + a + b + ab/100$. - Net change from 100 = $a + b + ab/100$.

- Plug in $a = +20$, $b = -10$: $20 - 10 + (20)(-10)/100 = 10 - 2 = +8\%$.
- No decimal multiplications. Done in 5 sec.

Traditional: $100 \rightarrow 120 \rightarrow 120 \times 0.9 = 108 \rightarrow 8\%$ net. Same answer, slower.

Variations: - Three successive: apply pairwise. a, b combined $\rightarrow$ c. Then c, x combined. - Equal-and-opposite: $+r\%$ then $-r\%$ gives net = $-r^2/100$ (**always a loss**). Classic trap. - Discount-on-discount: two successive discounts of 10 % and 20 % are equivalent to a single discount of **28 %**, not 30 %.

Self-check: Price up 30 %, then down 20 %. Net? $\rightarrow 30 - 20 - 6 = +4\%$.

Type 6 — Population / value after n years (compound growth)

Cue: "Population 50,000; grows at 10 % annually; population after 3 years?"

Insight: $P \cdot (1 + r/100)^n$. This IS compound interest — memorise once, apply twice.

Shortcut: Multiply $(1.1)^3 = 1.331$. So answer = $50,000 \times 1.331 = 66,550$.

Traditional: Year 1 $\rightarrow 55,000$; Year 2 $\rightarrow 60,500$; Year 3 $\rightarrow 66,550$. Same number, slower.

Variations: - **Declining** population: $50,000 \times (0.9)^3 = 50,000 \times 0.729 = 36,450$. - **Mixed** rates (Year 1: $+10\%$, Year 2: $+5\%$, Year 3: -8%): chain-multiply. - **Find the original:** given final, divide instead of multiply. "Population is 66,550 after 3 years at 10 % — original?" $\rightarrow 66,550 / 1.331 = 50,000$.

Self-check: ₹8000 at 25 % p.a. CI for 2 years $\rightarrow 8000 \times (5/4)^2 = 8000 \times 25/16 = ₹12,500$.

Type 7 — Forward vs reverse percentage relationship

Cue: "If A's salary is 25 % more than B's, by what % is B's salary less than A's?"

Insight: Forward $r \rightarrow$ reverse is $r / (100 + r) \times 100$ %. Signs flip: "more" becomes "less".

Shortcut: $r = 25 \rightarrow 25 / 125 \times 100 = 20$ % less.

Traditional: Let $B = 100 \rightarrow A = 125 \rightarrow (A - B)/A = 25/125 = 20$ %. Same.

Variations: - "25 % less" $\rightarrow$ other one is $25/75 \times 100 = 33.33$ % more. - Remember the pair table: 20 % $\leftrightarrow$ 25 %, 25 % $\leftrightarrow$ 33.33 %, 10 % $\leftrightarrow$ 11.11 %, 50 % $\leftrightarrow$ 100 %, 9.09 % $\leftrightarrow$ 10 %, 14.28 % $\leftrightarrow$ 16.67 %. These sit inside 80 % of exam problems.

Self-check: A is 50 % more than B $\rightarrow$ B is how much less than A? $\rightarrow 50/150 = 33.33$ %.

Type 8 — Income $\rightarrow$ Expenditure $\rightarrow$ Savings

Cue: "Raj spends 75 % of his income. If income rises 20 % and expenditure rises 10 %, find % change in savings."

Insight: Savings = Income – Expenditure. Fix Income = 100 (because we only want percentages), let Expense = 75, Savings = 25. Apply changes to both, recompute savings.

Shortcut: New income = 120. New exp = $75 \times 1.1 = 82.5$. New savings = 37.5. Change = $37.5 - 25 = 12.5$. Percent = $12.5/25 \times 100 = +50$ %.

Traditional: Same algebra — no faster long route here.

Variations: - Given the % change in savings, find the unknown % change in expenditure. - "Saved 40 % of income" — base moves to "savings" side.

Self-check: Income +10 %, Expense +20 %, Expense was 80 % of income. New savings? $\rightarrow 110 - 96 = 14$ vs old 20 $\rightarrow \Delta = -6 \rightarrow -30$ %.

Type 9 — Pass / fail marks

Cue: "Pass mark is 40 %. A scored 220 and failed by 20 marks. Find max marks."

Insight: Pass mark = A's marks + his deficit. $220 + 20 = 240$. And $240 = 40$ % of max $\rightarrow$ max = $240 \times 100 / 40 = 600$.

Shortcut: $240 \rightarrow \div 4 \times 10 = 600$. Mental.

Traditional: Same, just written out.

Variations: - Two students, one passes, the other fails — two-equation system. Subtract to eliminate max. - "Exceeds pass by" vs "short of pass by" — sign matters.

Self-check: Pass mark = 33 %. Scored 152, failed by 19 $\rightarrow 171 = 33$ % of max $\rightarrow$ max = **~518.18** (actually a good example that not all exams give round answers — watch the exam's intent).

Type 10 — Elections (valid vs invalid votes)

Cue: "In an election 10 % of votes are invalid. Winner got 60 % of valid votes, which was 1800 more than loser. Total votes?"

Insight: Two stages: total $\rightarrow$ valid $\rightarrow$ winner/loser.

Shortcut: - Valid votes = 90 % of total = $0.9T$. - Winner – Loser = 60 % – 40 % = 20 % of valid = $0.2 \times 0.9T = 0.18T$. - $0.18T = 1800 \rightarrow T = \mathbf{10,000}$.

Traditional: Same equation, just set up step by step.

Variations: - Only two candidates $\rightarrow$ 60 % and 40 % split, difference = 20 % of valid. - Three candidates $\rightarrow$ different vote shares; sum to 100 % of valid.

Type 11 — Mixture-style percentage (alcohol, salt, milk)

Cue: "30 L of a mixture has 20 % alcohol. How much water to add to reduce alcohol to 15 %?"

Insight: Quantity of pure alcohol stays the same (we only add water). Set up on alcohol.

Shortcut: - Pure alcohol = 20 % of 30 = 6 L. - After adding x L water, total = $(30 + x)$ and alcohol still 6 L. - $6 / (30 + x) = 15/100 = 3/20 \rightarrow 30 + x = 40 \rightarrow x = \mathbf{10 \text{ L}}$.

Traditional: Same, equation written longer.

Variations: - Evaporation / remove $\rightarrow$ removing water CONCENTRATES alcohol %. Same invariant (alcohol qty). - Replacing a part of the mixture with water repeatedly $\rightarrow$ falls under Successive replacement formula: Final pure = Initial $\cdot (1 - x/V)^n$, where x is removed each time, V is total.

Self-check: 40 L of 25 % milk. Water added to make 20 % milk $\rightarrow$ milk qty 10 L stays $\rightarrow 10/(40+x) = 1/5 \rightarrow x = \mathbf{10 \text{ L}}$.

Type 12 — Price $\leftrightarrow$ Quantity inverse (consumption) problems

Cue: "Price of sugar rises 25 %. By what % must consumption drop to keep expenditure unchanged?"

Insight: Expenditure = Price $\times$ Quantity. If Expenditure is fixed, Price and Quantity are inversely proportional.

Shortcut: Reverse-percent trick (Type 7). Price up $r \rightarrow$ quantity down $r/(100+r) \times 100$. $r = 25 \rightarrow 25/125 = \mathbf{20 \%}$.

Traditional: Let price 100 $\rightarrow$ 125. Quantity 100 $\rightarrow 100/1.25 = 80$. So 20 % drop. Same number, slower.

Variations: - Price FALLS $r \% \rightarrow$ consumption can rise by $r/(100-r) \times 100 \%$. - Two-step: price change followed by budget change.

Self-check: Price up 10 % $\rightarrow$ consumption must drop by $10/110 = \mathbf{9.09 \%}$.

Type 13 — Successive discounts (at shop)

Cue: "Listed ₹2000. Two successive discounts of 20 % and 10 %. Selling price?"

Insight: Successive % changes (Type 5), both negative.

Shortcut: $-20 -10 + (20 \cdot 10)/100 = -30 + 2 = \mathbf{-28 \%}$. So SP = 72 % of 2000 = **₹1440**.

Traditional: 2000 $\rightarrow$ 1600 (after 20 % off) $\rightarrow$ 1440 (after 10 % off). Identical answer.

Variations: - Single equivalent discount of 20 % and 10 % = 28 % (NOT 30 %). - Three discounts: chain pairwise.

Self-check: 20 % + 25 % discount $\rightarrow$ equivalent single? $\rightarrow -20 - 25 + 5 = \mathbf{-40 \%}$.

Type 14 — Error in measurement (geometry-meets-percentage)

Cue: "Side of square measured 3 % in excess. Error % in area?"

Insight: Area $\propto$ side². If side is $(1 + r)$, area is $(1 + r)^2 \approx 1 + 2r + r^2$. With $r = 3\% = 0.03$, area error = $6 + 0.09 \approx$ **6.09 %**.

Shortcut: Successive-% with $a = b = r$. Net = $2r + r^2/100$. $r = 3 \rightarrow 6 + 9/100 =$ **6.09 %**.

Traditional: Same algebra explicitly.

Variations: - Cube / volume: three successive $r \rightarrow 3r + 3r^2/100 + r^3/1000 \approx 3r$. - One side $+r\%$, other side $-s\%$: apply signed successive.

Self-check: Radius $+4\% \rightarrow$ area of circle error? $\rightarrow 2(4) + 16/100 =$ **8.16 %**.

Type 15 — "x % more than y % of a number" chain problems

Cue: "Find the number whose 25 % is 40 % less than 60 % of 750."

Insight: Work right to left. Compute 60 % of 750, apply 40 % less, then reverse-divide by 25 %.

Shortcut: - 60 % of 750 = 450. - 40 % less = $450 \times 0.6 = 270$. - 25 % of $N = 270 \rightarrow N = 270 \times 4 =$ **1080**.

Traditional: Same sequence.

Variations: Add another nesting level; swap "less" for "more"; introduce ratios.

Self-check: 30 % of $N = 20\%$ more than 80 % of 500 $\rightarrow N = 480/0.3 =$ **1600**.



PART 2 — PROFIT, LOSS, DISCOUNT

2.0 Opener

Why: Shop/market context; 2-4 Qs guaranteed in Banking Prelims and SSC Quantitative.

Mental-math arsenal: percent-fraction table + "CP-SP-MP" vocabulary.

Memorised formulas: 1. Profit = SP – CP (both on the same currency). 2. Profit % = Profit/CP × 100 — **base is always CP** unless explicitly said. 3. Loss % = Loss/CP × 100. 4. SP = CP × (100 + P%)/100. SP = CP × (100 – L%)/100. 5. Discount = MP – SP. Discount % = Discount/MP × 100 — **base is MP**. 6. MP = CP × (100 + markup%)/100. 7. When two articles sold at same SP, one at +x % and the other at –x % → **always a net loss** of $x^2/100$ %.

Generic attack plan: read "gain/profit/loss" in terms of **CP**; read "discount" in terms of **MP**. Never confuse the bases. If both percentages appear, translate to a common number (set CP = 100).

Type 1 — Basic profit / loss percent

Cue: "CP = 400, SP = 460. Profit %?"

Shortcut: $(460 - 400)/400 = 60/400 = 3/20 = 15$ %.

Traditional: $60 \times 100 / 400 = 15$.

Variation: SP < CP → loss %, base still CP.

Type 2 — Find SP given CP and profit %

Cue: "CP = 800, wants 12.5 % profit. SP?"

Shortcut: 12.5 % = 1/8. So profit = 100. SP = **900**.

Traditional: SP = $800 \times 112.5/100 = 900$.

Variation: Loss instead of profit. 12.5 % loss → SP = $7/8 \times 800 = 700$.

Type 3 — Find CP given SP and profit %

Cue: "Sold at 1080 for 20 % profit. CP?"

Shortcut: SP is 120 % of CP → CP = $1080 \times 100/120 = 900$. The common trap is dividing by 1.2 vs subtracting 20 %.

Traditional: Let CP = x. $1.2x = 1080 \rightarrow x = 900$.

Type 4 — Two articles at same SP, opposite %

Cue: "A man sells two watches at ₹2400 each, gaining 20 % on one, losing 20 % on the other. Net gain/loss %."

Insight: **Always a net LOSS** of $x^2/100$ %. (Constant — cache this.)

Shortcut: $20^2/100 = 4$ % **LOSS**.

Traditional: - Gainer's CP = $2400 \times 100/120 = 2000$. - Loser's CP = $2400 \times 100/80 = 3000$. - Total CP = 5000, total SP = 4800 → loss $200/5000 = 4\%$.

The shortcut saves the whole CP derivation.

Type 5 — Marked price, discount, and profit combined

Cue: "Marked at 25 % above CP. Discount 10 %. Find profit %."

Shortcut: Successive % (Type 5 of percent): +25 then -10. Net = $15 - 2.5 = +12.5\%$ profit.

Traditional: CP 100 → MP 125 → SP $125 \times 0.9 = 112.5$ → profit 12.5.

Type 6 — Discount % required to clear stock at desired profit

Cue: "Shopkeeper marks goods 40 % above CP. Wants 12 % profit. Discount %?"

Shortcut: MP/CP ratio = 1.4. Desired SP/CP = 1.12. Discount = $1 - (SP/MP) = 1 - 1.12/1.40 = 1 - 0.8 = 20\%$.

Traditional: CP 100 → MP 140 → SP 112 → Discount = 28 on 140 = 20 %.

Type 7 — False weight / dishonest shopkeeper

Cue: "A dealer claims to sell at cost price but uses a 900 g weight for 1 kg. Profit %."

Insight: He charges for 1000 g but gives 900 g. Gain = $(1000 - 900)/900 \times 100 = 11.11\%$ (base = what he *actually* gave).

Variation: Uses BOTH a false weight AND marks up the price. Combine via successive %.

Self-check: 950 g instead of 1 kg → profit = $50/950 = 5.26\%$.

Type 8 — Dishonest shopkeeper buys too

Cue: "While buying uses 1100 g weight, while selling uses 900 g; charges CP either way. Overall profit %."

Shortcut: - On buying, gets 1100 g by paying for 1000 g → effective CP / g is reduced to 10/11 of claimed. - On selling, delivers 900 g for 1000 g price → SP / g is increased to 10/9. - Ratio SP/CP per gram = $(10/9) / (10/11) = 11/9$ → profit = $2/9 \times 100 = 22.22\%$.

Type 9 — Cost-price-unknown combined puzzles

Cue: "By selling at ₹810, a man loses 10 %. At what price should he sell to gain 10 %?"

Shortcut: 810 is 90 % of CP → CP = 900 → desired SP = 110 % = ₹990.

Traditional: Step-by-step CP recovery.

Type 10 — Partnership-style gain share (sneaks in)

Covered under Part 9 (Partnership). The link is the same "per unit" approach.

Type 11 — Successive discounts (equivalent single)

Already in Type 5 (percentage). Recapping for SP/MP context: $-10\% + 20\% \rightarrow$ equivalent 28% . $-10\% + 20\% + 5\% \rightarrow$ pairwise: 28% then with $5\% \rightarrow 28 + 5 - 1.4 = 31.6\%$.

Type 12 — Percentage profit on SP instead of CP (trap)

Cue: "A shopkeeper's profit is 20% of SP. Find profit % on CP."

Insight: If profit = 20% of SP, then CP = 80% of SP. Profit/CP = $20/80 = 25\%$.

Variation: Some exams ask "20 % loss on SP" $\rightarrow$ loss/CP = $20/120 = 16.67\%$.

PART 3 — SIMPLE & COMPOUND INTEREST

3.0 Opener

Why: Banking exams always carry 1-3 Qs of SI/CI, often combined with % and ratios.

Mental-math arsenal: $(1 + r/100)^n$ values for $r = 5, 10, 12.5, 15, 20, 25$ at $n = 2, 3, 4$.

Memorised formulas: 1. $SI = PRT/100$. 2. Amount (SI) = $P(1 + RT/100)$. 3. $CI = P \cdot (1 + R/100)^n - P$. 4. **CI – SI for 2 years = $P \cdot (R/100)^2$** (very important shortcut). 5. **CI – SI for 3 years = $P \cdot (R/100)^2 \cdot (3 + R/100)$** (slightly less used). 6. When compounded half-yearly: rate becomes $R/2$ %, time $2n$ periods. Quarterly: $R/4$ %, $4n$. 7. CI rate $\equiv$ successive % growth $\rightarrow$ use successive-% formula for 2-3 years.

Type 1 — Straight SI

Cue: P, R, T given. SI or Amount.

Shortcut: $PRT/100$. Keep everything as a fraction; cancel early.

Variation: Find R or T given SI. Same formula, solve for the unknown.

Type 2 — Principal doubling / tripling (SI)

Cue: "At what rate will a sum double in 8 years (SI)?"

Insight: Doubling means $SI = P \rightarrow PRT/100 = P \rightarrow RT = 100$. So $R \times 8 = 100 \rightarrow R = 12.5$ %.

Shortcut: For SI, $RT = 100 \times (\text{multiplier} - 1)$. Tripling $\rightarrow RT = 200$.

Type 3 — Compound interest for 2 / 3 years

Cue: "P = 10,000, R = 10 %, n = 2. CI?"

Shortcut: Successive % growth: +10 %, +10 % $\rightarrow$ net +21 %. CI = 21 % of 10,000 = **₹2100**.

Traditional: $A = 10000 \times 1.1^2 = 12100 \rightarrow CI = 2100$.

Variation (3 years, 10 %): successive of +10, +10, +10 $\rightarrow$ first pair = +21, then with 10 = +33.1. CI = 33.1 % of P.

Type 4 — CI – SI difference

Cue: "Difference of CI and SI on ₹5000 at 8 % for 2 years."

Shortcut: $P \cdot (R/100)^2 = 5000 \times (8/100)^2 = 5000 \times 64/10000 = \mathbf{₹32}$.

Traditional: Compute $SI = 800$. $CI = 5000 \times 1.08^2 - 5000 = 5832 - 5000 = 832$. Difference = 32. Same.

Type 5 — Compound frequency (half-yearly / quarterly)

Cue: "P = 8000, R = 10 % p.a., compounded half-yearly, for 1 year."

Shortcut: New rate = 5 %, periods = 2. $A = 8000 \times 1.05^2 = 8000 \times 1.1025 = \mathbf{₹8820}$. CI = 820.

Variation: Quarterly at 12 % p.a. for 1 year $\rightarrow$ rate 3 %, $n = 4 \rightarrow A = P \times 1.03^4 = P \times 1.1255$.

Type 6 — Instalment-style loan repayment

Cue: "Loan ₹2500 at 4 % p.a. CI, repaid in 2 equal annual instalments. Find instalment."

Shortcut: Each instalment's present value discounts by $(1 + R/100)^k$. $-x/(1.04) + x/(1.04)^2 = 2500$. $-x \cdot [1/1.04 + 1/1.0816] = 2500$. $-x \cdot (1.0816 + 1.04)/(1.04 \times 1.0816) = 2500$. $-x = 2500 \times 1.04 \times 1.0816 / (1.0816 + 1.04) \approx$
₹1325.03.

Traditional: Same algebra.

Type 7 — Find rate when CI for 2 years is given

Cue: "CI for 2 years on ₹6000 = ₹1260. Rate?"

Shortcut: $1260 = 6000 \times ((1 + r)^2 - 1)$ where $r = R/100$. $\rightarrow (1 + r)^2 = 1.21 \rightarrow r = 0.1 \rightarrow$ **R = 10 %.**

Type 8 — Ratios that recur

Cue: "A sum becomes ₹4840 in 2 years and ₹5324 in 3 years (CI). Rate and sum."

Insight: In CI, ratio of consecutive amounts is $(1 + R/100)$.

Shortcut: $5324 / 4840 = 1.1 \rightarrow R = 10 \%$. Then $P = 4840 / 1.21 =$ **₹4000.**

PART 4 — RATIO, PROPORTION, VARIATION

4.0 Opener

Memorised: 1. $a : b = c : d \Leftrightarrow ad = bc$. 2. $a/b = c/d = e/f = (a + c + e)/(b + d + f)$ — *addendo* rule. 3. Direct variation: $y = kx$. Inverse: $y = k/x$. Joint: $y = kxz / w$. 4. Compound ratio: $(a:b) \times (c:d) = ac : bd$.

Attack plan: convert every word problem into "per unit of ratio" (let ratio constant = k).

Type 1 — Split a quantity in ratio $a:b:c$

"Divide ₹2400 in 3:5:4 $\rightarrow$ parts = 600, 1000, 800."

Type 2 — Age/quantity ratio change after some years

"A:B = 3:5 now. After 6 years, 4:6. Find present ages."

Solve algebraically — set $3k$ and $5k$, translate "+6" condition.

Type 3 — Mixture ratio change

Detailed under Mixtures (Part 7).

Type 4 — Proportion (fourth proportional, mean, third)

- Fourth proportional of $a, b, c \rightarrow x = bc/a$.
- Mean proportional of $a, c \rightarrow x = \sqrt{ac}$.
- Third proportional of $a, b \rightarrow x = b^2/a$.

Type 5 — Direct/Inverse variation problems

"12 men finish a job in 8 days. How long will 16 men take?" $\rightarrow$ inverse: $12 \times 8 = 16 \times t \rightarrow t = 6$.

Type 6 — "Two numbers in ratio 3:5, their HCF is 4. Find numbers."

Numbers = $3k, 5k$ where $k = \text{HCF} \rightarrow 12, 20$.

Type 7 — Partners' investment ratio across different durations

"A invests 4000 for 6 months, B invests 5000 for 8 months" $\rightarrow$ ratio = $4000 \times 6 : 5000 \times 8 = 3:5$.

Type 8 — Alligation-linked ratio (bridge to Part 7)



PART 5 — AVERAGES

5.0 Opener

Memorised: Average = Sum / Count. Sum = Average × Count. If one value changes, new sum shifts by the change — new average = old + change/count.

Type 1 — Plain arithmetic mean.

Type 2 — New member joins / leaves, average changes.

"11 students' avg = 60. 12th student joins, avg drops by 1. 12th student's marks?" - New sum = $59 \times 12 = 708$. Old sum = 660. 12th = 48.

Type 3 — Replacement of a member.

"A class of 10 had avg 75. A student of 50 is replaced by new one; avg rises by 2." → new student - 50 = $2 \times 10 = 20$ → new student scored **70**.

Type 4 — Weighted average / mixture of groups.

Used heavily in DI.

Type 5 — Bowling/batting averages (cricket-style).

"Batsman avg 30 in 20 innings. Scores 50 in 21st. New avg?" - New avg = $(30 \times 20 + 50) / 21 = 650 / 21 \approx 30.95$.

Type 6 — Average speed (harmonic mean trap).

Not the arithmetic mean of two speeds. If two equal distances at speeds $u, v \rightarrow$ avg speed = $2uv/(u+v)$. See Part 8.

Type 7 — Consecutive integers.

Avg of n consecutive integers = middle term (for odd n) or average of two middles (even n).

Type 8 — Missing value from known average.



PART 6 — TIME & WORK, PIPES & CISTERNS

6.0 Opener

Memorised: 1. Work = Rate $\times$ Time. Rate is "fraction of job per unit time". 2. If A alone = a days, rate = $1/a$ per day. 3. Combined rate = sum of individual rates. 4. For pipes: inflow rates positive, outflow negative.

Attack plan: use LCM as total "units of work". Avoid fractions until the last step.

Type 1 — A alone a days, B alone b days $\rightarrow$ A+B together.

"A: 12 days, B: 18 days. Together?" $\rightarrow$ LCM 36 units; A 3/day, B 2/day; together 5/day $\rightarrow 36/5 = 7.2$ days.

Type 2 — Three persons, one joins/leaves midway.

"A and B together 12 days. After 4 days, B leaves; A finishes in another 10 days. A alone?" - Work done by A+B in 4 days = $4/12 = 1/3$. - Remaining $2/3$ done by A alone in 10 days $\rightarrow$ A alone = 15 days.

Type 3 — Efficiency ratio given.

"A is twice as efficient as B. Together in 10 days. A alone?" - Rates A:B = 2:1 $\rightarrow$ sum 3. Together 10 days $\rightarrow$ total work = 30 units. A alone = $30/2 = 15$ days.

Type 4 — Men-days / women-days work.

"15 men build in 20 days. How many men for 10 days?" $\rightarrow 15 \times 20 = n \times 10 \rightarrow n = 30$.

Type 5 — Mixed workforce.

"3 men or 5 women build in 20 days. 2 men + 3 women in ?" - M:W rates = $1/3 : 1/5$ (job per day per person), normalised. - Build the per-day team rate, then days.

Type 6 — Pipes filling with outflow (leak).

"Pipe A fills in 10 h, leak empties in 15 h. With leak, time to fill?" - Rates: $+1/10 - 1/15 = 1/30 \rightarrow 30$ hours.

Type 7 — Alternate / shift work.

"A works 1st day, B 2nd day alternately. A alone 8 d, B alone 12 d. Total?" - Per 2 days, work done = $1/8 + 1/12 = 5/24$. - After 4 such pairs (8 days), work = $20/24$. Remaining $4/24 = 1/6$. - Day 9 is A's, A does $1/8$ in a day, needs $8 \times (1/6) / (1/8) = 8/6 = 4/3$ days $> 1 \rightarrow$ use day 9 (A) fully $\rightarrow 1/8$ more; still left $4/24 - 3/24 = 1/24$. Day 10 is B's $\rightarrow 1/12$ per day = $2/24$. Takes half a day. Total = **9.5 days**.

Type 8 — Wages split when A and B worked together.

Wages $\propto$ work done $\propto$ (1/time). If together for 12 days and A alone 20 d, B alone 30 d, split 1800 wages $\rightarrow$ A:B = $30:20 = 3:2 \rightarrow 1080:720$.



PART 7 — MIXTURES & ALLIGATION

7.0 Opener

Memorised (alligation rule): For mixing two substances of cost / concentration C1 and C2 to get mean M:

$$\text{Ratio of C1 : C2} = \frac{(M - C2)}{(C1 - M)}$$

(Cheaper is on the C1 slot if $C1 < C2$; the differences are ALWAYS taken as positive.)

Type 1 — Two-component price mixing.

"Rice @ ₹30/kg and ₹45/kg mixed to sell at ₹36/kg. Ratio?" - $(45 - 36) : (36 - 30) = 9 : 6 = 3 : 2$.

Type 2 — Milk-water composition.

"Vessel has 40 L milk. Replace 10 L with water. Do it twice. Final milk %?" - Each time milk becomes $30/40 = 3/4$ of before. After 2 times: $(3/4)^2 = 9/16$ of pure milk. Final milk qty = $40 \times 9/16 = 22.5$ L → **56.25 %**. - Formula: Final pure = Initial $\cdot (1 - x/V)^n$.

Type 3 — Mixing three substances.

Apply alligation pairwise twice.

Type 4 — Gain % via mixing (link to P&L).

"CP 20/kg mixed with water (free) to sell at 25/kg." → Profit % based on CP of actual content.

Type 5 — Fraction of pure liquid after repeated replacement.

Type 6 — "How much water to add" / evaporate.

Already in Part 1 Type 11.



PART 8 — TIME, SPEED, DISTANCE (TSD)

8.0 Opener

Memorised: 1. $D = S \times T$. Units alert: km/h $\leftrightarrow$ m/s conversion: $\times 5/18$ (km/h $\rightarrow$ m/s), $\times 18/5$ (m/s $\rightarrow$ km/h). 2. Avg speed for equal DISTANCES at $u, v \rightarrow 2uv/(u + v)$ (harmonic mean). 3. Avg speed for equal TIMES at $u, v \rightarrow (u + v)/2$ (arithmetic mean). 4. Relative speed: same direction $\rightarrow |u - v|$; opposite direction $\rightarrow u + v$. 5. Train passing: length of train / relative speed = time.

Type 1 — Direct $D=ST$ lookups.

Type 2 — Avg speed (equal distance).

"Goes at 40 km/h, returns at 60 km/h. Avg speed?" $\rightarrow 2 \cdot 40 \cdot 60 / (100) = 48$ km/h. *Not 50.*

Type 3 — Relative speed (same/opposite direction).

"Two trains 100 km apart approach each other at 40 and 60 km/h. Time to meet?" $\rightarrow 100/100 = 1$ h.

Type 4 — Train passing a man / pole / platform / another train.

- Passing a man/pole: length / speed.
- Passing a platform: (length_train + length_platform) / speed.
- Passing another train (same direction): $(L1 + L2) / |u - v|$.

Type 5 — Boats and streams.

- Downstream speed = boat + stream. Upstream = boat - stream.
- If u = downstream, v = upstream $\rightarrow$ boat = $(u + v)/2$, stream = $(u - v)/2$.

Type 6 — Circular track problems.

- First meet at starting point: LCM of individual laps.
- First meet anywhere: track / |relative speed|.

Type 7 — Variable speed / broken journey.

"Walks at 4 km/h — 15 min late. Walks at 5 km/h — 15 min early. Distance?" - Let distance = d . $d/4 - d/5 = 30$ min = 0.5 h $\rightarrow d/20 = 0.5 \rightarrow d = 10$ km.

Type 8 — Race / head-start.

"A beats B by 100 m in a 500 m race." $\rightarrow$ in the same TIME, A runs 500, B runs 400. Ratio of speeds = 5:4.

Type 9 — Escalator problems.

"Man takes 50 steps down a moving up escalator; 125 steps up a moving up escalator (stationary escalator) when escalator is moving up." Treat escalator as a stream; steps visible = escalator + man (or - for opposing).

Type 10 — Clock problems (minute vs hour hand angle).

- Minute hand $6^\circ/\text{min}$; hour hand $0.5^\circ/\text{min}$; relative $5.5^\circ/\text{min}$.
 - Overlap every $65 \frac{5}{11}$ minutes.
-

PART 9 — PARTNERSHIP

9.0 Opener

Profit share $\propto$ (Capital $\times$ Time).

Type 1 — Equal time, different capital.

Ratio of profit = ratio of capital.

Type 2 — Different times (someone joins late / leaves early).

Weight each capital by the months it stayed.

Type 3 — Capital changed midway.

Break into sub-periods, sum C·T for each.

Type 4 — Working vs sleeping partner.

Working partner gets a flat salary first; remainder split by capital $\times$ time.

Type 5 — Loss sharing.

Same formula — negatives. A fixed-interest partner earns even in loss years (per contract).

PART 10 — AGES

Most age problems reduce to linear equations.

Type 1 — "Present age ratio" + "future/past age ratio" → solve simultaneously.

Type 2 — Sum/Difference of ages.

Type 3 — Average age of a group; one member replaced.

Same as Part 5 Type 3.

Type 4 — Parents-children age problems.

"Father is $3\times$ son's age. 10 years hence, $2\times$."

Type 5 — Multi-sibling + parent problems.

Combine averages + differences.

PART 11 — MENSURATION (2D & 3D)

Diagram accuracy rule: Geometric illustrations here must be to-scale SVG/TikZ, not hand-wave sketches. In the exam mode, every diagram is rendered from an exact coordinate spec. If a diagram cannot be made exact (e.g., specific irregular quadrilateral), the question is deferred to a visual-asset batch, not shipped with a crude ASCII box.

11.0 Opener

Must memorise — area & perimeter of square, rectangle, triangle ($\frac{1}{2} \cdot b \cdot h$), equilateral triangle ($\frac{\sqrt{3}}{4} \cdot a^2$), circle (πr^2), sector ($\frac{1}{2} \cdot r^2 \cdot \theta$), trapezium ($\frac{1}{2} \cdot (a+b) \cdot h$), parallelogram ($b \cdot h$), rhombus ($\frac{1}{2} \cdot d_1 \cdot d_2$); volume + TSA + CSA of cube, cuboid, cylinder, cone, sphere, hemisphere, frustum.

Types list (covered with SVG figures in the rendered PDF):

1. Direct substitution (basic area/perimeter).
2. Optimisation (largest square inscribed in a circle, etc.).
3. Path/road around a rectangle or garden.
4. Shaded-region problems (rectangle minus inner circle; overlapping circles).
5. Combined solids (cone on hemisphere; cylinder with conical lid).
6. Similar figures — ratio of areas is square of ratio of sides; volumes cube.
7. Frustum of cone (bucket).
8. Rolling / rotation — wheel covers πD per revolution.
9. Effect of changing one dimension on area/volume (link to Percentage Type 14).
10. Triangle-inside-triangle, midpoint theorem.

Each type has accurate diagrams authored as SVG; not rendered inline in this md file to avoid ASCII art that misleads the student. See [study_notes/_assets/mensuration/](#) for the figure set that the pandoc build pipeline embeds.

∠ PART 12 — GEOMETRY

Same diagram-accuracy rule applies. Triangles, circles, quadrilaterals, tangents — every figure is to-scale SVG.

Types:

1. Triangle basics: angle sum, exterior angle, triangle inequality.
 2. Similarity & congruence criteria (AA, SAS, SSS, RHS).
 3. Midpoint theorem; basic proportionality theorem (Thales).
 4. Pythagorean theorem + triples (3-4-5, 5-12-13, 8-15-17, 7-24-25, 20-21-29).
 5. Centroid (2:1), circumcentre, incentre, orthocentre positions.
 6. Circle theorems: angle at centre = $2 \times$ angle at circumference; angles in same segment; cyclic quadrilateral (opposite angles 180°).
 7. Tangent properties (tangent $\perp$ radius; two tangents from external point equal).
 8. Length relations for intersecting chords / secants.
 9. Polygon angle sums: interior = $(n-2) \cdot 180^\circ$; exterior sum = 360° .
 10. Coordinate geometry: distance, section, slope, collinearity.
-



PART 13 — TRIGONOMETRY (for SSC-CGL / RRB NTPC level)

13.0 Opener

Standard angle values — the non-negotiable table:

θ	$\sin$	$\cos$	$\tan$
0°	0	1	0
30°	$1/2$	$\sqrt{3}/2$	$1/\sqrt{3}$
45°	$1/\sqrt{2}$	$1/\sqrt{2}$	1
60°	$\sqrt{3}/2$	$1/2$	$\sqrt{3}$
90°	1	0	∞

- $\sin^2\theta + \cos^2\theta = 1$; $1 + \tan^2\theta = \sec^2\theta$; $1 + \cot^2\theta = \operatorname{cosec}^2\theta$.
- Complementary: $\sin(90^\circ - \theta) = \cos \theta$, etc.

Types:

- Standard-angle substitution.
 - Complementary-angle simplifications.
 - Identity-based simplifications.
 - Height & distance (angle of elevation / depression) — triangle + tan.
 - Two observers / two angles of observation.
-

PART 14 — ALGEBRA, PERMUTATION, PROBABILITY

14.1 Algebra — types

1. Linear equation in one / two variables.
2. Quadratic — discriminant, sum/product of roots ($\alpha + \beta = -b/a$; $\alpha \cdot \beta = c/a$).
3. Polynomial factor/remainder theorem.
4. Simultaneous equations (elimination vs substitution; cross-multiplication).
5. Inequalities (one-variable; watch sign flip on multiplication/division by negative).
6. Quadratic comparison (x & y) — banking favourite; solve both, tabulate sign.

14.2 Permutations & Combinations

Must memorise: - $nPr = n! / (n-r)!$. - $nCr = n! / (r! \cdot (n-r)!)$. - Arrangements with repetitions = $n! / (p! \cdot q! \cdot \dots)$.

Types: arrangements, combinations, circular $(n-1)!$, word-problems, committee formation.

14.3 Probability

- $P(A)$ = favourable / total.
- Complement: $P(A^c) = 1 - P(A)$.
- Addition: $P(A \cup B) = P(A) + P(B) - P(A \cap B)$.
- Independent: $P(A \cap B) = P(A) \cdot P(B)$.
- Conditional: $P(A|B) = P(A \cap B) / P(B)$.

Types: dice, cards (52-deck layout must be cached), coins, balls-in-urn, pick-without-replacement.



PART 15 — DATA INTERPRETATION (DI)

DI isn't a separate topic — it's an assembly of percent, ratio, average applied to visuals.

Types (question styles):

1. Table DI.
2. Bar graph DI (single, double, stacked).
3. Line graph DI.
4. Pie chart DI.
5. Caselet / paragraph DI (text-only).
6. Mixed chart (table + bar + line).
7. Missing DI (some cells blank; solve with given %/avg clues).

Each DI set has 4-5 sub-questions that translate into the core types. The speed gain is **reading the chart once, storing totals mentally, answering 5 Qs with that one pass.**

APPENDIX A — The 30-Second Diagnostic

Before solving any arithmetic question, run this 30-second classifier:

1. Is there a **percentage or ratio** word? → Parts 1-4.
2. Is there a **₹, cost, SP, MP, profit, discount**? → Part 2.
3. **Rate / time / interest / years / compound**? → Part 3.
4. **Days, hours, pipes, men, work, job**? → Part 6.
5. **Speed, distance, train, boat, stream, escalator, clock**? → Part 8.
6. **Mixture, alloy, milk, water, alcohol, ratio**? → Part 7.
7. **Partners, investment, capital, months**? → Part 9.
8. **Ages, father, son, years hence, years ago**? → Part 10.
9. **Area, perimeter, volume, surface**? → Parts 11-12.
10. **Triangle, circle, angle, tangent, chord**? → Part 12.
11. **sin, cos, tan, height, angle of elevation**? → Part 13.
12. **Letters arranged, committee, dice rolled, cards picked**? → Part 14.
13. Has a **chart / table / pie / bar**? → Part 15 + routing to 1-5.

This is the discipline. **Classify → shortcut → verify.** Every time.

APPENDIX B — Image / Diagram Pipeline

This book intentionally does **not** embed ASCII diagrams (they mislead). Instead, the pandoc build (`_build/build_pdfs.sh`) consumes two folders:

- `_build/figures/` — hand-authored SVG/TikZ for geometry, mensuration, reasoning visuals.
- `_build/mermaid/` — Mermaid source for flowcharts and type-trees.

Each diagram is **verified to-scale** before commit. No question is shipped with a sketch that misrepresents angle / ratio. For reasoning visuals (dice, mirror, water, paper-fold) the pipeline requires an accurate net / coordinate spec per item. Text-only questions that cannot be diagrammed faithfully are deferred rather than shipped with a rough approximation.

Companion books: `reasoning.md` (types-first), `english.md` (rules + traps). Quiz pairings live under topic codes `QNT_*`, `REAS_*`, `ENG_*`.

EXAM-FOCUS APPENDIX — BANKS (PO, Clerk, RRB Bank, RBI)

Use this appendix when reading this same book under the title "**Quantitative Aptitude — for Banks**". Banks weight DI + Caselet + Approximation + Quadratic comparison far more than SSC; geometry / mensuration / trigonometry are NEAR-ZERO weight.

Banks-priority chapter map

Topic in this book	Banks weight	Drill priority
Part 1 — Percentage	High	● Day 1
Part 2 — Profit & Loss	Mid	● Day 2
Part 3 — Simple + Compound Interest	High	● Day 3
Part 4 — Ratio + Average	High	● Day 3
Part 5 — Number system + LCM/HCF	Low	● skim
Part 6 — Time & Work + Pipes	Mid	● Day 4
Part 7 — Mixture & Alligation	Mid	● Day 4
Part 8 — Time, Speed, Distance + Boats + Trains	Mid	● Day 5
Part 9 — Partnership	Low	● skim
Part 10 — Ages	Low	● skim
Part 11 — Mensuration 2D	Low	● skim
Part 12 — Geometry	Negligible	○ skip
Part 13 — Trigonometry / Heights	Negligible	○ skip
Part 14 — Permutation / Probability	Mid	● Day 6
Part 15 — DI + Caselet	Highest	● Day 7-15
Part 17 (this) — Approximation + Quadratic + Quantity comparison	High	● Day 16-20

NEW Part — Approximation (Banks-only chapter)

What it tests: find the closest integer / "roughly equal to" value of a complex arithmetic expression in 30 seconds.

Method (drill): 1. Round each number to the nearest "easy" value (2-significant-figure, half, quarter, etc.). 2. Apply BODMAS on rounded values. 3. Compare with options.

Worked example. $? \approx 39.97\% \text{ of } 1899.98 + 24.04 \times 17.97$.

Round: $40\% \text{ of } 1900 = 760$. $24 \times 18 = 432$. Sum ≈ 1192 . Pick option closest to 1192.

Hot-list of round-able quantities:

Difficult	Round to
14.97	15
24.04	24
39.97	40
11.95	12

Difficult	Round to
16.66 %	1/6
33.33 %	1/3
12.5 %	1/8
14.28 %	1/7
$\sqrt{26}$	≈ 5.1
$\sqrt{48}$	≈ 6.93
12 % of 250	30

Drill: 50 sums daily for a week. Time-per-sum should drop from 60 s $\rightarrow$ 25 s.

NEW Part — Quadratic comparison (Banks specialty)

What it tests: two quadratics in x and y ; solve both, choose: $x > y$ / $x \geq y$ / $x < y$ / $x \leq y$ / $x = y$ / no relation.

Method: 1. Factor or use formula on each. 2. Get all 4 roots: x_1, x_2, y_1, y_2 . 3. Compare the **smallest x** with the **largest y** , and **largest x** with the **smallest y** . 4. If $\text{smallest}_x \geq \text{largest}_y \rightarrow x \geq y$. If $\text{largest}_x \leq \text{smallest}_y \rightarrow x \leq y$. Else $\rightarrow$ no relation.

Worked example. $x^2 - 7x + 12 = 0 \Rightarrow x \in \{3, 4\}$. $y^2 - 9y + 20 = 0 \Rightarrow y \in \{4, 5\}$.

$\text{smallest}_x = 3$, $\text{largest}_y = 5 \rightarrow 3 < 5$ (so NOT $x \geq y$). $\text{largest}_x = 4$, $\text{smallest}_y = 4 \rightarrow 4 = 4$ (so NOT $x < y$). $\rightarrow$ **No relation can be established.**

NEW Part — Quantity comparison

Given Quantity I and Quantity II as expressions; choose: $I > II$ / $I \geq II$ / $I < II$ / $I \leq II$ / $I = II$ / no relation.

Method: evaluate both quantities; compare directly. Often one is a rate problem and the other a percentage / formula.

DI deep-drill (high yield — re-read Part 15)

Banks DI types reorder: 1. **Tabular** — straight-up tables with multi-row, multi-column. 2. **Bar chart** — single, double, stacked. 3. **Pie chart** — % distribution; total often given. 4. **Line graph** — trend across years. 5. **Mixed combo** — bar + line, table + pie. 6. **Caselet** — DI in pure paragraph form, no chart. **Banks Mains specialty.** 7. **Missing-value DI** — table with one cell missing; deduce from constraints. 8. **Radar chart** — RBI & SBI Mains.

For each DI set (5 questions): 1. Read the chart caption + axis labels — 30 s. 2. Compute totals / row sums / column sums in margin if helpful — 30 s. 3. Then attack each of the 5 questions in 60-90 s.

Total budget per DI set: 5-7 min. With 4 sets in IBPS PO Mains DI section, that's 25-30 min for 20 marks — high ROI.

EXAM-FOCUS APPENDIX — SSC + RRB

Use this appendix when reading this same book under the title "**Arithmetic — for SSC & RRB**". SSC + RRB weight Geometry + Mensuration + Trigonometry HEAVILY (15-20 marks combined), and DI is shorter / simpler.

SSC + RRB priority chapter map

Topic in this book	SSC weight	RRB weight	Drill priority
Part 1 — Percentage	High	High	● Day 1
Part 2 — Profit, Loss, Discount	High	High	● Day 1
Part 3 — SI + CI	Mid	Mid	● Day 2
Part 4 — Ratio + Proportion + Average	High	High	● Day 2
Part 5 — Number system + LCM/HCF + Divisibility	High	High	● Day 3
Part 6 — Time & Work + Pipes	Mid	Mid	● Day 4
Part 7 — Mixture / Alligation	Mid	Mid	● Day 4
Part 8 — Time-Speed-Distance + Boats + Trains	High	High	● Day 5
Part 9 — Partnership	Mid	Low	● Day 6
Part 10 — Ages	Mid	Mid	● Day 6
Part 11 — Mensuration 2D	High	High	● Day 7
Part 12 — Geometry (incl. Triangles, Circles)	Highest	High	● Day 8-10
Part 13 — Trigonometry + Heights	Highest	Mid	● Day 11-13
Part 14 — Permutation / Probability	Low	Low	● skim
Part 15 — DI	Mid (smaller table/bar)	Low	● Day 14
Algebra (basic identities)	High	Mid	● Day 15

Key SSC algebra identities to memorise

- $(a + b)^2 = a^2 + b^2 + 2ab$
- $(a - b)^2 = a^2 + b^2 - 2ab$
- $a^2 - b^2 = (a + b)(a - b)$
- $a^3 + b^3 = (a + b)(a^2 - ab + b^2)$
- $a^3 - b^3 = (a - b)(a^2 + ab + b^2)$
- $(a + b)^3 = a^3 + b^3 + 3ab(a + b)$
- $(a - b)^3 = a^3 - b^3 - 3ab(a - b)$
- $a^3 + b^3 + c^3 - 3abc = (a + b + c)(a^2 + b^2 + c^2 - ab - bc - ca)$
- if $a + b + c = 0 \Rightarrow a^3 + b^3 + c^3 = 3abc$
- $a + 1/a = k \Rightarrow a^2 + 1/a^2 = k^2 - 2$; $a^3 + 1/a^3 = k^3 - 3k$
- $a^2 + b^2 \geq 2ab$ (AM $\geq$ GM)

Trigonometry: identities you MUST recall in <2 s

- $\sin^2\theta + \cos^2\theta = 1$
- $1 + \tan^2\theta = \sec^2\theta$
- $1 + \cot^2\theta = \operatorname{cosec}^2\theta$
- $\sin(90 - \theta) = \cos \theta$; $\cos(90 - \theta) = \sin \theta$; $\tan(90 - \theta) = \cot \theta$

- $\sin(A + B) = \sin A \cos B + \cos A \sin B$
- $\cos(A + B) = \cos A \cos B - \sin A \sin B$
- $\tan(A + B) = (\tan A + \tan B) / (1 - \tan A \cdot \tan B)$
- $\sin 2\theta = 2 \sin \theta \cos \theta$; $\cos 2\theta = \cos^2\theta - \sin^2\theta = 1 - 2 \sin^2\theta = 2 \cos^2\theta - 1$
- standard angles: $\sin 0 = 0$, $\sin 30 = \frac{1}{2}$, $\sin 45 = 1/\sqrt{2}$, $\sin 60 = \sqrt{3}/2$, $\sin 90 = 1$
- height & distance: $\tan(\text{angle of elevation}) = \text{opposite/adjacent}$

Geometry — the 25 must-know facts

1. Sum of interior angles of a triangle = 180° .
2. Sum of exterior angles of any polygon = 360° .
3. Sum of interior angles of an n-gon = $(n - 2) \times 180^\circ$.
4. Each interior angle of a regular n-gon = $(n - 2) \times 180^\circ / n$.
5. Area of triangle = $\frac{1}{2} \times \text{base} \times \text{height} = \frac{1}{2} ab \sin C = \sqrt{s(s-a)(s-b)(s-c)}$ (Heron's).
6. Area of equilateral triangle (side a) = $\sqrt{3}/4 \times a^2$.
7. Inradius (incircle): $r = \text{Area} / s$.
8. Circumradius: $R = abc / (4 \times \text{Area})$.
9. Pythagoras: $a^2 + b^2 = c^2$ in right triangle.
10. 30-60-90 triangle: sides 1 : $\sqrt{3}$: 2.
11. 45-45-90 triangle: sides 1 : 1 : $\sqrt{2}$.
12. Median, altitude, angle-bisector, perpendicular-bisector concur at distinct points: centroid (G), orthocentre (O), incentre (I), circumcentre (C).
13. Centroid divides median in 2 : 1 from vertex.
14. In right triangle, circumcentre = midpoint of hypotenuse.
15. Mid-segment theorem: line joining midpoints of two sides is parallel to third side and half its length.
16. Apollonius' theorem: $m_a^2 = (2b^2 + 2c^2 - a^2) / 4$.
17. Tangent from external point — two tangents are equal.
18. Tangent $\perp$ radius at point of contact.
19. Angle in same segment is equal; angle in semicircle = 90° .
20. Cyclic quadrilateral: opposite angles sum to 180° .
21. Power of a point: for two chords through P, $AP \cdot BP = CP \cdot DP$.
22. Two intersecting chords: $AP \cdot PB = CP \cdot PD$.
23. Angle subtended at centre = $2 \times$ angle at remaining circumference.
24. Inscribed angle theorem.
25. Ptolemy's theorem (cyclic quadrilateral): $AC \cdot BD = AB \cdot CD + AD \cdot BC$.

SSC speed rules (60 min, 25 questions)

- **Arithmetic word problems** (percentage / SI-CI / TSD / W&T / mixture): 12-14 questions, 80 s each.
- **Algebra identities + simplification**: 3-5 questions, 60 s each.
- **Geometry / mensuration**: 4-5 questions, 90 s each — diagram-driven.
- **Trigonometry**: 2-3 questions, 60 s each.
- **DI / Tabular**: 3-5 questions in one set, 75 s each.

Total $\approx$ 50 min for 25 questions, leaving 10 min buffer.

GLOBAL DIFFICULTY MATRIX (per Q to design study time)

Topic	SSC easy : medium : hard	Banks easy : medium : hard
Percentage	30 : 50 : 20	20 : 50 : 30
SI + CI	25 : 55 : 20	20 : 50 : 30
TSD	30 : 50 : 20	25 : 55 : 20
W & T	25 : 55 : 20	20 : 55 : 25
Mixture	30 : 55 : 15	25 : 55 : 20
Geometry	20 : 60 : 20	—
Trigonometry	25 : 50 : 25	—
DI	30 : 55 : 15	15 : 55 : 30
Caselet	—	10 : 50 : 40
Approximation	—	30 : 50 : 20
Quadratic	—	25 : 60 : 15

Use the same source `arithmetic.md` to build BOTH `arithmetic_ssc_rrb.pdf` and `quantitative_apptitude_banks.pdf` — the manifest in `_build/manifest.json` selects the title + cover; the appendix above tells the candidate which chapters to drill harder.

Practice Set (50 exam-style worked examples)

The questions below are modelled on the actual patterns asked in past papers across SSC, RRB, banking, and PSU exams. For each, you get: the stem → smart shortcut (with derivation) → answer → the trap option and why students pick it.

Section 1 — PERCENTAGE (5 worked + variations)

Q1. A's salary is 25 % more than B's. By what % is B's salary less than A's?

What kind of problem is this? "A is r % more than B" → asking the inverse. Standard pattern: **base shift problem**.

Solving it. Set B = 100 (always make the smaller / reference quantity = 100). Then A = 125.

Quick way: Compute the % decrease from A to B. - Difference = A – B = 25 - % decrease = Difference / A × 100 = 25 / 125 × 100 = **20 %**

Worth knowing: Memorise the shortcut formula. If A is r % more than B, then **B is $r / (100 + r) \times 100$ % less than A**. - Verify with r = 25: 25 / 125 × 100 = 20 % ✓

Watch out: "25 %" — students assume "25 % more" implies "25 % less" by symmetry. That is WRONG: percentage is a ratio, and the base changes when you flip direction.

A similar question: A is r % LESS than B, find by what % B is more than A. Use **B is $r / (100 - r) \times 100$ % more than A**. - Example: A is 20 % less than B → B is 20/80 × 100 = **25 % more** than A.

Another twist: Successive percentage relations. If A is 20 % more than B, and B is 25 % more than C, what % is A more than C? - A = 1.20 B = 1.20 × 1.25 C = 1.50 C → A is **50 % more** than C.

Q2. The price of sugar is increased by 25 %. By what % must consumption decrease so that monthly expenditure remains unchanged?

What kind of problem is this? Same base-shift formula as Q1. "Price up by r % → consumption must drop by $r/(100+r) \times 100$ % for expenditure to stay constant."

Solving it. Why the formula works (derivation in 1 line).** Expenditure = Price × Quantity. If Price becomes 1.25P, then new Quantity must satisfy $1.25P \times Q' = P \times Q \Rightarrow Q' = Q \times 1/1.25 = Q \times 0.80 \Rightarrow$ a 20 % drop.

Quick way: Apply. **$r = 25 \rightarrow 25 / 125 \times 100 = 20$ %**.**

Watch out: "25 %" — students assume "if price up by r %, cut consumption by r %". That UNDER-shoots — would result in $1.25 \times 0.75 = 0.9375 \times$ original expenditure (still down 6.25 %).

A similar question: Price up by 50 %, consumption cut needed = $50 / 150 \times 100 =$ **33.33 %**.

Another twist: Price DECREASES by r %, by how much can consumption increase to keep expenditure same? Use **$r / (100 - r) \times 100$ %**. - Example: Price down 20 % → consumption can go up by $20/80 \times 100 =$ **25 %**.

Q3. A number is increased by 20 % and then decreased by 20 %. Net change?

What kind of problem is this? Successive percentage changes (one up, one down).

Solving it. Apply the formula. **Net change = $a + b + (a \times b) / 100$** , where positive = increase, negative = decrease. - Here $a = +20$, $b = -20$. - **Net = $+20 + (-20) + (20 \times -20)/100 = 0 + (-400/100) = -4\%$** **

Quick way: Verify with concrete number. **Take 100 $\rightarrow +20\% \rightarrow 120 \rightarrow -20\% \rightarrow 96$. Final 96, original 100, net change = -4%** ** ✓.

Worth knowing: Memorise the universal pattern. **Whenever you do " $+r\%$ then $-r\%$ " (any equal r), net is ALWAYS a loss = $(r/10)^{2**}$.** For $r = 20$: $(20/10)^2 = 4\%$ loss. For $r = 30$: 9% loss. For $r = 10$: 1% loss.

Watch out: " 0% " — students assume $+20\%$ and -20% cancel. They DO NOT, because the second 20% is taken on a larger base (120, not 100).

A similar question: Increased by 30% , then decreased by 20% . **Net = $30 - 20 + (30)(-20)/100 = 10 - 6 = +4\%$** gain.

Another twist: Three successive changes (a, b, c). Apply the formula stepwise: combine $a + b$ first, then combine result with c . - Example: $+10\%$, $+20\%$, -10% . Step 1: $10 + 20 + 2 = +32\%$. Step 2: $32 + (-10) + (32 \times -10)/100 = 22 - 3.2 = +18.8\%$.

Q4. In an exam 60% students passed in Maths, 50% in English, 30% in both. What % failed in both?

What kind of problem is this? Two-set inclusion-exclusion problem (Venn diagram).

Solving it. Apply the inclusion-exclusion formula. - $n(M \cup E) = n(M) + n(E) - n(M \cap E) = 60 + 50 - 30 = 80\%$ passed at least one subject. - Failed in both = $100 - 80 = 20\%$.

Quick way: Visualise (mental Venn). **Draw two overlapping circles. - Pass Maths only = $60 - 30 = 30\%$ - Pass English only = $50 - 30 = 20\%$ - Pass both = 30% - Total pass-at-least-one = $30 + 20 + 30 = 80\%$ - Fail both = $100 - 80 = 20\%$** **

Watch out: " 10% " — students subtract 30% from 100 once and stop, forgetting that " 30% in both" is part of the 60% AND part of the 50% .

A similar question: Three subjects: 70% pass A, 60% pass B, 50% pass C; 30% pass all three; 40% pass A&B, 35% pass B&C, 25% pass A&C. Use 3-set formula: $n(A \cup B \cup C) = \text{sum} - \text{pairs} + \text{triple} = (70+60+50) - (40+35+25) + 30 = 180 - 100 + 30 = 110$. Cap at 100; rework if >100 (means data inconsistent).

Another twist: "Failed in both" stated $\rightarrow$ "passed at least one" = $100 - \text{failed-both}$. Useful inverse direction.

Q5. If 30% of $(X - Y) = 20\%$ of $(X + Y)$, find $X : Y$.

What kind of problem is this? Linear equation in two variables; the answer is a ratio (only one equation needed because the ratio is scale-invariant).

Solving it. Convert percentages to fractions. - $0.3(X - Y) = 0.2(X + Y)$

Quick way: Expand both sides. - $0.3X - 0.3Y = 0.2X + 0.2Y$

Worth knowing: Collect like terms. - $0.3X - 0.2X = 0.2Y + 0.3Y$ - $0.1X = 0.5Y$

Note: Take the ratio. - **$X / Y = 0.5 / 0.1 = 5 / 1$ - So $X : Y = 5 : 1$** **

Watch out: " $1 : 5$ " — students invert the ratio at the end.

A similar question: If 40% of $(X+Y) = 60\%$ of $(X-Y)$, find $X : Y$. Repeat the steps: $0.4X + 0.4Y = 0.6X - 0.6Y \rightarrow 1.0Y = 0.2X \rightarrow X/Y = 5/1$.

Another twist: 25% of A = 35% of B. Find A : B. $0.25A = 0.35B \rightarrow A/B = 0.35/0.25 = 7 : 5$.

Section 1A — ADVANCED PERCENTAGE (exam-level variations)

Q1.5 (Advanced) — Multi-step percentage chain with population. The population of a town was 50,000 in 2022. It increased by 10 % in 2023, then decreased by 5 % in 2024, then increased by 20 % in 2025. Find the population at the end of 2025.

Solving it. Apply each change as a multiplier (do NOT add/subtract percentages directly). - After 2023: $50,000 \times 1.10 = 55,000$ - After 2024: $55,000 \times 0.95 = 52,250$ - After 2025: $52,250 \times 1.20 = 62,700$

Worth knowing: Net multiplier = $1.10 \times 0.95 \times 1.20 = 1.254$. So final = $50,000 \times 1.254 = 62,700$ (same answer in one step).

Net % change = $(62,700 - 50,000) / 50,000 \times 100 = +25.4 \%$.

A similar question. Salary ₹40,000 increased by 15 %, then by 10 %, then decreased by 5 %. Final = $40,000 \times 1.15 \times 1.10 \times 0.95 = ₹48,070$ (net +20.18 %).

Q1.6 (Advanced) — Population growth/decay with given rate. A town's population grows at 4 % per annum compounded annually. If current population is 1,000,000, what will it be after 3 years? After 5 years?

Solving it. - Use compound formula: $P(t) = P_0 \times (1 + r)^t$. - After 3 years: $1,000,000 \times (1.04)^3 = 1,000,000 \times 1.124864 = 1,124,864$ - After 5 years: $1,000,000 \times (1.04)^5 = 1,000,000 \times 1.21665 = 1,216,653$

Quick mental approach for compounded growth (≤ 10 yrs at low rate). Use: $(1 + r)^t \approx 1 + r \times t + (\text{small higher-order terms})$. For 4 % over 3 yrs: $\approx 1 + 0.12 + 0.005 = 1.125$ (close to actual 1.1249).

A similar question — Decay (depreciation). A car worth ₹8,00,000 depreciates 15 % per year. Value after 3 years = $8,00,000 \times (0.85)^3 = 8,00,000 \times 0.614125 = ₹4,91,300$.

Q1.7 (Advanced) — Successive % with target outcome. A shopkeeper marks his goods 50 % above cost price (CP) and offers a 20 % discount on the marked price (MP). What is his net profit %?

Solving it. Set CP = 100. - MP = $100 \times 1.50 = 150$ - SP after 20 % discount = $150 \times 0.80 = 120$ - Profit = $120 - 100 = 20 \rightarrow$ Profit % = $(20/100) \times 100 = 20 \%$

Worth knowing: General formula. Net % = markup % - discount % - (markup $\times$ discount)/100. Here = $50 - 20 - 10 = 20 \%$ ✓.

A similar question. Mark up by 60 %, discount 25 %. Net = $60 - 25 - 15 = +20 \%$ profit.

Section 2 — PROFIT, LOSS, DISCOUNT (5)

Q6. A man sells two articles at ₹600 each. On one he gains 20 %, on the other loses 20 %. Net P/L %?

Step 1 — Identify the type. Two articles, **same SP, equal % gain on one and equal % loss on the other**. This is the classic "same SP, equal $\pm r\%$ " pattern.

Solving it. Apply the standard result (with derivation). - Article A: SP = ₹600 at 20% gain $\rightarrow CP_A = SP/(1+r) = 600/1.20 = ₹500$ - Article B: SP = ₹600 at 20% loss $\rightarrow CP_B = SP/(1-r) = 600/0.80 = ₹750$ - Total CP = $500 + 750 = ₹1,250$ - Total SP = $600 + 600 = ₹1,200$ - Net loss = $1,250 - 1,200 = ₹50$ - Loss % = $50 / 1,250 \times 100 = 4 \%$

Quick way: Memorise the shortcut. **Whenever same SP + equal $\pm r \%$, net result is always a loss = $(r/10)^2$.** - For **$r = 20 \%$: loss = $(20/10)^2 = 4 \%$** ✓ - For **$r = 10 \%$: loss = $(10/10)^2 = 1 \%$** - For **$r = 25 \%$: loss = $(25/10)^2 = 6.25 \%$** **

Watch out: "0 %" — students assume gain and loss cancel out. They DO NOT, because the bases (CP_A vs CP_B) are different.

Variation A. Same problem but SP not given (only ratios) — answer is still 4 % loss; SP is irrelevant when only % gain/loss are equal.

Variation B. Different rates (e.g., gain 20% on one, loss 10% on other, same SP) — formula doesn't apply; do CP_A + CP_B and compare to total SP individually.

Q7. MP is 40 % above CP. Two successive discounts of 10 % and 20 % are given. Find net P/L %.

Step 1 — Set CP = 100 (always do this for percentage problems — it makes mental math trivial).

Solving it. Compute MP. $MP = CP \times 1.40 = 140^{**}$.

Quick way: Apply discounts in order. - After 1st discount (10 % off MP): $140 \times 0.90 = 126$ - After 2nd discount (20 % off remaining): $126 \times 0.80 = 100.80$

Worth knowing: Final SP = 100.80, CP = 100. **So profit = 0.80, profit % = +0.80 %**** (small gain).

Memory peg for successive discounts. When two successive discounts of x % and y % are applied, the **single equivalent discount** = $x + y - (xy/100)$. For 10% and 20%: $10 + 20 - 2 = 28\%$ off MP. So SP = $140 \times 0.72 = 100.80$ ✓ (same answer).

Watch out: "-4 %" or "-10 %" — students forget to multiply discounts on top of MARKED PRICE, not on CP.

Variation A. MP is 50% above CP, two successive discounts 10% + 10% — try this: MP = 150, after discounts = $150 \times 0.9 \times 0.9 = 121.50$. Profit = 21.5%.

Variation B. Three successive discounts (a, b, c) — compute step by step OR use formula iteratively: equivalent of first two, then combine with third.

Q7. MP is 40 % above CP. Two successive discounts of 10 % and 20 % are given. Find net P/L %. *Method.* Let CP = 100. MP = 140. After discounts: $140 \times 0.9 \times 0.8 = 100.80$. P/L = +0.80 % gain. Watch out: "-4 %" or "-10 %" — students forget the markup direction.

Q8. A trader marks up the price by 30 %, gives a 10 % discount, AND uses a faulty meter that gives 900 g when the customer pays for 1 kg. Find his actual profit %.

What kind of problem is this? Combo problem: markup + discount + false weight. Each is a multiplication factor; chain them.

Solving it. Convert each effect into a multiplier. - **Markup** of 30 % means he marks at $1.30 \times CP$. - **Discount** of 10 % means he sells at $0.90 \times MP$, i.e., at $1.30 \times 0.90 = 1.17 \times CP$ (he gets paid $1.17 \times CP$ for the goods). - **False weight** — customer thinks they got 1,000 g (worth $1.17 \times CP$ at the marked rate) but actually received 900 g. So the trader effectively SOLD only 900 g of goods for the price of 1,000 g. Multiplier: $1,000 / 900 = 10/9 \approx 1.111$.

Quick way: Multiply all three multipliers. - Total revenue / cost = $1.17 \times (10/9) = 11.70 / 9 = 1.30$ - So profit = 30 % over CP. Answer = **30 %**.

Worth knowing: Sanity-check. **Take CP of 1 kg = ₹100. Marks to ₹130, sells at ₹130 × 0.9 = ₹117 (for what customer thinks is 1 kg). Customer actually receives 900 g, which cost the trader ₹90 (since 1 kg cost ₹100). So trader spent ₹90 of goods and received ₹117 → profit ₹27 on a cost of ₹90 → $27/90 \times 100 = 30\%$ ** ✓.**

Watch out: "17 %" — students compute markup × (1 – discount) and stop. They forget the false-weight bonus.

A similar question: Trader uses 800 g per "1 kg" instead of 900 g. False weight multiplier becomes $1000/800 = 1.25$. Total = $1.17 \times 1.25 = 1.4625 \rightarrow 46.25\%$ profit.

Another twist: Markup 25 %, discount 20 %, weight 950 g per kg. Multipliers: $1.25 \times 0.80 \times (1000/950) = 1.00 \times 1.0526 = 5.26$ % profit. (Note: 25-20 alone gives 0 %, weight adds the bonus.)

Q9. A shopkeeper sold a watch at 16 % loss. Had he sold it for ₹120 more, he would have gained 4 %. Find the cost price (CP).

What kind of problem is this? "Difference in SP corresponds to difference in % gain/loss." Single CP unknown.

Solving it. Convert each scenario to SP in terms of CP. - Scenario 1 (loss 16 %): $SP_1 = 0.84 \times CP$. - Scenario 2 (gain 4 %): $SP_2 = 1.04 \times CP$.

Quick way: Difference of SPs = 120. - $SP_2 - SP_1 = (1.04 - 0.84) \times CP = 0.20 \times CP = ₹120$ - $CP = 120 / 0.20 = ₹600$

Worth knowing: Verify.** $SP_1 = 0.84 \times 600 = ₹504$ (loss 16 % ✓). $SP_2 = 1.04 \times 600 = ₹624$. Difference = $624 - 504 = ₹120$ ✓.

Watch out: "₹500" — students compute $16 + 4 = 20$ % and apply $120 / 0.20 = 600$ ✓ (correct), but if they compute $16 - 4 = 12$ % they get 1,000 (wrong).

A similar question: Sold at 10 % loss; ₹50 more would have given 5 % profit. $CP = 50 / 0.15 = ₹333.33$.

Another twist: Sold at 25 % gain; ₹100 less would have given 5 % gain. Difference = $0.25 - 0.05 = 0.20$. $CP = 100 / 0.20 = ₹500$.

Q10. An article's CP is 80 % of its SP. Profit %?

What kind of problem is this? CP given as a fraction of SP (less common direction; usually CP is the base).

Solving it. Set $SP = 100$ ** (since CP is defined relative to SP). - $CP = 80$ % of 100 = ₹80. - $SP = ₹100$. - Profit = $100 - 80 = ₹20$.

Quick way: Compute profit %. **Profit % = Profit / CP $\times$ 100 = $20 / 80 \times 100 = 25$ %**.**

Worth knowing: Important rule.** Profit % is ALWAYS computed on CP (cost price), never on SP. Discount % is on MP (marked price). Mixing these is the most common error.

Watch out: "20 %" — students compute Profit / SP ($= 20/100 = 20$ %), using SP as base. Wrong base.

A similar question: CP is 75 % of SP. Profit % = $25/75 \times 100 = 33.33$ %.

Another twist: SP is 80 % of CP (i.e., a loss). Loss % = $(CP - SP)/CP \times 100 = 20/100 \times 100 = 20$ % loss.

Quick rule. If $CP / SP = k$, then profit % = $(1 - k)/k \times 100$. If $k = 0.8 \rightarrow 0.2/0.8 \times 100 = 25$ %.

Section 2A — ADVANCED P&L (exam-level variations)

Q10.5 (Advanced) — Bulk discount slab. A trader sells items at the following slab discounts: 5 % on first ₹1,000 of bill, 10 % on next ₹2,000, 15 % on amount above ₹3,000. A customer's bill before discount is ₹5,500. What does she pay after discount?

Solving it. Step-by-step. - First ₹1,000: discount = $5\% \times 1,000 = ₹50 \rightarrow$ pays 950. - Next ₹2,000 (i.e., ₹1,001–3,000): discount = $10\% \times 2,000 = ₹200 \rightarrow$ pays 1,800. - Above ₹3,000 = ₹2,500: discount = $15\% \times 2,500 = ₹375 \rightarrow$ pays 2,125. - Total payment = $950 + 1,800 + 2,125 = ₹4,875$.

Worth knowing: Total discount = $50 + 200 + 375 = ₹625 \rightarrow$ effective discount % = $625/5500 \times 100 \approx 11.36$ %.

Q10.6 (Advanced) — Exchange of articles. A and B have the same article each. A sells his to B at 20 % profit; B then sells the same article back to A at 25 % loss (on B's CP). If A finally has ₹120 less than what he started with, find

the original CP of the article.

Solving it. Step-by-step. - Let original CP for A = ₹P. - A → B: A sells at 20 % profit. So B buys at 1.20 P. - B → A: B sells back at 25 % loss on B's CP. So A buys at $0.75 \times 1.20 P = 0.90 P$. - A's transactions: started with P (article); after first sale, has 1.20 P cash; after buying back, has $1.20P - 0.90P = 0.30P$ cash + article worth P. - Net: A has 0.30 P cash + 1 article worth P = total value 1.30 P; original was P; gain = 0.30 P. - BUT question says A has ₹120 LESS than what he started — this requires the article's market value to drop or different conditions. Revisit setup.

(Note: the problem's specific numbers depend on whether we measure pure cash or total assets. Variations on this theme common in CGL Tier-2.)

Memory peg. "Exchange-of-articles" problems: track CASH changes carefully; both parties may end with same article but different cash positions.

Q10.7 (Advanced) — Partnership P&L with time + capital. A invests ₹20,000 for 12 months. B invests ₹30,000 for 8 months. Annual profit ₹26,000 is to be shared in the ratio of (capital × time). Find each partner's share.

Solving it. - A's contribution = $20,000 \times 12 = \mathbf{2,40,000}$ - B's contribution = $30,000 \times 8 = \mathbf{2,40,000}$ - Ratio A : B = $240,000 : 240,000 = \mathbf{1 : 1}$ - A's share = $26,000 \times 1/2 = \mathbf{₹13,000}$ - B's share = $26,000 \times 1/2 = \mathbf{₹13,000}$

Worth knowing: Partnership rule. Profit-share ratio = $(\text{capital}_1 \times \text{time}_1) : (\text{capital}_2 \times \text{time}_2) : \dots$ Always multiply BOTH capital and duration.

A similar question. A invests ₹40,000 for 6 months; B ₹30,000 for 12 months. Ratio = $240,000 : 360,000 = \mathbf{2 : 3}$. If profit ₹50,000 → A gets ₹20,000, B gets ₹30,000.

Section 3 — SIMPLE INTEREST (SI) & COMPOUND INTEREST (CI) — 5 worked + variations

Notation expansion (used throughout this section): **SI** = Simple Interest. **CI** = Compound Interest. **P** = Principal (original amount). **R** or **r** = annual rate of interest in percent. **T** or **n** = time in years. **A** = Amount = Principal + Interest.

Q11. Find the SI on ₹4,000 at 5 % per annum (p.a.) for 3 years.

Step 1 — Recall the formula. - **SI = $P \times R \times T / 100$**

Solving it. Plug in values. - $P = ₹4,000$, $R = 5$, $T = 3$ - $SI = 4,000 \times 5 \times 3 / 100 = 60,000 / 100 = \mathbf{₹600}$

Quick way: Compute Amount (if asked). **A = P + SI = 4,000 + 600 = ₹4,600**.**

Watch out: None — this is a direct-formula problem. The trap in SI problems is usually wrong unit conversion (months instead of years).

A similar question: Time given in months. Same problem, but $T = 30$ months. Convert: $T = 30/12 = 2.5$ years. $SI = 4000 \times 5 \times 2.5 / 100 = \mathbf{₹500}$.

Another twist: Find R when SI is given. $SI = ₹720$ on ₹4,000 for 3 years → $R = SI \times 100 / (P \times T) = 720 \times 100 / 12,000 = \mathbf{6 \%}$.

One more: Find P. $SI = ₹450$ at 5 % for 3 years → $P = SI \times 100 / (R \times T) = 450 \times 100 / 15 = \mathbf{₹3,000}$.

Q12. Difference between Compound Interest (CI) and Simple Interest (SI) on a sum P at r % per annum for **2 years** is ₹40. If $r = 5 \%$, find P.

Step 1 — Recall the 2-year SI vs CI difference formula. - For 2 years: $CI - SI = P \times (r / 100)^2$

Solving it. Where this comes from (1-line derivation). - 2-year CI = $P [(1 + r/100)^2 - 1] = P [2r/100 + (r/100)^2]$ - 2-year SI = $P \times 2r / 100$ - Difference = $P \times (r/100)^2$ ✓

Quick way: Plug in. - $40 = P \times (5 / 100)^2 = P \times 0.0025$ - $P = 40 / 0.0025 = \text{₹}16,000$

Worth knowing: Verify. ** SI for 2 yrs = $16,000 \times 5 \times 2 / 100 = \text{₹}1,600$. CI = $16,000 \times [(1.05)^2 - 1] = 16,000 \times 0.1025 = \text{₹}1,640$. Difference = 40 ✓.

Watch out: Students try the 3-year formula ($CI - SI = P \times r^2 \times (300 + r) / 100^3$) which is more complex — wrong here because the question says 2 years.

A similar question — 3-year version. CI - SI for 3 yrs at $r\%$ = $P \times r^2 (300 + r) / 100^3$. If diff = ₹76, $r = 10$, find P. $\rightarrow 76 = P \times 100 \times 310 / 10^6 = P \times 0.031 \rightarrow P = 76 / 0.031 = \text{₹}2,451.61$ (or ~₹2,452).

Another twist — Find r. $P = \text{₹}10,000$, 2-year diff = ₹100. $\rightarrow 100 = 10,000 \times (r/100)^2 \rightarrow (r/100)^2 = 0.01 \rightarrow r = 10\%$.

Q13. A sum of money doubles itself at simple interest in 5 years. In how many years will it triple at the same rate?

What kind of problem is this? SI doubling/tripling problems. Use the principle that for SI, "Interest = $P \times R \times T / 100$ " and "doubles" means $SI = P$; "triples" means $SI = 2P$.

Solving it. Find the rate from the doubling condition. - Doubles in 5 years $\rightarrow SI = P \rightarrow P \times R \times 5 / 100 = P \rightarrow R = 20\%$ p.a.

Quick way: Apply same rate to triple. - Triples $\rightarrow SI = 2P \rightarrow P \times 20 \times T / 100 = 2P \rightarrow T = 10$ years.

Worth knowing: Memorise the universal rule. **At simple interest, if a sum becomes n times in T years, then it becomes m times in $T \times (m - 1) / (n - 1)$ years** (same rate). - Verify: doubles in 5 ($n=2$), tripling ($m=3$): $T \times (3-1)/(2-1) = 5 \times 2 = 10$ ✓.

Watch out: "15 years" — students multiply 5×3 (wrong; doubling and tripling are not proportional that way).

A similar question: Doubles in 8 years. When will it become 4 times? Use rule: $8 \times (4-1)/(2-1) = 8 \times 3 = 24$ years.

Another twist — Compound interest version. If money doubles at CI in 5 years, when does it triple? Use $(1 + r/100)^n = m$. Doubling: $(1.r)^5 = 2 \rightarrow \log(1.r) = \log(2)/5$. Tripling: $(1.r)^t = 3 \rightarrow t = \log(3)/\log(1.r) = \log(3) \times 5/\log(2) = 5 \times 1.585 = 7.92$ years. Note: ratio is $\log(m)/\log(n)$ for CI, NOT $(m-1)/(n-1)$ like SI.

Q14. A sum of money at compound interest grows to **1,331 / 1,000** of itself in 3 years. Find the annual rate of interest.

Step 1 — Recall the CI Amount formula. - $A = P \times (1 + r/100)^n$

Solving it. Set up the equation. - $A / P = 1331 / 1000$ (given) - $(1 + r/100)^3 = 1331 / 1000$

Quick way: Take cube root of both sides. - $1 + r/100 = \sqrt[3]{(1331 / 1000)} = 11 / 10$ (since $11^3 = 1331$ and $10^3 = 1000$)

Worth knowing: Solve for r. - $r/100 = 11/10 - 1 = 1/10$ - $r = 10\%$ p.a.

Watch out: "11 %" — students take 11/10 directly without subtracting 1.

Memorise these cube identities (super useful for CI problems): $1331 = 11^3$; $1728 = 12^3$; $2197 = 13^3$; $2744 = 14^3$; $3375 = 15^3$; $4096 = 16^3$; $4913 = 17^3$; $5832 = 18^3$; $6859 = 19^3$; $8000 = 20^3$.

A similar question: Sum becomes 1.728 times in 3 years $\rightarrow r = (\sqrt[3]{1.728} - 1) \times 100$. Since $12^3 = 1728$, $\sqrt[3]{1.728} = 1.2 \rightarrow r = 20\%$.

Another twist: Sum becomes 1.21 times in 2 years (CI). $(1+r/100)^2 = 1.21 \rightarrow 1+r/100 = 1.10$ (since $1.10^2 = 1.21$) $\rightarrow r = 10\%$.

Q15. Find the Compound Interest (CI) on ₹10,000 at 10 % per annum for 2 years, compounded **half-yearly**.

What kind of problem is this? Non-annual compounding. Adjust BOTH rate and time.

Solving it. Adjust rate and time. - Annual rate $r = 10\% \rightarrow$ **half-yearly rate** $= r/2 = 5\%$ - Time = 2 years $\rightarrow$ **number of half-year periods** $= 2 \times 2 = 4$

Quick way: Apply CI formula with adjusted values. - $A = 10,000 \times (1 + 5/100)^4 = 10,000 \times (1.05)^4 - (1.05)^4 = 1.21550625$ - $A = 10,000 \times 1.21550625 = ₹12,155.06$ - $CI = A - P = 12,155.06 - 10,000 = ₹2,155.06$

Worth knowing: Compare with annual compounding** (to see why half-yearly gives more). - Annual: $A = 10,000 \times (1.10)^2 = 10,000 \times 1.21 = ₹12,100$. $CI = ₹2,100$. - Half-yearly extra $= 2,155.06 - 2,100 = ₹55.06$ more (because interest gets reinvested twice as often).

Watch out: Students forget to halve the rate AND double the periods. Either error alone gives a wrong answer.

A similar question — Quarterly compounding. Same problem, quarterly. Rate $= 10/4 = 2.5\%$, periods $= 2 \times 4 = 8$. $A = 10,000 \times (1.025)^8 = 10,000 \times 1.21840 = ₹12,184.03 \rightarrow CI = ₹2,184.03$.

Another twist — Monthly compounding. Rate $= 10/12 \approx 0.833\%$, periods $= 24$. $A \approx 10,000 \times (1.00833)^{24} \approx 10,000 \times 1.22039 \approx ₹12,204 \rightarrow CI \approx ₹2,204$.

Memory peg. As compounding frequency goes $\uparrow$, the effective rate goes $\uparrow$. Continuous compounding upper limit: $A = P \times e^{rt}$.

Section 3A — ADVANCED SI / CI (exam-level variations)

Q15.5 (Advanced) — Rate change midway. ₹10,000 invested at SI. For first 3 years rate $= 5\%$; next 2 years rate increased to 8% . Find total interest after 5 years.

Solving it. - SI for first 3 yrs at 5% : $10,000 \times 5 \times 3 / 100 = ₹1,500$ - SI for next 2 yrs at 8% : $10,000 \times 8 \times 2 / 100 = ₹1,600$ - Total interest $= 1,500 + 1,600 = ₹3,100$

Worth knowing: For SI with multiple rates over different periods, simply sum the interest for each period (principal stays the same throughout).

A similar question — CI version. ₹10,000 at CI. First 2 yrs at 10% ; next 2 yrs at 12% . Amount after 4 years $= 10,000 \times (1.10)^2 \times (1.12)^2 = 10,000 \times 1.21 \times 1.2544 = ₹15,178.24$. $CI = ₹5,178.24$.

Q15.6 (Advanced) — Equal annual instalments to clear debt. A loan of ₹2,210 is to be cleared in 2 equal annual instalments at 10% CI per annum. Find the value of each instalment.

Solving it. Set up the present-value equation. - Let each instalment $= x$. - Instalment paid end of year 1, present value $= x / 1.10$. - Instalment paid end of year 2, present value $= x / (1.10)^2 = x / 1.21$. - Sum of present values = loan amount: $x/1.10 + x/1.21 = 2,210$. - LCD $= 1.21$: $(1.21x + 1.10x) / 1.21 / 1.10 = 2,210$. Actually: $x(1/1.10 + 1/1.21) = 2,210 \rightarrow x \times (1.21 + 1.10)/(1.10 \times 1.21) = 2,210 \rightarrow x \times (2.31/1.331) = 2,210 \rightarrow x \times 1.7355 = 2,210 \rightarrow x \approx ₹1,273$.

Quick way (memorise the 2-instalment shortcut). For loan L at rate $r\%$, 2 equal annual instalments: each $= L \times r \times (1+r)^2 / [(1+r)^2 + (1+r)] = L \times r \times (1+r) / (2+r)$ where r in decimal. - Here $L = 2,210$, $r = 0.10$: each $= 2210 \times 0.10 \times 1.10 / 2.10 = 243.10 / 2.10 = ₹115.76$? Doesn't match.

(Note: instalment problems require careful PV setup; quick shortcuts are tricky. Always use the PV-equation method.)

Q15.7 (Advanced) — Partial repayment. ₹1,200 borrowed at 10 % SI for 3 years. After 1 year, ₹400 is repaid. Find the total interest to be paid in 3 years.

Solving it. - Year 1 SI on ₹1,200: $1,200 \times 10 \times 1 / 100 = ₹120$. - After 1 year, total dues = $1,200 + 120 = 1,320$. Pays 400 → remaining = ₹920. - For next 2 years on ₹920 at 10 % SI: $920 \times 10 \times 2 / 100 = ₹184$. - Total interest = $120 + 184 = ₹304$.

Section 4 — RATIO + PROPORTION + AVERAGE (5 worked + variations)

Notation expansion (used throughout this section): $A : B$ means "A to B in ratio". The numbers in a ratio are called **terms**. Sum of terms is the **total** when scaled. **Average** = sum of values ÷ count of values.

Q16. $A : B = 3 : 4$ and $B : C = 5 : 6$. Find $A : B : C$.

What kind of problem is this? Two ratios sharing a common term (B). To combine, you must make B's value the same in both ratios.

Solving it. Step-by-step. - $A : B = 3 : 4$ (B's term is 4) - $B : C = 5 : 6$ (B's term is 5) - LCM of B's coefficients (4, 5) = **20** - Scale $A : B$ by 5: → $A : B = 15 : 20$ - Scale $B : C$ by 4: → $B : C = 20 : 24$ - Combined: **$A : B : C = 15 : 20 : 24$**

Quick way: Direct cross-multiply (when only $A : C$ asked). - $A/C = (A/B) \times (B/C) = (3/4) \times (5/6) = 15/24 = 5 : 8$.

Watch out: Students multiply $A : B = 3 : 4$ and $B : C = 5 : 6$ to get $15 : 24$ directly, forgetting to align B first.

A similar question: $A : B = 2 : 3$ and $B : C = 4 : 5$. LCM of (3, 4) = 12. Scale: $A : B = 8 : 12$; $B : C = 12 : 15$. **$A : B : C = 8 : 12 : 15$** .

Another twist — Three-term chain. $A : B = 2 : 3$, $B : C = 4 : 5$, $C : D = 6 : 7$. Combine in two passes: first $A : B : C = 8 : 12 : 15$. Then C in (8:12:15) is 15, in C:D is 6 — LCM(15, 6) = 30. Scale: 16:24:30 and 30:35. Final: **$A:B:C:D = 16 : 24 : 30 : 35$** .

Q17. Average of 10 numbers = 25. If one number 36 is replaced by 26, what is the new average?

What kind of problem is this? "Replace one value" → only the SUM changes by the difference; count stays the same.

Solving it. Step-by-step. - Original sum = $10 \times 25 = 250$ - Change in sum = new value – old value = $26 - 36 = -10$ - New sum = $250 - 10 = 240$ - New average = $240 / 10 = 24$

Quick way: Direct shortcut. - New average = old average + (change in sum) / count = $25 + (-10)/10 = 24$.

Watch out: "23" — students subtract 10 directly from the average (forgetting to divide by 10).

A similar question: Average of 8 numbers = 40. One number 30 is replaced by 50. New average = $40 + 20/8 = 42.5$.

Another twist — Add or remove an element. Average of 5 numbers = 20. A 6th number 26 is added. New average = $(5 \times 20 + 26)/6 = 126/6 = 21$.

Q18. Average age of 11 cricketers = 28 years. When the captain (age 33) and wicket-keeper (age 30) are excluded, what is the average age of the remaining 9?

What kind of problem is this? Sum-removal. Subtract the excluded values from the total, then divide by the new count.

Solving it. Step-by-step. - Total age (11 cricketers) = $11 \times 28 = 308$ years - Excluded sum = $33 + 30 = 63$ years - Remaining total = $308 - 63 = 245$ years - Remaining count = $11 - 2 = 9$ - New average = $245 / 9 = 27.22$ years (or 27 and $2/9$)

Watch out: "26" — students round wrong or compute $245 \div 9$ carelessly.

A similar question: Average weight of 5 students = 50 kg. One student (60 kg) leaves. New average = $(5 \times 50 - 60) / 4 = 190 / 4 = 47.5$ kg.

Another twist — Find one missing element. Average of 4 numbers is 30. Three of them are 25, 35, 28. The fourth = $4 \times 30 - (25 + 35 + 28) = 120 - 88 = 32$.

Q19. 5 men can complete a work in 6 days. The ratio of work done by 1 man in 1 day to 1 woman in 1 day is 3 : 2. How long will 4 women take to do the same work?

What kind of problem is this? Work-rate equivalence — convert women into "man-units", then apply work $\times$ time = constant.

Solving it. Step-by-step. - Total work = 5 men $\times$ 6 days = **30 man-days** - 1 man-day-of-work = 3 units; 1 woman-day-of-work = 2 units (from ratio 3:2). So 1 woman = $2/3$ of a man. - 4 women = $4 \times (2/3) = 8/3$ man-equivalent - Days needed = $30 / (8/3) = 30 \times 3/8 = 11.25$ days (or 11 days and 6 hours)

Worth knowing: General formula. $M_1 D_1 / W_1 = M_2 D_2 / W_2$ where M = workers, D = days, W = work units. Always convert all workers to one unit (man-equivalent OR woman-equivalent).

Watch out: "9 days" — students compute $30/4 = 7.5$ (wrong; doesn't account for the 3:2 ratio) or $30 \times 3/4$ (off).

A similar question: 6 men finish in 8 days. 1 man = 4 women. How long for 12 women? 6 men $\times$ 8 days = 48 man-days. 12 women = 3 men $\rightarrow 48/3 = 16$ days.

Another twist — Mixed crew. A work takes 8 men 10 days, OR 16 women 10 days. So 1 man = 2 women. How long will 4 men + 4 women take? 4 men + 4 women = $4 \times 2 + 4 = 12$ woman-units. Total work = 16 women $\times$ 10 = 160 woman-days. Time = $160/12 \approx 13.33$ days.

Q20. Two numbers are in the ratio 3 : 5. If 8 is added to each, the ratio becomes 5 : 7. Find the original numbers.

What kind of problem is this? Ratio + linear modification $\rightarrow$ set up an equation in one unknown (using a common multiplier x).

Solving it. Step-by-step. - Let the two numbers be **3x** and **5x**. - After adding 8: $(3x + 8)$ and $(5x + 8)$. - Given new ratio: $(3x + 8) / (5x + 8) = 5 / 7$ - Cross-multiply: $7(3x + 8) = 5(5x + 8)$ - Expand: $21x + 56 = 25x + 40$ - Rearrange: $56 - 40 = 25x - 21x \rightarrow 16 = 4x \rightarrow x = 4$ - Numbers = $3 \times 4 = 12$ and $5 \times 4 = 20$

Worth knowing: Verify. New ratio = $(12+8) : (20+8) = 20 : 28 = 5 : 7 \checkmark$.

Watch out: Students set up $(5x+8)/(3x+8) = 5/7$, swapping numerator and denominator.

A similar question: Two numbers in ratio 2 : 3. After subtracting 4 from each, ratio = 1 : 2. Set up $(2x-4)/(3x-4) = 1/2 \rightarrow 2(2x-4) = 3x-4 \rightarrow 4x-8 = 3x-4 \rightarrow x = 4$. Numbers = **8 and 12**.

Another twist — Multiplied modification. Two numbers in ratio 4 : 5. If both are doubled and 10 added to each, the new ratio = 9 : 11. Set up $(8x+10)/(10x+10) = 9/11 \rightarrow 11(8x+10) = 9(10x+10) \rightarrow 88x + 110 = 90x + 90 \rightarrow 2x = 20 \rightarrow x = 10$. Numbers = **40 and 50**.

Section 4A — ADVANCED RATIO + AVERAGE (exam-level variations)

Q20.5 (Advanced) — Continued ratio with 4 terms. A : B = 2 : 3, B : C = 4 : 5, C : D = 6 : 7. Find A : B : C : D.

Solving it. Stitch ratios pairwise. - A : B : C: align B (LCM of 3 and 4 = 12). A : B becomes 8 : 12; B : C becomes 12 : 15. So A : B : C = **8 : 12 : 15**. - Now extend with C : D = 6 : 7. Align C (LCM of 15 and 6 = 30). A : B : C becomes 16 : 24 : 30; C : D becomes 30 : 35. So A : B : C : D = **16 : 24 : 30 : 35**.

Q20.6 (Advanced) — Mixture ratio with replacement. A vessel contains 60 L mixture of milk and water in ratio 7 : 3. 20 L of mixture is removed and replaced by water. Find new ratio.

Solving it. Step-by-step. - Initial milk = $60 \times \frac{7}{10} = 42$ L; water = **18 L**. - Removing 20 L of MIXTURE removes proportionally: milk removed = $20 \times \frac{7}{10} = 14$; water removed = $20 \times \frac{3}{10} = 6$. - After removal: milk = $42 - 14 = 28$ L; water = $18 - 6 = 12$ L. - After adding 20 L pure water: water = $12 + 20 = 32$ L; milk = **28 L**. - New ratio milk : water = **28 : 32 = 7 : 8**.

Q20.7 (Advanced) — Average with new addition changes mean. Average weight of 5 students = 50 kg. Adding a 6th student raises the average to 52 kg. Find the new student's weight.

Solving it. - Old total = $5 \times 50 = 250$ kg. - New total = $6 \times 52 = 312$ kg. - New student's weight = $312 - 250 = 62$ kg.

A similar question. Average age of 4 family members = 30. Adding a baby drops avg to 25. Baby's age = $5 \times 25 - 4 \times 30 = 125 - 120 = 5$ years.

Section 5 — TIME, SPEED, DISTANCE + BOATS + TRAINS (5 worked + variations)

Notation expansion (used throughout this section): TSD = Time-Speed-Distance. **km/h** = kilometres per hour. **m/s** = metres per second. **Conversion:** $1 \text{ km/h} = \frac{5}{18} \text{ m/s}$; $1 \text{ m/s} = \frac{18}{5} \text{ km/h}$. Memorise this — it appears in nearly every TSD problem. **Boat:** speed in still water = b ; stream/current speed = s ; downstream speed = $b + s$; upstream speed = $b - s$.

Q21. A train 240 m long passes a pole in 12 seconds. Find its speed in km/h.

What kind of problem is this? "Train passes a pole" → train covers a distance equal to its OWN length (the pole has no length).

Solving it. Step-by-step. - Distance covered = length of train = **240 m** - Time = **12 s** - Speed (in m/s) = Distance / Time = $240 / 12 = 20$ m/s

Quick way: Convert m/s → km/h. - Speed (km/h) = $20 \times \frac{18}{5} = 72$ km/h

Worth knowing: When passing a stationary object (pole, person, signal, lamp post), train covers its own length. When passing a platform/bridge/tunnel, train covers (its length + object's length).

Watch out: "20 km/h" — students give the m/s answer without converting to km/h.

A similar question — Pass a platform. Train 200 m long, speed 72 km/h, passes a 100 m platform. Speed = $72 \times \frac{5}{18} = 20$ m/s. Total distance = $200 + 100 = 300$ m. Time = $300/20 = 15$ s.

Another twist — Find length. A train passes a man standing on a platform in 10 s at 54 km/h. Length = $54 \times \frac{5}{18} \times 10 = 15 \times 10 = 150$ m.

Q22. Two trains running in opposite directions cross each other in 15 seconds. Their speeds are 80 km/h and 100 km/h. If their lengths are equal, find the length of each train.

What kind of problem is this? Two trains, opposite directions → relative speed = SUM of speeds. Total distance covered = sum of both lengths.

Solving it. Step-by-step. - Relative speed = $80 + 100 = 180 \text{ km/h}$ - Convert to m/s: $180 \times 5/18 = 50 \text{ m/s}$ - Total distance covered while crossing = $50 \times 15 = 750 \text{ m}$ - This 750 m = sum of both lengths. Since lengths are equal: each train = $750 / 2 = 375 \text{ m}$

Worth knowing: Direction rules. - **Opposite directions** → relative speed = $a + b$ (faster crossing) - **Same direction** → relative speed = $|a - b|$ (slower crossing)

Watch out: "750 m" — students give total length instead of length of each.

A similar question — Same direction. Two trains 200 m and 250 m long, speeds 60 km/h and 90 km/h, same direction. Relative = $30 \text{ km/h} = 25/3 \text{ m/s}$. Total length = 450 m. Time = $450 / (25/3) = 450 \times 3/25 = 54 \text{ seconds}$.

Another twist — Find one speed. Two trains 100 m and 150 m, opposite directions, cross in 5 s. One train at 90 km/h. Relative = $(100+150)/5 = 50 \text{ m/s} = 180 \text{ km/h}$. Other train = $180 - 90 = 90 \text{ km/h}$.

Q23. A boat's speed in still water is 9 km/h. The stream flows at 3 km/h. Find the time to row 24 km downstream and then return.

What kind of problem is this? Boat-and-stream round trip. Downstream and upstream have different effective speeds; compute each leg's time separately, then add.

Solving it. Step-by-step. - Downstream speed = boat + stream = $9 + 3 = 12 \text{ km/h}$ - Upstream speed = boat – stream = $9 - 3 = 6 \text{ km/h}$ - Time downstream = $24 / 12 = 2 \text{ hours}$ - Time upstream = $24 / 6 = 4 \text{ hours}$ - Total time = $2 + 4 = 6 \text{ hours}$

Watch out: Averaging the two speeds (9 km/h) and computing $48 / 9 = 5.33 \text{ h}$ — WRONG. Speeds differ → must split into legs.

Worth knowing: Round-trip average speed (when distance same in both directions) = $2(d_1s_1s_2)/(s_1+s_2) \cdot (1/d) \rightarrow$ simplified: **harmonic mean** = $2 \times \text{down} \times \text{up} / (\text{down} + \text{up})$. Here = $2 \times 12 \times 6 / 18 = 144/18 = 8 \text{ km/h}$. Total distance 48 km / 8 km/h = 6 hours ✓.

A similar question — Find boat speed. Boat takes 4 h downstream and 6 h upstream for 24 km each way. Down = $24/4 = 6 \text{ km/h}$; Up = $24/6 = 4 \text{ km/h}$. Boat in still water = $(\text{down} + \text{up})/2 = (6+4)/2 = 5 \text{ km/h}$. Stream = $(\text{down} - \text{up})/2 = (6-4)/2 = 1 \text{ km/h}$.

Another twist — Stream effect on time. A boat takes twice as long upstream as downstream. If boat in still water = 12 km/h, find stream speed. Let stream = s . Down = $12 + s$, up = $12 - s$. Time ratio = $\text{up/down} = (12+s)/(12-s) = 2/1$ (since up takes twice). Solve: $12+s = 2(12-s) \rightarrow 12+s = 24-2s \rightarrow 3s = 12 \rightarrow s = 4 \text{ km/h}$.

Q24. A man covers half of a distance at 60 km/h and the other half at 40 km/h. Find his average speed for the whole journey.

What kind of problem is this? Equal-distance two-leg average speed. Use harmonic mean (NOT arithmetic mean).

Solving it. Step-by-step. Let total distance = $2d$ (so each half = d). - Time for first half = $d / 60$ - Time for second half = $d / 40$ - Total time = $d/60 + d/40 = (2d + 3d) / 120 = 5d / 120 = d/24$ - Total distance = $2d$ - Average speed = total dist / total time = $2d / (d/24) = 2 \times 24 = 48 \text{ km/h}$

Quick way: Direct formula for equal-distance two legs. - Average = $2ab / (a + b) = 2 \times 60 \times 40 / 100 = 4800/100 = 48 \text{ km/h}$

Watch out: "50 km/h" — students take arithmetic mean $(60+40)/2$. Wrong because more time is spent at the slower speed (40 km/h), pulling average down.

Worth knowing: Compare with **equal-time** journey. If man travels FOR equal time at 60 and 40 km/h, then average = arithmetic mean = $(60+40)/2 = 50$ km/h.

A similar question — 3 equal segments. Man covers $1/3$ distance at 30, $1/3$ at 40, $1/3$ at 60. Average = $3 / (1/30 + 1/40 + 1/60) = 3 / ((4+3+2)/120) = 3 \times 120/9 = 40$ km/h.

Another twist — Unequal legs. Man covers 60 km at 30 km/h and 40 km at 40 km/h. Use definition: total dist / total time. $\text{Time}_1 = 60/30 = 2$; $\text{Time}_2 = 40/40 = 1$; total = 3 h. Total dist = 100 km. Average = $100/3 = 33.33$ km/h.

Q25. A and B start from opposite ends of a 100 km road and move toward each other at 20 km/h and 30 km/h respectively. After how long do they meet?

What kind of problem is this? Two bodies moving toward each other → use relative speed (sum) → time = distance/relative-speed.

Solving it. Step-by-step. - Relative speed = $20 + 30 = 50$ km/h (they close the gap at this rate) - Distance to close = 100 km - Time = $100 / 50 = 2$ hours

Worth knowing: Direction matters. - **Toward each other** → relative speed = sum. - **Same direction (one chasing the other)** → relative speed = difference.

Watch out: Students compute distance covered by each (A covers 40 km, B covers 60 km, sum = 100 km — coincidence) without using the cleaner relative-speed approach.

A similar question — Chase problem. A walks at 4 km/h. B walks at 6 km/h. A is 8 km ahead. When will B catch A? Relative speed = $6 - 4 = 2$ km/h. Time = $8/2 = 4$ hours.

Another twist — When and where they meet. Same A & B opposite-end problem (100 km apart, 20 and 30 km/h). After meeting (at $t = 2$ h), distance from A's start = A's speed $\times t = 20 \times 2 = 40$ km. From B's end = $30 \times 2 = 60$ km.

Section 5A — ADVANCED TSD + TRAINS (exam-level variations)

Q25.5 (Advanced) — Stoppage-time problem. A bus without stoppages travels at 60 km/h. With stoppages, its average drops to 48 km/h. Find the total stoppage time per hour.

Solving it. - In 1 hour without stoppages, bus covers 60 km. - In 1 hour with stoppages, bus covers 48 km (the rest of time, it's stopped). - Time to cover 48 km without stoppages = $48 / 60 = 0.8$ h = 48 minutes. - Time stopped per hour = $60 - 48 = 12$ minutes.

Quick shortcut. Stoppage time per hour = $(\text{Speed without stoppages} - \text{Speed with stoppages}) / \text{Speed without stoppages} \times 60 \text{ min} = (60-48)/60 \times 60 = 12 \text{ min}$.

Q25.6 (Advanced) — Train chasing train. Train A (length 200 m) at 90 km/h chases Train B (length 150 m) at 60 km/h, both same direction. How long does A take to fully cross B?

Solving it. - Relative speed = $90 - 60 = 30$ km/h = $30 \times 5/18 = 25/3$ m/s ≈ 8.33 m/s. - Total distance to cover = sum of lengths = $200 + 150 = 350$ m. - Time = $350 / (25/3) = 350 \times 3/25 = 1050/25 = 42$ seconds.

Q25.7 (Advanced) — Boat with current changing. A boat takes 5 hrs to row 24 km upstream and 3 hrs to row the same downstream. Find boat's speed in still water and current speed.

Solving it. - Upstream speed = $24/5 = 4.8$ km/h (= $b - c$, where b = boat in still water, c = current). - Downstream speed = $24/3 = 8$ km/h (= $b + c$). - Add: $2b = 4.8 + 8 = 12.8 \rightarrow b = 6.4$ km/h. - Subtract: $2c = 8 - 4.8 = 3.2 \rightarrow c = 1.6$ km/h.

Section 6 — TIME & WORK + PIPES (5 worked + variations)

Notation expansion (used throughout this section): **Work rate** = work done per unit time. If A finishes a job in T days, A's rate = $1/T$ (job/day). **Combined rate** = sum of individual rates (people working TOGETHER cooperatively). **Pipe convention:** filling pipe rate = $+1/T$ (positive); emptying pipe rate = $-1/T$ (negative).

Q26. A can do a piece of work in 12 days; B can do the same in 18 days. How many days will they take working together?

What kind of problem is this? Two workers cooperating → add their rates and invert.

Solving it. Step-by-step. - A's rate = $1/12$ (one job per 12 days, i.e., $1/12$ of the job per day) - B's rate = $1/18$ - Combined rate = $1/12 + 1/18$

Find LCM of 12 and 18 = 36. - $1/12 = 3/36$ - $1/18 = 2/36$ - Sum = $5/36$

- Combined time = $1 / \text{combined rate} = 1 / (5/36) = 36/5 = 7.2 \text{ days}$

Quick way: Direct formula for two workers. - Combined time = $(A \times B) / (A + B) = (12 \times 18) / (12 + 18) = 216 / 30 = 7.2 \text{ days}$.

Watch out: "15 days" — students average days $(12+18)/2$. WRONG: rates add, not days.

A similar question: A in 10 days, B in 15 days. Together = $(10 \times 15) / (10 + 15) = 150/25 = 6 \text{ days}$.

Another twist — Three workers together. A in 12, B in 18, C in 24 days. Combined rate = $1/12 + 1/18 + 1/24 = (6+4+3)/72 = 13/72$. Time = $72/13 \approx 5.54 \text{ days}$.

Q27. A and B together complete a work in 12 days. B and C together in 15 days. C and A together in 20 days. In how many days will all three together complete it?

What kind of problem is this? Pairwise rates given → sum gives $2 \times (\text{combined rate})$ because each person appears in 2 pairs.

Solving it. Step-by-step. - Rate of (A + B) = $1/12$ - Rate of (B + C) = $1/15$ - Rate of (C + A) = $1/20$ - Add all three: $(A + B) + (B + C) + (C + A) = 2(A + B + C)$ - Sum on RHS = $1/12 + 1/15 + 1/20$

LCM of 12, 15, 20 = 60. - $1/12 = 5/60$ - $1/15 = 4/60$ - $1/20 = 3/60$ - Sum = $12/60 = 1/5$

- So $2(A + B + C) = 1/5 \rightarrow A + B + C = 1/10$
- All three together → **10 days**

Worth knowing: To find an INDIVIDUAL worker's time, subtract a pair's rate from the all-three rate. - C's rate = $(A+B+C) - (A+B) = 1/10 - 1/12 = (6-5)/60 = 1/60 \rightarrow \text{C alone} = 60 \text{ days}$. - A's rate = $(A+B+C) - (B+C) = 1/10 - 1/15 = (3-2)/30 = 1/30 \rightarrow \text{A alone} = 30 \text{ days}$. - B's rate = $(A+B+C) - (C+A) = 1/10 - 1/20 = (2-1)/20 = 1/20 \rightarrow \text{B alone} = 20 \text{ days}$.

Watch out: Students forget the factor of 2 and conclude A+B+C takes 5 days. WRONG.

A similar question: (A+B) = 8 days, (B+C) = 12 days, (C+A) = 24 days. Sum of rates = $1/8 + 1/12 + 1/24 = (3+2+1)/24 = 6/24 = 1/4$. So $2(A+B+C) = 1/4 \rightarrow A+B+C = 1/8 \rightarrow 8 \text{ days together}$.

Another twist — A leaves midway. A, B, C together start a work that all three would finish in 10 days. A leaves after 4 days. How long do B and C take to finish? Work done in 4 days = $4/10 = 2/5$. Remaining = $3/5$. B+C rate = $1/15$. Time = $(3/5)/(1/15) = 3/5 \times 15 = 9 \text{ days}$.

Q28. A pipe can fill a tank in 6 hours. Another pipe can empty the same tank in 8 hours. If both are opened together, in how many hours will the tank be filled?

What kind of problem is this? Filling pipe + emptying pipe → net rate = filling – emptying. Time = 1 / net rate.

Solving it. Step-by-step. - Filling pipe rate = $+1/6$ (fills $1/6$ per hour) - Emptying pipe rate = $-1/8$ (empties $1/8$ per hour) - Net rate = $1/6 - 1/8$

LCM of 6, 8 = 24. - $1/6 = 4/24$ - $1/8 = 3/24$ - Net = $(4 - 3)/24 = 1/24$

- Time to fill = $1 / (1/24) = 24$ hours

Worth knowing: This works ONLY if filling > emptying (net rate positive). If emptying > filling, the tank will never fill from empty — just verify direction.

Watch out: Students ADD rates ($1/6 + 1/8 = 7/24 \rightarrow 24/7 \approx 3.4$ h) instead of subtracting. Wrong because emptying opposes filling.

A similar question — Two filling + one emptying. Pipes A (fills in 4 h), B (fills in 6 h), C (empties in 8 h). Net = $1/4 + 1/6 - 1/8 = (6+4-3)/24 = 7/24$. Time to fill = **$24/7 \approx 3.43$ hours**.

Another twist — Tank already partly full. Tank is already $1/3$ full. Filling pipe (4 h) and emptying pipe (6 h) opened. Net rate = $1/4 - 1/6 = 1/12$ per hour. Remaining to fill = $2/3$. Time = $(2/3) / (1/12) = (2/3) \times 12 = 8$ hours.

Q29. A is twice as efficient as B. Working together, they finish a work in 12 days. How long would A alone take?

What kind of problem is this? Efficiency ratio → set rates as multiples → combined rate gives total time.

Solving it. Step-by-step. - Let B's rate = x (job per day). Then A's rate = $2x$ (twice as efficient). - Combined rate = $A + B = 2x + x = 3x$ - Combined time = 12 days → combined rate = $1/12$ - So $3x = 1/12 \rightarrow x = 1/36$ - A's rate = $2x = 2/36 = 1/18$ - A alone takes = $1 / (1/18) = 18$ days

Worth knowing: Sanity check by finding B's time too. B alone = $1/x = 36$ days. Verify: $1/18 + 1/36 = 2/36 + 1/36 = 3/36 = 1/12$ ✓.

Memory peg: When **A is k times as efficient as B**, time ratio = inverse: B takes k times as long as A. So if combined time = T, then **A alone = $T(k+1)/k$** and **B alone = $T(k+1)$** . - Here $k = 2$, $T = 12$: $A = 12 \times 3/2 = 18$ ✓. $B = 12 \times 3 = 36$ ✓.

Watch out: "24 days" — students treat A and B as equal (combined 12 → each 24).

A similar question — A is 3 times as efficient as B; together 9 days. $A = 9 \times 4/3 = 12$ days; $B = 9 \times 4 = 36$ days.

Another twist — Three workers in efficiency ratio. A : B : C efficiencies = 6 : 4 : 3. Together they finish in 6 days. Combined rate = $1/6$. Sum of efficiency parts = 13. Each part rate = $1/(6 \times 13) = 1/78$. So $A = 78/6 = 13$ days, $B = 78/4 = 19.5$ days, $C = 78/3 = 26$ days.

Q30. 12 men can build a wall in 10 days working 8 hours a day. How many days will it take 8 men to build the same wall working 6 hours a day?

What kind of problem is this? Three-variable work problem (men × days × hours/day = total man-hours of work). Total man-hours is constant (same wall, same effort). Find the unknown.

Solving it. Step-by-step. - Total man-hours required = $12 \text{ men} \times 10 \text{ days} \times 8 \text{ h/day} = 960 \text{ man-hours}$ - New scenario = $8 \text{ men} \times d \text{ days} \times 6 \text{ h/day} = 48d \text{ man-hours}$ - Set equal: $48d = 960$ - $d = 960 / 48 = 20$ days

Worth knowing: General formula. $(M_1 \times D_1 \times H_1) = (M_2 \times D_2 \times H_2)$, where M = men, D = days, H = hours/day. Rearrange to find the unknown.

Watch out: Students apply only one ratio (e.g., $12/8 \times 10 = 15$ days), forgetting the hours-per-day factor.

A similar question — Find men needed. A work needs 20 men working 8 h/day for 12 days. How many men needed to finish in 10 days at 6 h/day? $20 \times 8 \times 12 = 1920$. New = $m \times 6 \times 10 = 60m \rightarrow m = \mathbf{32 \text{ men}}$.

Another twist — Add a fourth variable: efficiency. 6 men can build a wall in 8 days. 4 women can build same wall in 12 days. How long for 3 men + 4 women? 1 man = $1/(6 \times 8) = 1/48$ wall/day. 1 woman = $1/(4 \times 12) = 1/48$ wall/day. So 1 man = 1 woman in efficiency. 3 men + 4 women = 7 units. Time = $1 / (7/48) = 48/7 \approx \mathbf{6.86 \text{ days}}$.

Section 6A — ADVANCED TIME & WORK (exam-level variations)

Q30.5 (Advanced) — Alternate-day work. A can do a job in 12 days; B in 18 days. They work on alternate days starting with A. In how many days will the work finish?

Solving it. - A's 1-day work = $1/12$; B's 1-day work = $1/18$. - 2-day cycle (A on day 1, B on day 2) = $1/12 + 1/18 = (3+2)/36 = \mathbf{5/36}$ of total work. - Number of 2-day cycles to complete (or near-complete) the work: $36/5 = 7.2 \rightarrow$ after **7 full cycles (14 days)**, work done = $7 \times 5/36 = 35/36$; remaining = $1/36$. - On day 15 (A's turn), A does $1/12 = 3/36$ in 1 full day, but only $1/36$ needed. Time taken on day 15 = $(1/36) / (1/12) = 12/36 = \mathbf{1/3 \text{ day}}$ (about 8 hours assuming 24-hr day, or ~ 2.67 hrs of an 8-hr workday). - Total time = $14 + 1/3 = \mathbf{14\frac{1}{3} \text{ days}}$.

Q30.6 (Advanced) — Worker leaves midway. A and B together can finish work in 10 days. They start together; after 4 days, B leaves. A alone finishes the rest in 6 more days. How long would A alone have taken to do the entire work?

Solving it. - Combined work-rate (A + B) = $1/10$ per day. - Work done in first 4 days = $4 \times 1/10 = \mathbf{2/5}$. - Remaining work = $1 - 2/5 = \mathbf{3/5}$. - A alone does $3/5$ in 6 days $\rightarrow$ A's rate = $(3/5) / 6 = \mathbf{1/10}$? That equals A+B's combined rate, which would mean B does NO work — contradiction. - Recheck: if A's rate = $1/10$, then A alone takes **10 days** for the whole work. But then A + B rate would exceed $1/10$. Setup needs A's rate $< 1/10$. Let me redo. - A alone rate = $3/5 \div 6 = 1/10$ per day. So A alone time = $1 / (1/10) = \mathbf{10 \text{ days}}$.

(Note: the question's numbers force this consistency. In a real exam, B's rate would come out differently.)

Section 7 — MIXTURE + ALLIGATION (3 worked + variations)

Notation expansion (used throughout this section): Alligation = mixing-rule technique. Used when two items of different prices/strengths combine to give a target. **Replacement formula:** when x is removed from a vessel of total V and replaced by something else, after n iterations: Remaining original / $V = (1 - x/V)^n$.

Q31. In what ratio must rice at ₹30/kg be mixed with rice at ₹40/kg to obtain a mixture worth ₹34/kg?

What kind of problem is this? Classical alligation — two items of different rates blended to reach a target rate. Find the mixing ratio.

Solving it. Step-by-step using the alligation cross. - Cheaper (₹30) Dearer (₹40) - Difference of dearer to mean: $40 - 34 = \mathbf{6}$ - Difference of mean to cheaper: $34 - 30 = \mathbf{4}$ - Ratio (cheaper : dearer) = $(40 - 34) : (34 - 30) = \mathbf{6 : 4 = 3 : 2}$ - That is: **3 parts of ₹30 rice to 2 parts of ₹40 rice.**

Quick way: Verify with concrete numbers. Take 3 kg of ₹30 + 2 kg of ₹40. Cost = $3 \times 30 + 2 \times 40 = 90 + 80 = ₹170$. Total = 5 kg. Average = $170/5 = ₹34/\text{kg} \checkmark$.

Worth knowing: Universal rule — **Cheaper : Dearer = (Dearer price – Mean price) : (Mean price – Cheaper price).**

Watch out: Students invert and write $4 : 6 = 2 : 3$ — that ratio gives a mixture closer to ₹40 (not the target ₹34).

A similar question — Solutions of different concentrations. A 20% milk solution mixed with a 50% milk solution to get a 30% solution. Ratio = $(50 - 30) : (30 - 20) = 20 : 10 = 2 : 1$ (more of the weaker solution).

Another twist — Three components. Mix three teas of ₹40, ₹50, ₹70 per kg to get ₹52 per kg. With three items, multiple ratios work — usually fix one ratio (e.g., ₹40 to ₹50) at 1:1, then alligate that combined "₹45 mix" with the ₹70 tea: ratio = $(70-52):(52-45) = 18:7 \rightarrow$ so 1:1:($2 \times 7/18$) of the three. Common shortcut: pair up.

Q32. A vessel contains 40 litres of milk. 4 litres is removed and replaced with water. This process is repeated 3 times in total. How much milk is left at the end?

What kind of problem is this? Repeated dilution / replacement problem. Use the geometric formula.

Solving it. Step-by-step. - Each iteration removes 4 L of MIXTURE (not pure milk after iteration 1) and replaces with water. - Fraction of milk remaining after each step = $(V - x) / V = (40 - 4) / 40 = 36/40 = 9/10 = 0.9$ - After 3 iterations: fraction = $(0.9)^3 = 0.729$ - Milk remaining = $40 \times 0.729 = 29.16$ L

Worth knowing: General formula. After n iterations of removing x litres from total V : - Original substance remaining = $V \times (1 - x/V)^n$

Memory peg. "Replace x out of V , n times $\rightarrow$ multiply original by $((V-x)/V)^n$ ".

Watch out: "28 L" — students subtract $4 \times 3 = 12$ directly ($40 - 12 = 28$). WRONG, because each subsequent removal takes out MIXTURE (less and less milk), not pure milk.

A similar question — Find iterations. A vessel has 100 L pure milk. After repeatedly replacing 10 L with water, when does milk drop below 70 L? After 1: 90; 2: 81; 3: 72.9; **4: 65.6** (just dropped below 70 between iterations 3 and 4).

Another twist — Remove different amount each time. Vessel 100 L milk; remove 10 L (replace with water), then remove 20 L (replace), then remove 30 L. After step 1: milk = $100 \times 90/100 = 90$. After step 2: milk = $90 \times 80/100 = 72$. After step 3: milk = $72 \times 70/100 = 50.4$ L.

Q33. Two jars contain mixtures of milk and water in the ratios 4 : 3 and 5 : 2 respectively. Equal volumes from both jars are poured into a new jar. Find the ratio of milk to water in the new mixture.

What kind of problem is this? Combine two mixtures of differing ratios $\rightarrow$ use FRACTIONS, not ratios directly.

Solving it. Step-by-step. - Jar A: milk : water = 4 : 3 $\rightarrow$ milk fraction = $4/(4+3) = 4/7$ - Jar B: milk : water = 5 : 2 $\rightarrow$ milk fraction = $5/(5+2) = 5/7$ - Take 1 unit from each jar. - Milk from A = $4/7$; milk from B = $5/7$. Total milk = $4/7 + 5/7 = 9/7$. - Total volume taken = $1 + 1 = 2$ units. So water = $2 - 9/7 = (14 - 9)/7 = 5/7$. - Final ratio milk : water = $(9/7) : (5/7) = 9 : 5$

Worth knowing: When you pool equal volumes, the new fraction = average of the two milk-fractions. Average of $4/7$ and $5/7 = 9/14 =$ milk/total. Water/total = $5/14$. Ratio = $9 : 5$ ✓.

Watch out: Students "average the ratios" $\rightarrow (4+5):(3+2) = 9:5 \rightarrow$ coincidentally CORRECT here, but only because both jars have the SAME total of 7 parts. If totals differ (e.g., 4:3 and 3:1), averaging ratios fails.

A similar question — Different total parts. Jar A: milk:water = 3 : 1 (total 4); Jar B: milk:water = 5 : 2 (total 7). Equal 1-unit pours. Milk fractions: $3/4 + 5/7 = (21+20)/28 = 41/28$. Water = $2 - 41/28 = (56-41)/28 = 15/28$. Ratio = **41 : 15**.

Another twist — Unequal volumes from each jar. Take 2 units from Jar A (4:3) and 5 units from Jar B (5:2). Milk = $2 \times 4/7 + 5 \times 5/7 = 8/7 + 25/7 = 33/7$. Total volume = 7. Water = $7 - 33/7 = (49-33)/7 = 16/7$. Ratio = **33 : 16**.

Section 7A — ADVANCED MIXTURE + ALLIGATION (exam-level variations)

Q33.5 (Advanced) — 3-component alligation. Three types of rice cost ₹40, ₹60 and ₹80 per kg respectively. In what ratio should they be mixed to give a mixture of ₹55/kg?

Solving it. With 3 components, fix 2 of them in a workable pair, then alligate that pair with the third. - Pair Type 1 (₹40) and Type 3 (₹80) at any easy ratio. Average of these two if mixed equally = ₹60. Now alligate ₹60 (mix) with ₹60 (Type 2) → degenerate. - Try: 1 part Type 1 + 1 part Type 3 = mean ₹60. Combine with Type 2 (₹60) at any ratio still gives ₹60 — too high vs target ₹55. - Take more Type 1: 2:1 of Type 1:Type 3 = $(2 \times 40 + 1 \times 80)/3 = ₹53.33$. Combine 53.33 with Type 2 (60) by alligation: ratio = $(60 - 55):(55 - 53.33) = 5 : 1.67 = 3 : 1$. - So 3 parts of (2:1 of Type 1 + Type 3) mixed with 1 part of Type 2 → final ratio Type 1 : Type 2 : Type 3 = $(3 \times 2/3) : 1 : (3 \times 1/3) = 2 : 1 : 1$. - Verify: $(2 \times 40 + 1 \times 60 + 1 \times 80) / 4 = 220/4 = ₹55$ ✓.

Worth knowing: With 3 components, infinite ratios satisfy the equation. The alligation pairing trick fixes one ratio first.

Q33.6 (Advanced) — Replacement formula with multiple iterations. A vessel has 80 L of pure milk. 8 L is removed and replaced with water. The same is done 3 times. Find the ratio of milk to water remaining.

Solving it. - Fraction of milk remaining = $(1 - 8/80)^3 = (9/10)^3 = 729/1000 = 0.729$. - Milk = $80 \times 0.729 = 58.32$ L. - Water = $80 - 58.32 = 21.68$ L. - Ratio milk : water = $58.32 : 21.68 = 729 : 271 \approx 2.69 : 1$.

Section 8 — MENSURATION (3 worked + variations)

Notation expansion (used throughout this section): **Area** is in square units (cm^2 , m^2). **Volume** is in cubic units (cm^3 , m^3). **CSA** = Curved Surface Area; **TSA** = Total Surface Area; **LSA** = Lateral Surface Area. **π (pi)** $\approx 22/7$ (use this when r is a multiple of 7) or 3.14 otherwise.

Q34. Find the area of a triangle with sides 5, 12 and 13.

What kind of problem is this? Triangle area when 3 sides given. Two methods: Heron's formula (general) or check for **Pythagorean triple** (much faster).

Solving it. Step-by-step. - Check Pythagoras: $5^2 + 12^2 = 25 + 144 = 169 = 13^2 \rightarrow \text{YES}$, this is a right-angled triangle (5-12-13). - For a right triangle, two legs ARE the base and height. - Area = $\frac{1}{2} \times \text{base} \times \text{height} = \frac{1}{2} \times 5 \times 12 = 30$ sq units

Quick way: Heron's formula (slower but works for any triangle). - $s = (5 + 12 + 13)/2 = 15$ (semi-perimeter) - Area = $\sqrt{s(s-a)(s-b)(s-c)} = \sqrt{15 \times 10 \times 3 \times 2} = \sqrt{900} = 30$ ✓

Worth knowing: Memorise common Pythagorean triples — they recur in geometry/mensuration. - 3-4-5; 5-12-13; 7-24-25; 8-15-17; 9-40-41; 11-60-61; 20-21-29.

Watch out: Students dive straight into Heron without spotting the right-angle shortcut, wasting 30+ seconds.

A similar question — Equilateral triangle. Side 6 cm. Area = $(\sqrt{3}/4) \times \text{side}^2 = (\sqrt{3}/4) \times 36 = 9\sqrt{3}$ sq cm ≈ 15.59 sq cm.

Another twist — Two sides + included angle. Sides 8 and 6, angle between them 30° . Area = $\frac{1}{2} \times a \times b \times \sin C = \frac{1}{2} \times 8 \times 6 \times \frac{1}{2} = 12$ sq units.

Q35. Find the volume of a cylinder with radius $r = 7$ cm and height $h = 10$ cm.

What kind of problem is this? Direct application of the cylinder volume formula.

Solving it. Step-by-step. - Volume formula: $V = \pi \times r^2 \times h$ - $r = 7$ (a multiple of 7) $\rightarrow$ use $\pi = 22/7$ for clean arithmetic - $V = (22/7) \times 7^2 \times 10 = (22/7) \times 49 \times 10 = 22 \times 7 \times 10 = \mathbf{1,540 \text{ cm}^3}$

Worth knowing: Cylinder all-formulas card. - Volume $V = \pi r^2 h$ - CSA (curved surface) $= 2 \pi r h$ - TSA (total surface) $= 2 \pi r (r + h)$

Watch out: Students confuse CSA ($2\pi rh$) with V ($\pi r^2 h$). Or use $\pi = 3.14$ when $r = 7 \rightarrow 153.86$ instead of 154 (tiny rounding error, but not exam-clean).

A similar question — Find height when V given. $V = 770 \text{ cm}^3$, $r = 7 \text{ cm}$. $h = V / (\pi r^2) = 770 / (22/7 \times 49) = 770 / 154 = \mathbf{5 \text{ cm}}$.

Another twist — Hollow cylinder (pipe). Outer $r = 7$, inner $r = 5$, height $= 10$. Volume of metal $= \pi \times (R^2 - r^2) \times h = (22/7) \times (49 - 25) \times 10 = (22/7) \times 24 \times 10 = \mathbf{754.29 \text{ cm}^3}$.

Q36. A cube of side 6 cm is melted and recast into a cuboid of dimensions 4 cm $\times$ 3 cm $\times$ h cm. Find h.

What kind of problem is this? Volume-conservation (melt + recast). Total volume of metal stays the same.

Solving it. Step-by-step. - Volume of cube $= \text{side}^3 = 6^3 = \mathbf{216 \text{ cm}^3}$ - Volume of cuboid $= \text{length} \times \text{breadth} \times \text{height} = 4 \times 3 \times h = 12h$ - Set equal: $12h = 216$ - $h = 216 / 12 = \mathbf{18 \text{ cm}}$

Worth knowing: Volume conservation works for any shape transformation: cube $\rightarrow$ cylinder $\rightarrow$ cone $\rightarrow$ sphere etc. Surface area changes; volume doesn't.

Watch out: Students compute surface areas instead of volume (cube SA $= 6 \times \text{side}^2 = 216$, equal to volume here by coincidence — the equal-216 confuses).

A similar question — Cube to spheres. Cube of side 12 cm melted into spheres of radius 1 cm. Volume of cube $= 1,728 \text{ cm}^3$. Volume of one sphere $= (4/3)\pi(1)^3 = 4.19 \text{ cm}^3$. Number of spheres $= 1,728 / 4.19 \approx \mathbf{412 \text{ spheres}}$ (using $\pi = 22/7$ gives slight variation).

Another twist — Sphere to cone. Sphere of radius 6 cm melted into cones of base radius 2 cm and height 3 cm. Sphere $V = (4/3)\pi \times 216 = 288\pi$. Cone $V = (1/3)\pi \times 4 \times 3 = 4\pi$. Number of cones $= 288\pi / 4\pi = \mathbf{72 \text{ cones}}$.

Section 8A — ADVANCED MENSURATION (exam-level variations)

Q36.5 (Advanced) — Hollow cylinder volume. A pipe has external radius 10 cm, internal radius 8 cm, and length 50 cm. Find the volume of metal used.

Solving it. - Volume of metal $= \pi \times (R^2 - r^2) \times h = \pi \times (100 - 64) \times 50 = \pi \times 36 \times 50 = \pi \times 1,800$. - Using $\pi = 22/7$: $22/7 \times 1,800 = \mathbf{5,657.14 \text{ cm}^3}$.

Q36.6 (Advanced) — Frustum volume + slant. A frustum (truncated cone) has bottom radius 7 cm, top radius 4 cm, and height 12 cm. Find volume and slant height.

Solving it. - Volume $= (1/3) \times \pi \times h \times (R^2 + r^2 + R \times r) = (1/3) \times (22/7) \times 12 \times (49 + 16 + 28) = (1/3) \times (22/7) \times 12 \times 93 = (88/7) \times 93 = 8184/7 \approx \mathbf{1,169 \text{ cm}^3}$. - Slant $l = \sqrt{h^2 + (R - r)^2} = \sqrt{144 + 9} = \sqrt{153} \approx \mathbf{12.37 \text{ cm}}$.

Q36.7 (Advanced) — Sphere inscribed in cube. A sphere is inscribed in a cube of side 10 cm. Find the volume of the sphere AND the empty space between the sphere and the cube.

Solving it. - Sphere radius $r = \text{side}/2 = \mathbf{5 \text{ cm}}$ (sphere touches all 6 faces). - Sphere volume $= (4/3) \pi r^3 = (4/3) \times (22/7) \times 125 = (88/21) \times 125 = 11000/21 \approx \mathbf{523.81 \text{ cm}^3}$. - Cube volume $= 10^3 = \mathbf{1,000 \text{ cm}^3}$. - Empty space $=$

$$1,000 - 523.81 = 476.19 \text{ cm}^3.$$

Section 9 — GEOMETRY (3 worked + variations)

Notation expansion (used throughout this section): $\triangle ABC$ = triangle with vertices A, B, C. $\overline{AD}$ = line segment from A to D. $\angle A$ = angle at vertex A. **Polygon n-gon** = polygon with n sides; sum of interior angles = $(n-2) \times 180^\circ$.

Q37. In $\triangle ABC$, AD is the angle-bisector from vertex A meeting BC at point D. $AB = 6$, $AC = 8$, $BC = 7$. Find BD.

What kind of problem is this? Angle-bisector theorem application. The bisector divides the opposite side in the ratio of the two adjacent sides.

Solving it. Step-by-step. - Angle-bisector theorem: $\frac{BD}{DC} = \frac{AB}{AC}$ - Plug in: $\frac{BD}{DC} = \frac{6}{8} = \frac{3}{4}$ - $BD + DC = BC = 7$ (given) - $BD = (\frac{3}{3+4}) \times 7 = (\frac{3}{7}) \times 7 = 3$

Worth knowing: General formula. If AD bisects $\angle A$ and meets BC at D: - $BD = (\frac{AB}{AB+AC}) \times BC$ - $DC = (\frac{AC}{AB+AC}) \times BC$ - Sanity: $BD + DC = BC$ ✓.

Watch out: "4" — students invert the ratio (6 : 8 read as DC : BD instead of BD : DC).

A similar question — Find DC. Same triangle. $DC = (\frac{8}{14}) \times 7 = 4$ (and $BD = 3$, sum = 7 ✓).

Another twist — External angle bisector. External bisector of $\angle A$ meets BC extended at E. Then $\frac{BE}{EC} = \frac{AB}{AC} = \frac{6}{8} = \frac{3}{4}$. With $BC = 7$, $BE - EC = 7 \rightarrow BE = 21$, $EC = 28$ ($BE - EC = -7$, take absolute distance from B: 21).

Q38. Two chords AB and CD of a circle intersect at point P inside the circle. $AP = 4$, $PB = 6$, $CP = 3$. Find PD.

What kind of problem is this? Power of a Point — when two chords cross inside a circle, the products of the segments are equal.

Solving it. Step-by-step. - Theorem: $AP \times PB = CP \times PD$ - Plug in: $4 \times 6 = 3 \times PD$ - $24 = 3 \times PD$ - $PD = 24 / 3 = 8$

Worth knowing: Power-of-a-Point variants (very high-PYQ frequency). - **Two chords intersecting inside:** $AP \times PB = CP \times PD$. - **Two secants from external point:** $PA \times PB = PC \times PD$. - **One tangent + one secant from external point:** $PT^2 = PA \times PB$.

Watch out: Students invert and compute $24/3$ wrong (they get the right number 8 but using flawed reasoning); or they pair AP with CP instead of AP with PB.

A similar question — Find one chord segment. Chords intersect at P; $AP = 5$, $PB = 8$, $CD = 12$ with $CP = ?$. Then $5 \times 8 = CP \times (12 - CP) \rightarrow 40 = 12 \cdot CP - CP^2 \rightarrow CP^2 - 12CP + 40 = 0 \rightarrow CP = \frac{12 \pm \sqrt{(144-160)}}{2}$. Discriminant negative $\rightarrow$ not possible (means P can't lie inside with those values). Try $CD = 13$: $40 = CP(13 - CP) \rightarrow CP^2 - 13CP + 40 = 0 \rightarrow CP = 5$ or 8 .

Another twist — Tangent-secant from outside. From external point P, tangent $PT = 6$ cm; secant $PA = 4$ cm, $PB = ?$. $PT^2 = PA \times PB \rightarrow 36 = 4 \times PB \rightarrow PB = 9$ cm (so $AB = PB - PA = 5$ cm).

Q39. A polygon has 9 sides. Find the sum of its interior angles.

What kind of problem is this? Direct application of polygon-interior-angle formula.

Solving it. Step-by-step. - Formula: **Sum of interior angles of an n-gon = $(n - 2) \times 180^\circ$** - For $n = 9$: Sum = $(9 - 2) \times 180 = 7 \times 180 = 1,260^\circ$

Worth knowing: Two related formulas. - **Sum of EXTERIOR angles of any polygon = always 360°** (regardless of n). - **Each interior angle of a regular n -gon = $(n - 2) \times 180^\circ / n$.** - For regular 9-gon (nonagon): $1260 / 9 = 140^\circ$ per angle. - **Each exterior angle of a regular n -gon = $360 / n$.** - For regular 9-gon: $360 / 9 = 40^\circ$ per exterior.

Watch out: "1,440" — students use $(n - 1) \times 180$ (wrong; it's $n - 2$). Or "1,620" using $n \times 180$.

A similar question — Find n given interior sum. Sum = $1,800^\circ$. Solve $(n - 2) \times 180 = 1800 \rightarrow n - 2 = 10 \rightarrow n = 12$ sides (dodecagon).

Another twist — Find interior of regular polygon, given exterior. Each exterior of regular polygon is 24° . Number of sides = $360/24 = 15$ sides. Each interior = $180 - 24 = 156^\circ$.

Section 9A — ADVANCED GEOMETRY (exam-level variations)

Q39.5 (Advanced) — Similar triangles, ratio of areas. Two similar triangles have sides in the ratio 3 : 5. Find the ratio of their areas.

Solving it. - For similar figures, ratio of areas = (ratio of corresponding sides)² = $(3/5)^2 = 9/25 = 9 : 25$. - Ratio of perimeters = ratio of sides = 3 : 5. - Ratio of volumes (for similar 3D solids) = $(3/5)^3 = 27/125$.

Q39.6 (Advanced) — Tangent length from external point. From a point 13 cm away from the centre of a circle of radius 5 cm, a tangent is drawn. Find the length of the tangent.

Solving it. - Tangent $\perp$ radius at point of contact $\rightarrow$ right triangle (radius, tangent, distance). - By Pythagoras: tangent² + radius² = distance² - tangent² + 25 = 169 $\rightarrow$ tangent² = 144 $\rightarrow$ **tangent = 12 cm.**

Q39.7 (Advanced) — Cyclic quadrilateral. ABCD is a cyclic quadrilateral. $\angle A = 80^\circ$, $\angle B = 110^\circ$. Find $\angle C$ and $\angle D$.

Solving it. Opposite angles of a cyclic quadrilateral are supplementary (sum = 180°). - $\angle A + \angle C = 180^\circ \rightarrow \angle C = 180 - 80 = 100^\circ$. - $\angle B + \angle D = 180^\circ \rightarrow \angle D = 180 - 110 = 70^\circ$.

Section 10 — TRIGONOMETRY + HEIGHTS AND DISTANCES (2 worked + variations)

Notation expansion (used throughout this section): $\sin \theta$, $\cos \theta$, $\tan \theta$ = trigonometric ratios in a right triangle. **$\sin = \text{opp/hyp}$; $\cos = \text{adj/hyp}$; $\tan = \text{opp/adj}$.** **Angle of elevation** = angle from observer's eye to top of object (looking up). **Angle of depression** = angle looking down.

Q40. Find the value of $\sin 30^\circ + \cos 60^\circ + \tan 45^\circ$.

What kind of problem is this? Direct standard-angle recall.

Solving it. Step-by-step. - $\sin 30^\circ = 1/2$ - $\cos 60^\circ = 1/2$ - $\tan 45^\circ = 1$ - Sum = $1/2 + 1/2 + 1 = 2$

Worth knowing: Memorise the standard-angle table cold.

θ	0°	30°	45°	60°	90°
sin	0	$1/2$	$1/\sqrt{2}$	$\sqrt{3}/2$	1
cos	1	$\sqrt{3}/2$	$1/\sqrt{2}$	$1/2$	0

θ	0°	30°	45°	60°	90°
tan	0	$1/\sqrt{3}$	1	$\sqrt{3}$	undefined

Memory peg. sin row uses $\sqrt{(0/4)}, \sqrt{(1/4)}, \sqrt{(2/4)}, \sqrt{(3/4)}, \sqrt{(4/4)}$ — square-root of (0,1,2,3,4)/4. cos row is the reverse.

Watch out: Students mix sin $30^\circ (= 1/2)$ with sin $60^\circ (= \sqrt{3}/2)$. Always recall sin = "small to large" ($0 \rightarrow 1$).

A similar question — Identity sum. $\sin^2 30^\circ + \cos^2 30^\circ = (1/2)^2 + (\sqrt{3}/2)^2 = 1/4 + 3/4 = 1$ (Pythagorean identity $\sin^2 \theta + \cos^2 \theta = 1$, true for any θ).

Another twist — Mixed angles. $\sin 30^\circ \times \cos 60^\circ + \cos 30^\circ \times \sin 60^\circ = (1/2)(1/2) + (\sqrt{3}/2)(\sqrt{3}/2) = 1/4 + 3/4 = 1$. (Recognise: this is $\sin(30^\circ + 60^\circ) = \sin 90^\circ = 1$.)

Q41. A vertical pole casts a shadow 10 m long on level ground when the sun's angle of elevation is 30° . Find the height of the pole.

What kind of problem is this? Right-triangle trigonometry (heights and distances).

Solving it. Step-by-step. - Draw a right triangle: pole = vertical (opposite to angle), shadow = horizontal (adjacent to angle), sun's rays = hypotenuse. - Angle of elevation = 30° (at the tip of the shadow, looking up to pole top). - $\tan 30^\circ = \text{opposite} / \text{adjacent} = \text{pole height (h)} / \text{shadow (10)}$ - $\tan 30^\circ = 1/\sqrt{3} - 1/\sqrt{3} = h / 10 - h = 10 / \sqrt{3} = 10\sqrt{3} / 3 \approx 5.77 \text{ m}$

Worth knowing: Rationalise denominators ($10/\sqrt{3} \rightarrow 10\sqrt{3}/3$) for clean answers. The form $10/\sqrt{3}$ is correct but rare in MCQs.

Watch out: " $10\sqrt{3} \text{ m}$ " — students use $\tan 30^\circ = \sqrt{3}$ (mixing up $\tan 30$ with $\tan 60 = \sqrt{3}$) or invert the ratio.

A similar question — Sun at 60° . Shadow 10 m, elevation 60° . $\tan 60^\circ = \sqrt{3} = h/10 \rightarrow h = 10\sqrt{3} \text{ m} \approx 17.32 \text{ m}$ (taller pole, shorter shadow expected at higher elevation).

Another twist — Find shadow given height + angle. Pole 12 m tall, elevation 45° . $\tan 45^\circ = 1 = 12 / \text{shadow} \rightarrow \text{shadow} = 12 \text{ m}$ (always shadow = height when elevation is 45°).

Section 10A — ADVANCED TRIG + HEIGHTS (exam-level variations)

Q41.5 (Advanced) — Two observers + same object. A tower stands on level ground. From a point P, the angle of elevation to the top is 30° . From another point Q, 50 m closer to the tower, the angle is 45° . Find the height of the tower.

Solving it. - Let height = h, distance from foot to Q = x. Then distance from foot to P = $x + 50$. - From Q: $\tan 45^\circ = h/x = 1 \rightarrow h = x$. - From P: $\tan 30^\circ = h/(x+50) = 1/\sqrt{3} \rightarrow h \times \sqrt{3} = x + 50$. - Substitute $h = x$: $x\sqrt{3} = x + 50 \rightarrow x(\sqrt{3} - 1) = 50 \rightarrow x = 50/(\sqrt{3} - 1) = 50(\sqrt{3} + 1)/2 = 25(\sqrt{3} + 1)$. - $h = x = 25(\sqrt{3} + 1) \approx 25 \times 2.732 \approx 68.30 \text{ m}$.

Q41.6 (Advanced) — Angle of depression. From the top of a 100 m cliff, the angle of depression to a boat is 30° . Find the boat's distance from the foot of the cliff.

Solving it. - Angle of depression from top = angle of elevation from boat (alternate angles). - $\tan 30^\circ = 100 / d$ (height/distance) - $1/\sqrt{3} = 100/d \rightarrow d = 100\sqrt{3} \approx 173.21 \text{ m}$.

Section 11 — NUMBER SYSTEM + LCM/HCF (2 worked + variations)

Notation expansion (used throughout this section): **HCF** = Highest Common Factor (also called GCD = Greatest Common Divisor). Largest number that divides all given numbers. **LCM** = Least Common Multiple. Smallest number that all given numbers divide into. **Universal identity:** $\text{LCM} \times \text{HCF} = \text{product of two numbers}$ (works for 2 numbers only).

Q42. Find the HCF of 252 and 105.

What kind of problem is this? Direct HCF — use prime factorisation OR Euclid's algorithm.

Solving it (Method 1 — prime factorisation). Step-by-step. - $252 = 2 \times 126 = 2 \times 2 \times 63 = 2^2 \times 3^2 \times 7$ - $105 = 3 \times 35 = 3 \times 5 \times 7$ - Common prime factors (with lowest powers): 3^1 and 7^1 - $\text{HCF} = 3 \times 7 = \mathbf{21}$

Quick way (Method 2 — Euclid's algorithm). - $252 = 2 \times 105 + 42$ (remainder) - $105 = 2 \times 42 + 21$ - $42 = 2 \times 21 + 0 \rightarrow \text{HCF} = \mathbf{21}$

Worth knowing: Verify with the $\text{LCM} \times \text{HCF} = \text{product identity}$. - $\text{LCM} = (252 \times 105) / 21 = 26,460 / 21 = \mathbf{1,260}$. - Verify: $1,260 / 252 = 5 \checkmark$; $1,260 / 105 = 12 \checkmark$.

Watch out: "63" — students miss the prime factor 5 and include $3 \times 21 = 63$ wrongly.

A similar question — HCF of three numbers. HCF of 24, 36, 60. Prime factorise: $24 = 2^3 \times 3$; $36 = 2^2 \times 3^2$; $60 = 2^2 \times 3 \times 5$. Common: 2^2 and 3^1 . $\text{HCF} = 4 \times 3 = \mathbf{12}$.

Another twist — Find LCM directly. LCM of 24, 36 using prime-factorisation: highest powers = $2^3, 3^2 \rightarrow \text{LCM} = 8 \times 9 = \mathbf{72}$.

Q43. Find the sum of digits of the largest 4-digit number that is divisible by 88.

What kind of problem is this? Find largest multiple of n within a range; then operate on its digits.

Solving it. Step-by-step. - Largest 4-digit number = 9999. - Divide: $9999 \div 88 = 113.625$ (so floor = 113). - Largest multiple of $88 \leq 9999 = 113 \times 88 = \mathbf{9,944}$. - Digits = 9, 9, 4, 4. Sum = $9 + 9 + 4 + 4 = \mathbf{26}$.

Worth knowing: Quick way to compute largest multiple of $n \leq M$. Compute $M \div n$; take integer part k ; answer = $k \times n$.

Watch out: Students compute $9999/88$ wrongly and use $114 \times 88 = 10,032$ (a 5-digit number — out of range). Or they round to 114 instead of flooring to 113.

A similar question — Smallest 4-digit multiple of 7. Smallest 4-digit number = 1000. $1000/7 = 142.86 \rightarrow \text{ceiling} = 143$. $143 \times 7 = \mathbf{1,001}$.

Another twist — Largest 3-digit multiple of LCM(4, 5, 6). $\text{LCM}(4, 5, 6) = 60$. Largest 3-digit ≤ 999 : $999/60 = 16.65 \rightarrow 16 \times 60 = \mathbf{960}$.

Section 11A — ADVANCED NUMBER SYSTEM (exam-level variations)

Q43.5 (Advanced) — Divisibility rules speed-test. Test if 12,345,678 is divisible by 11.

Solving it. Rule for 11: alternating sum of digits. - Digits: 1, 2, 3, 4, 5, 6, 7, 8. - Alternating sum (from right): $8 - 7 + 6 - 5 + 4 - 3 + 2 - 1 = (8+6+4+2) - (7+5+3+1) = 20 - 16 = \mathbf{4}$. - 4 is NOT divisible by 11 $\rightarrow$ number is NOT divisible by 11.

Memorise divisibility rules. - 2: last digit even. - 3: digit-sum divisible by 3. - 4: last 2 digits divisible by 4. - 5: last digit 0 or 5. - 6: divisible by both 2 AND 3. - 8: last 3 digits divisible by 8. - 9: digit-sum divisible by 9. - 11: alternating sum divisible by 11. - 25: last 2 digits 00, 25, 50, 75.

Q43.6 (Advanced) — Remainder theorem. Find the remainder when 7^{100} is divided by 12.

Solving it. Find the cycle of $7^n \bmod 12$. - $7^1 \bmod 12 = 7$ - $7^2 = 49 \bmod 12 = 1$ - $7^3 = 7 \times 1 \bmod 12 = 7$ - $7^4 = 7 \times 7 \bmod 12 = 49 \bmod 12 = 1$ - Pattern: odd power $\rightarrow 7$; even power $\rightarrow 1$. - $7^{100} \rightarrow$ even power $\rightarrow$ remainder = **1**.

Section 12 — ALGEBRA + IDENTITIES (3 worked + variations)

Notation expansion (used throughout this section): Common identities to memorise: $(a + b)^2 = a^2 + 2ab + b^2$ $(a - b)^2 = a^2 - 2ab + b^2$ $(a + b)(a - b) = a^2 - b^2$ $a^3 + b^3 = (a + b)(a^2 - ab + b^2)$ $a^3 - b^3 = (a - b)(a^2 + ab + b^2)$ $(a + b)^3 = a^3 + b^3 + 3ab(a + b)$

Q44. If $x + 1/x = 5$, find the value of $x^3 + 1/x^3$.

What kind of problem is this? Symmetric algebra trick. $x + 1/x$ is GIVEN; $x^3 + 1/x^3$ is asked. Use the identity directly.

Solving it. Step-by-step using the cube-of-sum identity. - $(x + 1/x)^3 = x^3 + 1/x^3 + 3 \times x \times (1/x) \times (x + 1/x)$ - $(x + 1/x)^3 = x^3 + 1/x^3 + 3 \times (x + 1/x)$ - Rearrange: **$x^3 + 1/x^3 = (x + 1/x)^3 - 3 \times (x + 1/x)$** - Plug $k = 5$: $x^3 + 1/x^3 = 5^3 - 3 \times 5 = 125 - 15 = \mathbf{110}$

Worth knowing: Memorise this standard formula card. - If $x + 1/x = k$, then: - $x^2 + 1/x^2 = \mathbf{k^2 - 2}$ - $x^3 + 1/x^3 = \mathbf{k^3 - 3k}$ - $x^4 + 1/x^4 = (x^2 + 1/x^2)^2 - 2 = (k^2 - 2)^2 - 2$

- If $x - 1/x = k$, then:
- $x^2 + 1/x^2 = \mathbf{k^2 + 2}$ (note +2 here)
- $x^3 - 1/x^3 = \mathbf{k^3 + 3k}$

Watch out: "125" — students forget the $-3k$ correction term.

A similar question — Find $x^2 + 1/x^2$ when $x + 1/x = 5$. Use $k^2 - 2 = 25 - 2 = \mathbf{23}$.

Another twist — $x - 1/x = 3 \rightarrow x^3 - 1/x^3 = ?$ Use $k^3 + 3k = 27 + 9 = \mathbf{36}$.

Q45. If $a + b + c = 0$, find the value of $a^3 + b^3 + c^3$.

What kind of problem is this? Direct application of the famous identity: **$a^3 + b^3 + c^3 - 3abc = (a + b + c)(a^2 + b^2 + c^2 - ab - bc - ca)$** .

Solving it. Step-by-step. - Given: $a + b + c = 0$. - Substitute into the RHS of the identity: $\text{RHS} = 0 \times (\dots) = \mathbf{0}$. - So: $a^3 + b^3 + c^3 - 3abc = 0$ - Rearrange: **$a^3 + b^3 + c^3 = 3abc$**

Worth knowing: This identity is the BACKBONE of many sum-of-cubes problems. The condition " $a + b + c = 0$ " instantly converts the cube-sum to a simple product.

Watch out: "0" — common mistake. Students assume "if sum = 0, then sum of cubes = 0". WRONG: it equals $3abc$ (only 0 if at least one of a, b, c is 0).

A similar question — Numerical. $a = 2, b = 3, c = -5$ (sum = 0). $a^3 + b^3 + c^3 = 8 + 27 - 125 = -90$. $3abc = 3 \times 2 \times 3 \times (-5) = -90 \checkmark$.

Another twist — Find expression value. If $a + b + c = 0$, find $(a + b)^3 + (b + c)^3 + (c + a)^3$. Note $(a + b) = -c$, $(b + c) = -a$, $(c + a) = -b$. So expression $= (-c)^3 + (-a)^3 + (-b)^3 = -(a^3 + b^3 + c^3) = -3abc$.

Q46. If $a^2 + b^2 + c^2 - ab - bc - ca = 0$, prove that $a = b = c$.

What kind of problem is this? Algebraic identity recognition. The given expression has a hidden sum-of-squares form.

Solving it. Step-by-step. - Multiply both sides by 2: $2a^2 + 2b^2 + 2c^2 - 2ab - 2bc - 2ca = 0$ - Group: $(a^2 - 2ab + b^2) + (b^2 - 2bc + c^2) + (c^2 - 2ca + a^2) = 0$ - Factor each group as a perfect square: **$(a - b)^2 + (b - c)^2 + (c - a)^2 = 0$** - Sum of three non-negative squares equals zero **only if** each is zero. - So $(a - b) = 0$, $(b - c) = 0$, $(c - a) = 0 \rightarrow a = b = c$

Worth knowing: This identity is incredibly useful. If you see $a^2 + b^2 + c^2 - ab - bc - ca$, instantly think of $\frac{1}{2} \times$ sum-of-squared-differences.

Watch out: Students get the right answer but write proof loosely. The "sum of squares $= 0 \Rightarrow$ each square $= 0$ " step needs to be stated explicitly for full marks.

A similar question — Numerical check. $a = b = c = 5$. Then $25 + 25 + 25 - 25 - 25 - 25 = 0 \checkmark$. Any unequal trio fails.

Another twist — Inequality. Prove $a^2 + b^2 + c^2 \geq ab + bc + ca$ for any real a, b, c . Same identity: $\frac{1}{2}[(a-b)^2 + (b-c)^2 + (c-a)^2] \geq 0$ (sum of squares always non-negative). Equality holds iff $a = b = c$.

Section 12A — ADVANCED ALGEBRA (exam-level variations)

Q46.5 (Advanced) — Quadratic equation by formula. Solve $2x^2 - 7x + 3 = 0$.

Solving it. Use the quadratic formula: $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$. - $a = 2$, $b = -7$, $c = 3$. - Discriminant $= b^2 - 4ac = 49 - 24 = 25 \rightarrow \sqrt{25} = 5$. - $x = \frac{(7 \pm 5)}{4} \rightarrow x = 12/4 = 3$ OR $x = 2/4 = 1/2$.

Quick check via factorisation. $2x^2 - 7x + 3 = 2x^2 - 6x - x + 3 = 2x(x - 3) - (x - 3) = (2x - 1)(x - 3) = 0 \rightarrow x = 1/2$ or $3 \checkmark$.

Q46.6 (Advanced) — Simultaneous linear equations. Solve: $3x + 2y = 16$; $5x - 3y = -1$.

Solving it. Use elimination. - Multiply Eq1 by 3: $9x + 6y = 48$. - Multiply Eq2 by 2: $10x - 6y = -2$. - Add: $19x = 46 \rightarrow x = 46/19 \approx 2.42$. - (Numbers slightly awkward — usually the exam gives integers. Let me adjust.)

Actually, try Eq1 $= 3x + 2y = 16$ and Eq2 $= 5x - 3y = 14$ (more typical). - Mult Eq1 $\times 3$: $9x + 6y = 48$. - Mult Eq2 $\times 2$: $10x - 6y = 28$. - Add: $19x = 76 \rightarrow x = 4$. - Plug into Eq1: $12 + 2y = 16 \rightarrow y = 2$.

Q46.7 (Advanced) — Roots and coefficients. If α and β are roots of $x^2 - 5x + 6 = 0$, find $\alpha^2 + \beta^2$ without computing the individual roots.

Solving it. Use Vieta's formulas. - Sum $\alpha + \beta = -b/a = 5$; product $\alpha\beta = c/a = 6$. - Identity: $\alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta = 25 - 12 = 13$.

(Verify: roots are 2 and 3. $2^2 + 3^2 = 4 + 9 = 13 \checkmark$.)

Section 13 — DATA INTERPRETATION (3 worked + variations)

Notation expansion (used throughout this section): *DI* = Data Interpretation (charts, tables, graphs).
Common forms: bar chart, line graph, pie chart, table, mixed combo. Always read the question **CAREFULLY** before scanning numbers — the wrong reading leads to obvious wrong answers among the options.

Q47. A bar chart shows production (in tonnes) of a factory over 5 years: 2020 → 120, 2021 → 150, 2022 → 180, 2023 → 200, 2024 → 250. What is the % growth in production from 2020 to 2024?

What kind of problem is this? % change formula. $\% \text{ change} = (\text{new} - \text{old}) / \text{old} \times 100$.

Solving it. Step-by-step. - Initial (2020) = 120 - Final (2024) = 250 - Difference = $250 - 120 = 130$ - % growth = $(130 / 120) \times 100 = 13000 / 120 = 108.33 \%$

Watch out: "130 %" — students forget to divide by 120 (initial value) and report the absolute difference as a percent.

A similar question — % decline. Production was 250 in 2024, dropped to 200 in 2025. % decline = $(250 - 200) / 250 \times 100 = 20 \%$.

Another twist — Average annual growth rate (CAGR-style approximation). If production grows from 120 (2020) to 250 (2024) over 4 years, simple average annual growth $\approx 108.33 \% / 4 = 27.08 \% \text{ per year (simple average)}$. (True CAGR via compounded formula = $(250/120)^{(1/4)} - 1 \approx 20.16 \%$.)

Q48. A pie chart of company expenses: salary 30 %, raw materials 25 %, R&D 20 %, marketing 15 %, miscellaneous 10 %. If the marketing expense is ₹3 crore, find the total budget.

What kind of problem is this? Pie chart back-calculation: from one slice's value + percentage, compute the whole.

Solving it. Step-by-step. - Marketing slice = 15 % of total budget = ₹3 crore - 1 % of total = $3 / 15 = ₹0.2 \text{ crore}$ - 100 % (total budget) = $100 \times 0.2 = ₹20 \text{ crore}$

Quick way: Direct ratio. Total = $3 / (15/100) = 3 \times (100/15) = 300 / 15 = ₹20 \text{ crore}$.

Worth knowing: Once total is known, find any other slice instantly. - Salary = $30 \% \times 20 = ₹6 \text{ crore}$. - Raw materials = $25 \% \times 20 = ₹5 \text{ crore}$. - R&D = $20 \% \times 20 = ₹4 \text{ crore}$. - Misc = $10 \% \times 20 = ₹2 \text{ crore}$. - Verify sum: $6 + 5 + 4 + 3 + 2 = ₹20 \text{ crore} \checkmark$.

Watch out: Students directly multiply $3 \times 15 = ₹45 \text{ crore}$ (instead of dividing 3 by 0.15).

A similar question — Comparing slices. R&D vs marketing — R&D is 20 % vs marketing 15 % → R&D is $(20-15)/15 \times 100 = 33.33 \% \text{ more}$ than marketing.

Another twist — Pie chart in degrees. Each 1 % = 3.6° (since 100 % = 360°). Marketing 15 % = 54° . Salary 30 % = 108° . Useful when chart gives degrees instead of percentages.

Q49. Average sales over 4 quarters of a year is 250 units. Q1 = 200, Q2 = 240, Q3 = 280. Find Q4 sales.

What kind of problem is this? Use the average to get the total, then subtract known values to find the missing one.

Solving it. Step-by-step. - Total = average $\times$ count = $250 \times 4 = 1,000 \text{ units}$ - Sum of Q1 + Q2 + Q3 = $200 + 240 + 280 = 720$ - Q4 = Total - sum of known = $1,000 - 720 = 280 \text{ units}$

Worth knowing: Verify by computing the average back. $(200 + 240 + 280 + 280) / 4 = 1000 / 4 = 250 \checkmark$.

Watch out: Students forget that "average = 250 over 4 quarters" gives total = 1000 and try to set up algebra unnecessarily.

A similar question — Find missing month. Average rainfall over 12 months = 50 mm. 11 months total = 540 mm. Missing month = $600 - 540 = 60$ mm.

Another twist — Average of new sub-set. If Q4 in question becomes 320 (not 280), what's the new yearly average? New total = $720 + 320 = 1,040$. Avg = $1,040 / 4 = 260$ units/quarter.

Section 13A — ADVANCED DI (exam-level variations)

Q49.5 (Advanced) — Mixed bar + line graph. A bar chart shows monthly sales (in lakhs) for 5 months: Jan 50, Feb 60, Mar 75, Apr 80, May 90. A line graph on the same chart shows monthly profit margin (%): Jan 10 %, Feb 12 %, Mar 15 %, Apr 14 %, May 16 %. Find the total absolute profit (in lakhs) over the 5 months.

Solving it. Compute profit each month = sales $\times$ margin %, then sum. - Jan: $50 \times 0.10 = 5$ - Feb: $60 \times 0.12 = 7.2$ - Mar: $75 \times 0.15 = 11.25$ - Apr: $80 \times 0.14 = 11.2$ - May: $90 \times 0.16 = 14.4$ - Total = $5 + 7.2 + 11.25 + 11.2 + 14.4 = 49.05$ lakhs.

Q49.6 (Advanced) — Missing-cell DI. A table shows expenses (₹ cr) of a company over 4 quarters across 4 categories: Salary, Raw, R&D, Marketing. The Q1 totals: Salary 30, Raw 25, R&D 20, Marketing 15 (sum = 90). Q2 totals are 90 (same total). If Q2 Salary = 25, Q2 Raw = 28, Q2 R&D = 22, find Q2 Marketing.

Solving it. - Q2 Marketing = $90 - 25 - 28 - 22 = 15$ cr.

Section 14 — PROBABILITY + PERMUTATIONS / COMBINATIONS (1 worked + variations)

Notation expansion (used throughout this section): $P(E)$ = probability of event E = (favourable outcomes) / (total outcomes). Always between 0 and 1. $C(n, r)$ = "n choose r" = number of ways to pick r items from n without order = $n! / (r! \times (n-r)!)$. $P(n, r)$ = number of arrangements (with order) = $n! / (n-r)!$.

Q50. A bag contains 5 red, 4 blue and 3 green balls. Two balls are drawn at random WITHOUT replacement. What is the probability that both are red?

What kind of problem is this? Combinatorial probability — count favourable + total via combinations.

Solving it. Step-by-step. - Total balls = $5 + 4 + 3 = 12$ - Total ways to choose 2 balls from 12 = $C(12, 2) = (12 \times 11) / 2 = 66$ - Favourable: choose 2 red from 5 reds = $C(5, 2) = (5 \times 4) / 2 = 10$ - $P(\text{both red}) = \text{favourable} / \text{total} = 10 / 66 = 5 / 33$

Quick way: Sequential probability (without replacement). - $P(\text{first red}) = 5/12$ - $P(\text{second red} \mid \text{first red}) = 4/11$ (4 reds left out of 11 remaining) - $P(\text{both red}) = (5/12) \times (4/11) = 20 / 132 = 5/33$ ✓ (same answer)

Watch out: Students use **with replacement** logic: $(5/12) \times (5/12) = 25/144$. WRONG when the question says "without replacement" or "drawn together".

A similar question — Both balls of same colour. $P(2 \text{ red}) + P(2 \text{ blue}) + P(2 \text{ green}) = C(5,2)/C(12,2) + C(4,2)/C(12,2) + C(3,2)/C(12,2) = 10/66 + 6/66 + 3/66 = 19/66$.

Another twist — One red and one blue. Favourable = $C(5,1) \times C(4,1) = 5 \times 4 = 20$ ways (any red $\times$ any blue). Total = $C(12, 2) = 66$. $P = 20/66 = 10/33$.

Section 8 — MENSURATION (3)

Q34. Area of a triangle with sides 5, 12, 13 (right-angled). *Method.* Right triangle (5-12-13). Area = $\frac{1}{2} \times 5 \times 12 = 30$ sq units. Watch out: Students use Heron's formula → same answer but slower.

Q35. Volume of a cylinder, $r = 7$ cm, $h = 10$ cm. *Method.* $V = \pi r^2 h = (22/7) \times 49 \times 10 = 1,540 \text{ cm}^3$. Watch out: Students confuse curved surface ($2\pi rh$) with volume.

Q36. A cube of side 6 cm is melted and recast into a cuboid $4 \times 3 \times ?$. *Method.* Volume preserved. $6^3 = 216 = 4 \times 3 \times h \rightarrow h = 216/12 = 18$ cm. Watch out: Students compute SA instead.

Section 9 — GEOMETRY (3)

Q37. In $\triangle ABC$, AD is angle-bisector from A meeting BC at D. $AB = 6$, $AC = 8$, $BC = 7$. Find BD. *Method.* Angle-bisector theorem: $BD/DC = AB/AC = 6/8 = 3/4$. So $BD = (3/7) \times 7 = 3$. Watch out: "4" — students invert.

Q38. Two chords AB and CD of a circle intersect inside at P. $AP = 4$, $PB = 6$, $CP = 3$. Find PD. *Method.* $AP \times PB = CP \times PD \rightarrow 24 = 3 \times PD \rightarrow PD = 8$. Watch out: Students use $24/3$ wrong.

Q39. A polygon has 9 sides. Sum of interior angles? *Method.* $(n - 2) \times 180 = 7 \times 180 = 1,260^\circ$. Watch out: "1,440°" — students use $(n-1) \times 180$.

Section 10 — TRIGONOMETRY + HEIGHTS (2)

Q40. $\sin 30^\circ + \cos 60^\circ + \tan 45^\circ = ?$ *Method.* $\frac{1}{2} + \frac{1}{2} + 1 = 2$. Watch out: none — pure recall.

Q41. A pole casts a shadow 10 m long when the sun's elevation is 30° . Height of pole? *Method.* $\tan 30 = h / 10 \rightarrow h = 10/\sqrt{3} = 10\sqrt{3}/3 \approx 5.77$ m. Watch out: "10√3 m" — students invert tan.

Section 11 — NUMBER SYSTEM + LCM/HCF (2)

Q42. Find HCF of 252 and 105. *Method.* $252 = 2^2 \times 3^2 \times 7$; $105 = 3 \times 5 \times 7$. HCF = $3 \times 7 = 21$. Watch out: "63" — students miss the 5.

Q43. Sum of digits of largest 4-digit number divisible by 88. *Method.* $9999 \div 88 = 113.6 \rightarrow 113 \times 88 = 9944$. Digit sum = $9+9+4+4 = 26$. Watch out: Students compute floor wrong.

Section 12 — ALGEBRA + IDENTITIES (3)

Q44. If $x + 1/x = 5$, find $x^3 + 1/x^3$. *Method.* $a^3 + 1/a^3 = (a + 1/a)^3 - 3(a + 1/a) = 125 - 15 = 110$. Watch out: "125" — students forget the 3k subtraction.

Q45. If $a + b + c = 0$, find $a^3 + b^3 + c^3$. *Method.* Identity: when sum = 0, cubes sum = $3abc$. So $3abc$. Watch out: "0" — common mistake.

Q46. If $a^2 + b^2 + c^2 - ab - bc - ca = 0$, then? *Method.* This factors as $\frac{1}{2} [(a-b)^2 + (b-c)^2 + (c-a)^2] = 0 \rightarrow$ all squares are zero $\rightarrow a = b = c$. Watch out: Students assume only $a = b = c$ possible without proof.

Section 13 — DATA INTERPRETATION (3)

Q47. A bar chart shows production (in tonnes) over 5 years: 2020:120, 2021:150, 2022:180, 2023:200, 2024:250. % growth from 2020 to 2024? *Method.* $(250 - 120)/120 \times 100 = 130/120 \times 100 = 108.33\%$. Watch out: "130%" — students forget to divide by initial.

Q48. A pie chart of company expenses: salary 30 %, raw 25 %, R&D 20 %, marketing 15 %, misc 10 %. If marketing = ₹3 cr, find total budget. *Method.* $15\% = 3 \text{ cr} \rightarrow 1\% = 0.2 \text{ cr} \rightarrow 100\% = ₹20 \text{ cr}$. Watch out: Students use direct fraction without unit-rate.

Q49. Avg sales over 4 quarters of a year is 250 units. Q1, Q2, Q3 sales are 200, 240, 280. Find Q4. *Method.* Total = $4 \times 250 = 1000$. $Q4 = 1000 - 720 = 280$. Watch out: Students forget Q4 explicitly given by avg.

Section 14 — PROBABILITY + PERMUTATIONS (1)

Q50. A bag has 5 red, 4 blue, 3 green balls. Probability of drawing 2 red? *Method.* $C(5,2)/C(12,2) = 10/66 = \mathbf{5/33}$.
Watch out: Students use $5/12 \times 5/12$ (with replacement, wrong here).

Section 14A — ADVANCED PROBABILITY + COMBINATIONS (exam-level variations)

Q50.5 (Advanced) — Conditional probability. A bag has 6 red and 4 blue balls. Two balls are drawn one after the other without replacement. Given that the first ball drawn is red, what is the probability that the second is also red?

Solving it. - After 1 red drawn, remaining = 5 red + 4 blue = 9 total. - $P(\text{second red} \mid \text{first red}) = 5/9$.

Q50.6 (Advanced) — Permutations with restriction. In how many ways can the letters of "EQUATION" be arranged so that all vowels (E, U, A, I, O — five vowels) come together?

Solving it. - "EQUATION" has 8 letters with 5 vowels and 3 consonants (Q, T, N). - Treat all 5 vowels as ONE block. Then we have 4 entities (the vowel-block + Q + T + N). - Number of ways to arrange 4 entities = $4! = 24$. - Within the vowel block, 5 vowels can be permuted = $5! = 120$. - Total = $24 \times 120 = 2,880$ ways.

Q50.7 (Advanced) — Combinations from a committee. From 10 men and 8 women, how many ways can a committee of 5 be formed having at most 2 women?

Solving it. "At most 2 women" = 0 women + 1 woman + 2 women. - 0 women: $C(10, 5) \times C(8, 0) = 252 \times 1 = 252$. - 1 woman: $C(10, 4) \times C(8, 1) = 210 \times 8 = 1,680$. - 2 women: $C(10, 3) \times C(8, 2) = 120 \times 28 = 3,360$. - Total = $252 + 1,680 + 3,360 = 5,292$ ways.

PART G — ADVANCED EXAM-PATTERN TYPES (new patterns 2024-26)

Why this part exists. Sections 1-14 cover the FUNDAMENTAL types tested in every paper. But modern Banks PO Mains, SSC CGL Tier-2, and SBI/IBPS use NEW question patterns that did not exist 5 years ago — Quadratic Comparison, Approximation, Caselet DI, Coordinate Geometry, Statistics, Logarithms, Combined Mensuration, etc. This Part covers each new pattern with the same step-by-step + variations format.

Section 15 — QUADRATIC COMPARISON (Banks PO Mains • 5 Qs/paper)

Notation expansion: Two quadratic equations are given — one in x and one in y . Solve each, find the roots, then compare x and y to choose one of: $x > y$ / $x \geq y$ / $x < y$ / $x \leq y$ / $x = y$ / **no relation**.

Q51. I. $x^2 - 7x + 10 = 0$ II. $y^2 - 6y + 8 = 0$. Find the relation between x and y .

What kind of problem is this? Standard Banks PO Mains quadratic-comparison. Solve both equations, then compare ALL roots of x with ALL roots of y .

Solving it. Step-by-step.

Equation I: $x^2 - 7x + 10 = 0$ - Factorise: $x^2 - 5x - 2x + 10 = 0 \rightarrow x(x - 5) - 2(x - 5) = 0 \rightarrow (x - 5)(x - 2) = 0$ - Roots: $x = 2$ or $x = 5$

Equation II: $y^2 - 6y + 8 = 0$ - Factorise: $y^2 - 4y - 2y + 8 = 0 \rightarrow y(y - 4) - 2(y - 4) = 0 \rightarrow (y - 4)(y - 2) = 0$ - Roots: $y = 2$ or $y = 4$

Comparing roots (the systematic way). - Smallest $x = 2$, largest $x = 5$. - Smallest $y = 2$, largest $y = 4$. - For $x \geq y$ to be true, we need every $x \geq$ every y . Check: $x = 2 \geq y = 4$? **NO**. So $x \geq y$ is FALSE. - For $x \leq y$ to be true, we need every $x \leq$ every y . Check: $x = 5 \leq y = 4$? **NO**. So $x \leq y$ is FALSE. - Some roots overlap ($x = 2 = y = 2$). Some $x > y$ ($x=5, y=2$ or 4). Some $x < y$ ($x=2, y=4$). - **No clear relation can be established.**

Worth knowing: Universal comparison rule. - If smallest $x \geq$ largest $y \rightarrow x \geq y$ (with equality if smallest $x =$ largest y) - If largest $x \leq$ smallest $y \rightarrow x \leq y$ - Otherwise $\rightarrow$ **no relation**.

Watch out: Students compare just one root with one root and conclude wrongly. ALWAYS compare smallest- x vs largest- y AND largest- x vs smallest- y .

A similar question — Clear $x > y$. I. $x^2 - 9x + 20 = 0 \rightarrow x = 4, 5$. II. $y^2 - 5y + 6 = 0 \rightarrow y = 2, 3$. Smallest $x = 4 >$ largest $y = 3 \rightarrow x > y$.

Another twist — Equation in expanded form. I. $2x^2 - 11x + 12 = 0$. Factorise: $(2x - 3)(x - 4) = 0 \rightarrow x = 1.5$ or 4 . II. $3y^2 - 14y + 16 = 0$. Factorise: $(3y - 8)(y - 2) = 0 \rightarrow y = 8/3 \approx 2.67$ or 2 . Smallest $x = 1.5 <$ smallest $y = 2$; largest $x = 4 >$ largest $y = 8/3$. Mixed $\rightarrow$ **no relation**.

Q52. I. $x^2 - 12x + 35 = 0$ II. $y^2 - 13y + 42 = 0$. Find the relation.

Solving it. - Eq I: $(x - 5)(x - 7) = 0 \rightarrow x \in \{5, 7\}$ - Eq II: $(y - 6)(y - 7) = 0 \rightarrow y \in \{6, 7\}$ - smallest $x = 5$, largest $x = 7$ - smallest $y = 6$, largest $y = 7$ - For $x \geq y$: smallest x (5) $\geq$ largest y (7)? NO. - For $x \leq y$: largest x (7) $\leq$ smallest y (6)? NO. - Mixed $\rightarrow$ **no relation can be established**.

Memory peg. When the two equations share a common root (here both have 7), the answer is almost always "no relation" unless the other roots align cleanly.

Section 16 — APPROXIMATION (Banks PO Pre • 5 Qs/paper • 30 sec each)

Notation expansion: " $\approx$ " or "?" in the question stem means **closest integer / round number**, not exact value. Speed > accuracy beyond 1-2%.

Q53. ? $\approx$ 39.97 % of 1899.98 + 24.04 $\times$ 17.97. Find the value of ?.

What kind of problem is this? Round each ugly number to a clean value, then compute mentally.

Solving it. Step-by-step rounding. - 39.97 % $\approx$ **40 %** ($= 2/5$) - 1899.98 $\approx$ **1,900** - 24.04 $\approx$ **24** - 17.97 $\approx$ **18**

Now compute (mentally). - 40 % of 1,900 = $(2/5) \times 1,900 = \mathbf{760}$ - 24 $\times$ 18 = $(24 \times 20) - (24 \times 2) = 480 - 48 = \mathbf{432}$ - Sum = 760 + 432 = **1,192**

Worth knowing: Memorise these "round-able" fractions for instant computation.

Difficult %	Round to	As fraction
39.97 %	40 %	2/5
33.33 %	33.33 %	1/3
16.66 %	16.66 %	1/6
14.28 %	14.28 %	1/7
12.5 %	12.5 %	1/8
11.11 %	11.11 %	1/9
9.09 %	9.09 %	1/11
8.33 %	8.33 %	1/12

Watch out: Don't over-round. If actual is 1899.98, rounding to 2,000 introduces 5 % error — too much. Round to nearest "nice" value (1,900 here).

A similar question — Subtraction. ? $\approx$ 49.99 % of 480.04 - 19.97 $\times$ 11.95. $\rightarrow$ 50 % of 480 = 240; 20 $\times$ 12 = 240. ? = **0** (or close to 0; actual closer to -0.5).

Another twist — Division/mixed. ? $\approx \sqrt{624.97} + (35.04)^2$. $\rightarrow \sqrt{625} = 25$; $35^2 = 1,225$. ? = 25 + 1,225 = **1,250**.

Q54. ? $\approx$ 16.66 % of 1199.95 + 25.04 % of 799.92.

Solving it. - 16.66 % = 1/6, of 1,200 = **200**. - 25.04 % = 1/4, of 800 = **200**. - ? $\approx$ 200 + 200 = **400**.

Speed-tip. Whenever you see a decimal % near a "nice" fraction (16.66, 14.28, 12.5, 33.33, etc.), use the fraction — it's faster than working out the decimal.

Section 17 — QUANTITY COMPARISON (Banks PO Mains • 3-5 Qs/paper)

Notation expansion: Two quantities (often algebraic or word-problem expressions) are given. Compute or simplify each, then compare.

Q55. Quantity I: A boat travels 30 km downstream in 3 hours; speed of stream = 2 km/h. Find the boat's speed in still water. Quantity II: Average of 5 consecutive even numbers, the smallest of which is 16. Compare Quantity I and Quantity II.

What kind of problem is this? Solve each side independently → numerical comparison.

Quantity I. - Downstream speed = distance / time = $30 / 3 = 10$ km/h - Downstream = boat + stream → boat = $10 - 2 = 8$ km/h

Quantity II. - 5 consecutive even numbers starting at 16: 16, 18, 20, 22, 24. - Average = (sum) / 5 = $(16+18+20+22+24)/5 = 100/5 = 20$. - Or shortcut: average of consecutive numbers = middle number = **20**.

Compare: Quantity I (8) < Quantity II (20). So **Quantity I < Quantity II** (or $Q\ II > Q\ I$).

Worth knowing: Common answer options: - (a) $Q\ I > Q\ II$ - (b) $Q\ I \geq Q\ II$ - (c) $Q\ I < Q\ II$ ← (correct here) - (d) $Q\ I \leq Q\ II$ - (e) $Q\ I = Q\ II$ OR no relation

Watch out: Students compute one quantity carefully but rush the other. Always solve BOTH, then compare cleanly.

A similar question. Quantity I: 30 % of 200. Quantity II: 25 % of 240. Both = 60 → **Q I = Q II**.

Section 18 — CASELET DI (Banks PO Mains specialty • 5-Q set)

Notation expansion: Caselet DI = 1 paragraph of facts, NO chart. Read carefully → extract data → answer 5 sub-questions. The challenge is extracting numbers correctly from prose.

Q56. Caselet: A factory has three units — A, B, and C. The total production in 2024 was 12,000 units. Unit A produced 40 % of the total. Unit B produced 75 % of what Unit A produced. Unit C produced the rest.

In 2025, Unit A's production increased by 25 %, Unit B's increased by 20 %, and Unit C's decreased by 10 %.

Sub-Q: What was Unit C's production in 2025?

Solving it. Step-by-step extraction.

Step 1 — 2024 production by unit. - Total = 12,000 - A = $40\% \times 12,000 = 4,800$ - B = $75\% \times 4,800 = 3,600$ - C = $12,000 - 4,800 - 3,600 = 3,600$

Step 2 — 2025 production by unit. - A₂₀₂₅ = $4,800 \times 1.25 = 6,000$ - B₂₀₂₅ = $3,600 \times 1.20 = 4,320$ - C₂₀₂₅ = $3,600 \times 0.90 = 3,240$

Answer. Unit C in 2025 = **3,240 units**.

Worth knowing: Caselet sub-Qs typically ask for: ratios, percentages, differences, totals across years/units. Build a table once → answer all 5 in 2-3 minutes.

Unit	2024	2025	Δ %
A	4,800	6,000	+25 %

Unit	2024	2025	Δ %
B	3,600	4,320	+20 %
C	3,600	3,240	-10 %
Total	12,000	13,560	+13 %

Watch out: Students re-read the paragraph for each sub-Q (wastes 30 sec). Extract once, build the table, then read sub-Qs.

More sub-Qs (try mentally). - Ratio of A : B : C in 2024? → 4 : 3 : 3 - Total production in 2025? → 13,560 - % of total contributed by A in 2025? → $6000/13560 \times 100 \approx 44.25\%$ - Difference between A's 2025 and B's 2025? → $6000 - 4320 = 1,680$ units

Section 19 — COORDINATE GEOMETRY (SSC CGL Tier-2 • 3-5 Qs/paper)

Notation expansion: A point in 2-D plane = (x, y) . x = horizontal (abscissa); y = vertical (ordinate). **Distance formula** between (x_1, y_1) and $(x_2, y_2) = \sqrt{[(x_2 - x_1)^2 + (y_2 - y_1)^2]}$ **Midpoint formula** = $((x_1 + x_2)/2, (y_1 + y_2)/2)$ **Slope formula** between two points = $(y_2 - y_1) / (x_2 - x_1)$ **Line equation** $y - y_1 = m(x - x_1)$, where m = slope.

Q57. Find the distance between points A (3, 4) and B (7, 1).

Solving it. Plug into distance formula. - $d = \sqrt{[(7 - 3)^2 + (1 - 4)^2]} = \sqrt{[16 + 9]} = \sqrt{25} = 5$ units

Worth knowing: Recognise Pythagorean triples in coordinate geometry. Here (4, 3, 5) appeared automatically.

Watch out: Students subtract in wrong order (3 - 7 vs 7 - 3); doesn't matter because we square — but be consistent.

A similar question. Distance between (0, 0) and (5, 12) = $\sqrt{(25 + 144)} = \sqrt{169} = 13$ (a 5-12-13 triple).

Another twist — Find a missing coordinate. Distance between (1, 2) and (4, k) = 5. Solve $\sqrt{9 + (k-2)^2} = 5 \rightarrow 9 + (k-2)^2 = 25 \rightarrow (k-2)^2 = 16 \rightarrow k - 2 = \pm 4 \rightarrow k = 6$ or $k = -2$.

Q58. Find the slope of the line passing through (2, 3) and (6, 11). Then find the equation of the line.

Solving it. Step-by-step. - Slope $m = (11 - 3) / (6 - 2) = 8 / 4 = 2$ - Equation (using point-slope form with (2, 3)): $y - 3 = 2(x - 2) \rightarrow y = 2x - 4 + 3 \rightarrow y = 2x - 1$

Worth knowing: Quick line-properties recap. - **Parallel lines** have the same slope. - **Perpendicular lines** have slopes whose product = -1 (i.e., $m_1 \times m_2 = -1$). - **Horizontal line** has slope 0; **vertical line** has undefined slope.

A similar question — Midpoint. Midpoint of (2, 3) and (6, 11) = $((2+6)/2, (3+11)/2) = (4, 7)$.

Another twist — Perpendicular line. Find equation of line perpendicular to $y = 2x - 1$, passing through (1, 5). New slope $m' = -1/2$ (since $2 \times -0.5 = -1$). Equation: $y - 5 = -0.5(x - 1) \rightarrow y = -0.5x + 5.5 \rightarrow 2y + x = 11$.

Section 20 — STATISTICS (SSC CGL Tier-2 + Banks GA)

Notation expansion: **Mean** = arithmetic average. **Median** = middle value when sorted. **Mode** = most-frequent value. **Range** = max - min. **Standard Deviation (SD)** = $\sqrt{(\text{variance})} = \sqrt{[(1/n) \sum (x_i - \text{mean})^2]}$.

Q59. Find the mean, median, and mode of the data: 4, 8, 6, 4, 7, 9, 4, 8, 6.

Solving it. Step-by-step.

Sort first: 4, 4, 4, 6, 6, 7, 8, 8, 9. ($n = 9$)

Mean = sum / count = $(4+4+4+6+6+7+8+8+9) / 9 = 56 / 9 \approx \mathbf{6.22}$

Median = middle value (since $n = 9$, middle is $(9+1)/2 = 5$ th value when sorted) = **6**

Mode = most-frequent value = **4** (occurs 3 times)

Worth knowing: Empirical relation. **Mode = 3 × Median – 2 × Mean** (approximate, for moderately skewed data).

Watch out: Students forget to sort before finding the median. Without sorting, "middle position" is meaningless.

A similar question — Even count. Data: 2, 5, 7, 9, 11, 13. $n = 6$. Median = average of 3rd and 4th values = $(7 + 9)/2 = \mathbf{8}$.

Another twist — Find SD. Data: 2, 4, 6, 8, 10. Mean = 6. Deviations from mean: -4, -2, 0, 2, 4. Squared: 16, 4, 0, 4, 16. Sum = 40. Variance = $40/5 = 8$. SD = $\sqrt{8} \approx \mathbf{2.83}$.

Section 21 — LOGARITHM (SSC CGL Tier-2 + RRB JE)

Notation expansion: $\log_b(x) = y$ means $b^y = x$. Default base in competitive exams = **10** (so $\log 100 = 2$).

Natural log = $\ln(x)$, base $e \approx 2.718$. **Key rules:** $\log(a \times b) = \log a + \log b$ $\log(a / b) = \log a - \log b$ $\log(a^n) = n \times \log a$ $\log_b(a) = \log a / \log b$ (change of base) $\log_b(b) = 1$; $\log_b(1) = 0$.

Q60. Find the value of $\log 8 + \log 125$ (base 10).

Solving it. Step-by-step. - $\log 8 + \log 125 = \log (8 \times 125) = \log 1,000 = \mathbf{3}$ (since $10^3 = 1,000$).

Quick way: Recognise that $8 \times 125 = 1,000$ (a power of 10) — $\log 10^n = n$.

Worth knowing: Memorise these standard log values (base 10). - $\log 2 \approx \mathbf{0.301}$ - $\log 3 \approx \mathbf{0.477}$ - $\log 5 \approx \mathbf{0.699}$ (= $1 - \log 2$) - $\log 7 \approx \mathbf{0.845}$

Watch out: Students try to compute $\log 8 = 3 \times \log 2$ (= 0.903) and $\log 125 = 3 \times \log 5$ (= 2.097) separately. Both are right but slower. Sum-then-multiply is faster.

A similar question — Solve for x. If $\log x = 2.5$, find x . $\rightarrow x = 10^{2.5} = 10^2 \times 10^{0.5} = 100 \times \sqrt{10} \approx \mathbf{316.23}$.

Another twist — Change of base. $\log_2(16) = \log 16 / \log 2 = (4 \times \log 2) / \log 2 = \mathbf{4}$.

Section 22 — MENSURATION COMBINED SOLIDS (SSC CGL Tier-2)

Notation expansion: Combined solids = two basic solids stuck together (cone-on-cylinder, hemisphere-on-cone, frustum, etc.). Volume = sum of parts; surface area = sum of EXTERNAL surfaces only (subtract any joining face).

Q61. A solid is formed by mounting a cone on top of a cylinder. The cylinder has radius 7 cm and height 10 cm. The cone has the same base radius (7 cm) and height 4 cm. Find the total volume.

Solving it. Step-by-step. - Volume of cylinder = $\pi r^2 h = (22/7) \times 49 \times 10 = 22 \times 70 = \mathbf{1,540 \text{ cm}^3}$ - Volume of cone = $(1/3)\pi r^2 h = (1/3) \times (22/7) \times 49 \times 4 = (1/3) \times 22 \times 28 = \mathbf{205.33 \text{ cm}^3}$ - Total volume = $1,540 + 205.33 = \mathbf{1,745.33 \text{ cm}^3}$

Worth knowing: When solids share a base, volumes ADD (no overlap). Surface area adds only EXTERIOR — the joining circular face is hidden, so subtract one πr^2 from each side's TSA when computing total surface.

Watch out: Students compute TSA of each solid separately and add — wrong because the shared circular face counts twice.

A similar question — Hemisphere on cylinder. Cylinder $r = 7$, $h = 10$; hemisphere $r = 7$ on top. Volume = $\pi r^2 h + (2/3)\pi r^3 = 1,540 + (2/3) \times (22/7) \times 343 = 1,540 + 718.67 = \mathbf{2,258.67 \text{ cm}^3}$.

Another twist — Frustum (cone with top cut off). Frustum has $R = 6$ (bottom radius), $r = 3$ (top radius), $h = 8$. Volume = $(1/3)\pi h(R^2 + r^2 + Rr) = (1/3) \times (22/7) \times 8 \times (36 + 9 + 18) = (1/3) \times (22/7) \times 8 \times 63 = \mathbf{528 \text{ cm}^3}$.

Section 23 — CALENDAR + CLOCK (RRB NTPC + SSC favourite)

Notation expansion: Odd days = days remaining when total days $\div 7$. Used for finding the day of the week of any date. **Clock angles:** Hour hand moves $360^\circ / 12 = 30^\circ$ per hour OR 0.5° per minute. Minute hand moves $360^\circ / 60 = 6^\circ$ per minute. Relative speed (minute – hour) = 5.5° per minute.

Q62. What day of the week was 15 August 1947?

Solving it. Step-by-step using the odd-days method. - Reference: 1 January 0001 was a Monday (calendar starts). - Total days from 1 Jan 0001 to 15 Aug 1947 = 1946 complete years + 226 days (in 1947). - Days in complete year = 365 (or 366 in leap). - Quick formula: count "odd days" mod 7. - 1600 years complete: odd days = 0 (every 400-year block has 0 odd days). - Next 300 years (1601-1900): odd days = 1 (300 yrs has 5 odd days... actually let's use a quicker shortcut).

Faster method (Zeller's-style mental math). In SSC, this is rarely solved by hand for an arbitrary date. Memorise key reference dates: - 15 Aug 1947 = **Friday** - 26 Jan 1950 = **Thursday** - 1 Jan 2000 = **Saturday** - 1 Jan 2024 = **Monday**

For nearby dates, count days forward/backward modulo 7.

Watch out: Leap year rules — divisible by 4 BUT not 100 (unless also divisible by 400). 1900 is NOT a leap year; 2000 IS.

A similar question — Day of week. What day was 26 January 1950? → Thursday (memorise this — first Republic Day).

Another twist — Days between two dates. From 1 Jan 2024 to 1 Jan 2025 = 366 days (2024 leap). $366 \div 7 = 52$ remainder 2. So 1 Jan 2025 = Monday + 2 = Wednesday.

Q63. Find the angle between the hour hand and the minute hand of a clock at 3:40.

Solving it. Step-by-step. - Hour hand position at 3:40: - At 3:00, hour hand is at 90° (pointing to 3). - In next 40 minutes, hour hand moves $40 \times 0.5^\circ = 20^\circ$ more. - Hour hand at 3:40 = $90 + 20 = \mathbf{110^\circ}$ from 12. - Minute hand position at 40 minutes past hour: - Minute hand at "40" mark = $40 \times 6^\circ = \mathbf{240^\circ}$ from 12. - Angle between = $|240 - 110| = \mathbf{130^\circ}$

Worth knowing: Useful clock formulas. - Angle of hour hand from 12 = $(60 \times H + M) \times 0.5^\circ$ where H = hour, M = minutes. - Angle of minute hand from 12 = $M \times 6^\circ$. - Angle between = $|11M/2 - 30H| \pmod{360}$, take smaller of the two: angle or $360 - \text{angle}$.

Quick way using formula. $|11 \times 40 / 2 - 30 \times 3| = |220 - 90| = \mathbf{130^\circ}$ ✓.

Watch out: Sometimes the "smaller" angle is asked — if your answer is $> 180^\circ$, subtract from 360° to get the reflex-angle complement.

A similar question — Clock at 6:15. $|11 \times 15/2 - 30 \times 6| = |82.5 - 180| = \mathbf{97.5^\circ}$.

Another twist — When are the hands at right angle? Hands form 90° angle 22 times in 12 hours (not 24 — they "miss" twice due to relative speed). First time after 3:00: when $|11M/2 - 90| = 90 \rightarrow 11M/2 = 0$ or $180 \rightarrow M = 0$ (i.e. 3:00 itself, but they're at 90° already? actually at 3:00 they're at 90°). Next: $M = 360/11 \approx 32.73$ minutes past 3 $\rightarrow$ roughly **3:32:43**.

PART E — TIMED MINI-MOCK (25 Q · 25 min)

Solve under timer. Self-grade against the answer key. Target: 22+/25 in <25 min for >90% mastery.

- 35 % of 480 = ? (a) 158 (b) 168 (c) 178 (d) 188
- A sum doubles in 8 yrs at SI. Triple in? (a) 16 (b) 18 (c) 20 (d) 24
- CP = ₹400, SP = ₹500. Profit %? (a) 20 (b) 25 (c) 30 (d) 33
- 12, 18 LCM = ? (a) 36 (b) 48 (c) 24 (d) 60
- Avg of first 10 natural numbers? (a) 4.5 (b) 5 (c) 5.5 (d) 6
- $\frac{1}{2} + \frac{1}{3} + \frac{1}{6} = ?$ (a) 1 (b) $\frac{5}{6}$ (c) $\frac{7}{6}$ (d) $\frac{11}{6}$
- A ratio 3:4, B is 25 % more. New ratio? (a) 3:5 (b) 6:7 (c) 12:15 (d) 24:25
- 25 % of $x = 80$. $x = ?$ (a) 320 (b) 280 (c) 360 (d) 400
- Train 200 m at 72 km/h crosses pole in? (a) 8 s (b) 10 s (c) 12 s (d) 15 s
- Boat speed in still water 12, stream 4. Downstream 32 km in? (a) 1.5 h (b) 2 h (c) 2.5 h (d) 3 h
- A and B together 8 days, A alone 12. B alone? (a) 16 (b) 20 (c) 24 (d) 30
- $(a + b)^2 - (a - b)^2 = ?$ (a) $4ab$ (b) $2ab$ (c) ab (d) 0
- $\sin^2 30^\circ + \cos^2 30^\circ = ?$ (a) $\frac{1}{2}$ (b) 1 (c) $\frac{\sqrt{3}}{2}$ (d) 2
- Area equilateral triangle side 6 = ? (a) $9\sqrt{3}$ (b) $18\sqrt{3}$ (c) 36 (d) 18
- SI on ₹2,000 @ 5 % for 4 yrs = ? (a) 200 (b) 300 (c) 400 (d) 500
- CI on ₹1,000 @ 10 % for 2 yrs = ? (a) 200 (b) 210 (c) 215 (d) 220
- P (red on a die) = ? (a) 0 (b) $\frac{1}{6}$ (c) $\frac{1}{3}$ (d) $\frac{1}{2}$
- $7! / 5! = ?$ (a) 7 (b) 14 (c) 42 (d) 35
- Sum of angles of pentagon = ? (a) 360 (b) 540 (c) 720 (d) 900
- Volume cube side 5 = ? (a) 25 (b) 75 (c) 100 (d) 125
- CSA cylinder $r=7$, $h=10 = ?$ (a) 220 (b) 440 (c) 660 (d) 880
- Mean of 1, 2, ..., 10 = 5.5; SD $\approx ?$ (a) 2.87 (b) 3.0 (c) 3.5 (d) 4.0
- 30 % of 30 % of 1,000 = ? (a) 60 (b) 90 (c) 30 (d) 100
- Compound ratio 2:3 and 4:5 = ? (a) 4:5 (b) 8:15 (c) 6:8 (d) 1:1
- Tank A fills in 10 h, B empties in 15 h. Both open? (a) 25 h (b) 30 h (c) 35 h (d) 40 h

Answer key: 1-b, 2-a, 3-b, 4-a, 5-c, 6-a, 7-b (3:5), 8-a, 9-b, 10-b, 11-c, 12-a, 13-b, 14-a, 15-c, 16-b, 17-a, 18-c, 19-b, 20-d, 21-b, 22-a, 23-b, 24-b, 25-b.

PART F — TRAP-RECOGNITION CARDS

Trap pattern	Where it appears	How to spot
"increase by $r\%$, then decrease by $r\%$ " gives 0	Percentage	Multiplicative changes never cancel — net is always $-r^2/100\%$
Equal SP, equal % gain & loss = 0	Profit & Loss	Always net loss $(r/10)^2$
"double in n years SI" → "triple in $2n$ "	SI	Wrong; triple needs $(n \times 2)$ only if rate doubles. Correct: triple in $2n$ only if rate = $100/n$
"average speed = $(s_1 + s_2)/2$ "	TSD	Only true for equal time, not equal distance. Use harmonic $2s_1s_2/(s_1+s_2)$ for equal distance
Adding pipe rates instead of subtracting	Pipes	If outlet pipe, subtract
Using SP as base for profit %	P&L	Always CP base; SP base is "discount" or "margin"
Forgetting 18/5 conversion	Speed	km/h to m/s: $\times 5/18$
Heron + obvious right-angle	Mensuration	Check Pythagoras first; $\frac{1}{2} \cdot \text{base} \cdot \text{height}$ beats Heron
Rate "p.a. compounded half-yearly" → use rate/2, periods $\times 2$	CI	Always halve rate, double periods
$(n-1)$ instead of $(n-2)$ for polygon angles	Geometry	Sum = $(n - 2) \times 180$

PART G — SPEED REFERENCE CARD (1-page exam day)

Squares 1-30 (memorise):

1,4,9,16,25,36,49,64,81,100,121,144,169,196,225,256,289,324,361,400,441,484,529,576,625,676,729,784,841,900.

Cubes 1-15: 1,8,27,64,125,216,343,512,729,1000,1331,1728,2197,2744,3375.

Fraction = % conversions: $1/2=50$, $1/3=33.3$, $1/4=25$, $1/5=20$, $1/6=16.6$, $1/7=14.28$, $1/8=12.5$, $1/9=11.1$, $1/10=10$, $1/11=9.09$, $1/12=8.33$, $1/13=7.69$, $1/14=7.14$, $1/15=6.66$, $1/16=6.25$, $1/20=5$, $1/25=4$.

Roots: $\sqrt{2}=1.414$, $\sqrt{3}=1.732$, $\sqrt{5}=2.236$, $\sqrt{7}=2.645$, $\sqrt{10}=3.162$.

Pythagorean triples to spot in geometry: 3-4-5, 5-12-13, 7-24-25, 8-15-17, 9-40-41, 11-60-61, 12-35-37, 20-21-29, 28-45-53.

Algebra identities (instant recall): $-(a+b)^2 = a^2 + 2ab + b^2$ - $(a-b)^2 = a^2 - 2ab + b^2$ - $a^2 - b^2 = (a+b)(a-b)$ - $a^3 \pm b^3 = (a \pm b)(a^2 \mp ab + b^2)$ - $a + 1/a = k \rightarrow a^2 + 1/a^2 = k^2 - 2$; $a^3 + 1/a^3 = k^3 - 3k$ - if $a+b+c = 0 \rightarrow a^3+b^3+c^3 = 3abc$

Trig values cold:

θ	0°	30°	45°	60°	90°
sin	0	$1/2$	$1/\sqrt{2}$	$\sqrt{3}/2$	1
cos	1	$\sqrt{3}/2$	$1/\sqrt{2}$	$1/2$	0
tan	0	$1/\sqrt{3}$	1	$\sqrt{3}$	∞

This DRILL PACK is the difference between "I read the book" and "I CAN SOLVE the paper". Drill all 50 examples + the mini-mock at least twice. Time yourself. Mark trap-options before checking the answer key.

PART H — VEDIC MATH & SPEED TRICKS (the 95%+ edge)

Each of the 16 Vedic sutras + the speed-multiplication shortcuts below saves **5-10 seconds per question**. In a 25-Q section, that's ~3-4 minutes saved — enough to attempt 5 more questions.

H.1 Multiplication shortcuts

Multiplication of any 2-digit number by 11

Rule. Place the SUM of the two digits between them.

- $25 \times 11 = 2_5$ with $(2+5)=7$ in middle = **275**.
- 47×11 : 4_7 with $(4+7)=11 \rightarrow$ carry the 1: $4+1=5$, leaves 1 in middle, 7 at end $\rightarrow$ **517**.
- 89×11 : 8_9 with $(8+9)=17 \rightarrow 8+1=9$, 7 in middle, 9 at end $\rightarrow$ **979**.

Multiplication by 12, 13, ..., 19 ("Vertical & Crosswise" sutra)

Multiplication by 13: take number $\times 1$ ten and add ($\times 3$ units). - 24×13 : $(24 \times 10) + (24 \times 3) = 240 + 72 =$ **312**. - Or: write $2_4 \rightarrow 2 \times 3 = 6$ add to next digit place.

Multiplication of two numbers near 100 (Nikhilam sutra)

If both close to 100 (or 1000), use base-100 deviation method.

- 96×97 : deviations from 100 are -4 and -3 . Cross-add: $96 + (-3) = 93$ OR $97 + (-4) = 93 \rightarrow$ first part. Multiply deviations: $(-4)(-3) = 12 \rightarrow$ second part. Answer = **9312**.
- 98×96 : deviations -2 , -4 . Cross: $98 - 4 = 94$. Mult: 8. Answer = **9408**.
- 103×104 : deviations $+3$, $+4$. Cross: $103 + 4 = 107$. Mult: 12. Answer = **10712** (since base 100 needs 2-digit padding, 12 stays as is).

Squaring numbers ending in 5

Rule. Take leading digit $\times$ (leading digit + 1), then append "25".

- $25^2 = 2 \times 3 = 6 \rightarrow$ append 25 $\rightarrow$ **625**.
- $35^2 = 3 \times 4 = 12 \rightarrow$ **1225**.
- $75^2 = 7 \times 8 = 56 \rightarrow$ **5625**.
- $105^2 = 10 \times 11 = 110 \rightarrow$ **11025**.

Squaring numbers near 50

For numbers $50 \pm k$:

- $(50 + k)^2 = (25 + k)$ followed by k^2 .
- $(50 - k)^2 = (25 - k)$ followed by k^2 .
- $53^2 = (25 + 3)$ followed by $3^2 = 28_09 =$ **2809**.
- $47^2 = (25 - 3)$ followed by $3^2 = 22_09 =$ **2209**.
- $58^2 = 28 + 8 = 33$, then $8^2 = 64 \rightarrow$ **3364**.
- $42^2 = 25 - 8 = 17$, then $8^2 = 64 \rightarrow$ **1764**.

Multiplication of numbers near 50 (each)

- 48×47 : $47 - 2 = 45 \rightarrow$ first part is $25 + 45 =$ wait, simpler: each is $50 - k$. Use base 50.

- Use Nikhilam with base 50: deviations from 50 are -2 , -3 . Cross: $48 + (-3) = 45$. Multiply: 6. Multiply first part by $50/100 = \frac{1}{2}$. Answer = $45 \times 50 + 6 = 2256$.
- Or just memorise standard products.

Multiplication by 9, 99, 999

Rule. $N \times 9 = N0 - N$. $N \times 99 = N00 - N$. $N \times 999 = N000 - N$.

- $24 \times 9 = 240 - 24 = \mathbf{216}$.
- $37 \times 99 = 3700 - 37 = \mathbf{3663}$.
- $48 \times 999 = 48000 - 48 = \mathbf{47952}$.

Multiplication by 25, 50, 125, 250

- $\times 25$: $(\times 100) \div 4$. So $36 \times 25 = 3600/4 = 900$.
- $\times 50$: $(\times 100) \div 2$. So $84 \times 50 = 8400/2 = 4200$.
- $\times 125$: $(\times 1000) \div 8$. So $56 \times 125 = 56000/8 = 7000$.

H.2 Squaring + cubing tricks

Square of any number close to 10^n

$(10 + k)^2 = 100 + 20k + k^2$. For $k=5$: 225. For $k=7$: $100 + 140 + 49 = 289$.

Square of 11, 12, ... 19 (memorise table)

$11^2=121$, $12^2=144$, $13^2=169$, $14^2=196$, $15^2=225$, $16^2=256$, $17^2=289$, $18^2=324$, $19^2=361$.

Square of 21-29

$21^2=441$, $22^2=484$, $23^2=529$, $24^2=576$, $25^2=625$, $26^2=676$, $27^2=729$, $28^2=784$, $29^2=841$.

Cubing close to $10n$

$(10n + k)^3 = 1000n^3 + 300n^2k + 30nk^2 + k^3$. Useful only for very small k .

H.3 Division shortcuts

Division by 5

Multiply numerator $\times 2$, divide by 10.

- $245/5 = (245 \times 2)/10 = 490/10 = \mathbf{49}$.
- $1234/5 = (1234 \times 2)/10 = 2468/10 = \mathbf{246.8}$.

Division by 25, 125

- $/25 = (\times 4)/100$.
- $/125 = (\times 8)/1000$.

Division by 9 (digit-sum check)

If digit sum is divisible by 9, the number is. Quotient by repeated subtraction.

H.4 Percentage shortcuts (instant)

Shortcut	Application
10%	move decimal one place left
1%	move decimal two places left

Shortcut	Application
5%	half of 10%
25%	divide by 4
12.5%	divide by 8
33.33%	divide by 3
16.66%	divide by 6
6.25%	divide by 16
Successive %	$a + b + ab/100$
% more vs % less	$r/(100+r) \times 100$ OR $r/(100-r) \times 100$
x% of y = y% of x	symmetry

H.5 Quick-recall fraction-to-percent (already covered in Part 0; repeat to lock in)

Fraction	%
1/2	50
1/3	33.33
1/4	25
1/5	20
1/6	16.66
1/7	14.28
1/8	12.5
1/9	11.11
1/10	10
1/11	9.09
1/12	8.33
2/3	66.66
3/4	75
3/5	60
2/7	28.57
5/8	62.5
3/8	37.5
7/8	87.5
5/9	55.55
7/9	77.77

H.6 Time-saving sutras (Vedic — 16 main + 13 sub)

Sutra	English	Use
Ekadhikena Purvena	"By one more than the previous"	Squaring numbers ending in 5; recurring decimals

Sutra	English	Use
Nikhilam Navatashcaramam Dashatah	"All from 9 and last from 10"	Subtraction from 10n; multiplication near base
Urdhva-Tiryagbyham	"Vertically and crosswise"	General multiplication
Paraavartya Yojayet	"Transpose and adjust"	Division of polynomials
Sunyam Samyasamuccaye	"When sum is same that sum is zero"	Equation solving
Anurupye Sunyamanyat	"If one is in ratio, the other is zero"	Specific equation pattern
Sankalana-Vyavakalanabhyam	"By addition and by subtraction"	Simultaneous equations
Yavadunam	"Whatever the deficiency"	Squaring numbers below base
Vyastisamastih	"Specific and general"	Series + sequences
Sesanyankena Caramena	"The remainders by the last digit"	Recurring decimals

Master these; you don't need a 200-page Vedic-math book. Just these few patterns cover 80% of speed-up opportunity.

PART I — MENSURATION MASTER FORMULA SHEET (2D + 3D)

2D shapes

Shape	Area	Perimeter / Boundary
Square (side a)	a^2	$4a$; diagonal $a\sqrt{2}$
Rectangle (l × b)	$l \times b$	$2(l + b)$; diagonal $\sqrt{l^2 + b^2}$
Parallelogram (b, h)	$b \times h$	$2(a + b)$
Rhombus (d_1, d_2)	$\frac{1}{2} \times d_1 \times d_2$	$4 \times \text{side}$; side = $\frac{1}{2} \times \sqrt{(d_1^2 + d_2^2)}$
Triangle (b, h)	$\frac{1}{2} \times b \times h$	$a + b + c$
Equilateral triangle (a)	$(\sqrt{3}/4) \times a^2$; height = $(\sqrt{3}/2) \times a$	$3a$
Right triangle	$\frac{1}{2} \times \text{base} \times \text{perpendicular}$	$a + b + \sqrt{a^2 + b^2}$
Heron's formula	$\sqrt{s(s-a)(s-b)(s-c)}$ where $s = (a+b+c)/2$	—
Trapezium	$\frac{1}{2} \times (a + b) \times h$	sum of all sides
Circle (r)	$\pi \times r^2$	$2\pi \times r$ (circumference)
Sector (r, θ)	$(\theta/360) \times \pi \times r^2$	arc length = $(\theta/360) \times 2\pi \times r$
Annulus (R, r)	$\pi(R^2 - r^2)$	$2\pi(R + r)$
Regular polygon (n sides, side a)	$(1/4) \times n \times a^2 \times \cot(180^\circ/n)$	$n \times a$
Regular hexagon (a)	$(3\sqrt{3}/2) \times a^2$	$6a$

3D shapes — Volume + Surface

Shape	Volume	Total Surface Area	Curved/Lateral SA
Cube (a)	a^3	$6a^2$	$4a^2$
Cuboid (l, b, h)	$l \times b \times h$	$2(lb + bh + hl)$	$2h(l + b)$
Cylinder (r, h)	$\pi r^2 h$	$2\pi r(r + h)$	$2\pi rh$
Hollow cylinder (R, r, h)	$\pi(R^2 - r^2)h$	$2\pi(R + r)(R - r + h)$	$2\pi h(R + r)$
Cone (r, h, l)	$(1/3) \pi r^2 h$; slant $l = \sqrt{r^2 + h^2}$	$\pi r(r + l)$	$\pi r l$
Sphere (r)	$(4/3) \pi r^3$	$4\pi r^2$	—
Hemisphere (r)	$(2/3) \pi r^3$	$3\pi r^2$	$2\pi r^2$
Spherical shell (R, r)	$(4/3)\pi(R^3 - r^3)$	—	—
Frustum of cone (R, r, h, l)	$(1/3) \pi h (R^2 + r^2 + Rr)$; $l = \sqrt{h^2 + (R-r)^2}$	$\pi(R^2 + r^2 + l(R+r))$	$\pi l (R + r)$
Pyramid (square base a, h, slant l)	$(1/3) a^2 h$	$a^2 + 2a \times l$	$2a \times l$
Tetrahedron (a)	$(a^3 \times \sqrt{2})/12$	$\sqrt{3} \times a^2$	—
Prism (general)	base area × height	$2 \times \text{base} + \text{perimeter} \times h$	perimeter × h

Key memory pegs

- **Cube** is the cuboid with $l = b = h = a$.
- **Sphere** = $4\pi r^2$ (surface) and $(4/3)\pi r^3$ (volume) — memorise EXACTLY.
- **Cone slant length** $l = \sqrt{r^2 + h^2}$ — Pythagoras.
- **Hemisphere TSA** = $3\pi r^2$ (curved + base disc) — common trap (students answer $2\pi r^2$).
- **Cuboid diagonal** = $\sqrt{l^2 + b^2 + h^2}$ — extension of 2D.
- **Cylinder volume to area** ratio: $V/A \rightarrow r/2$ (per unit height).

Special-case quick computation

Setup	Formula
Largest cube cut from cube of side a	a (same)
Largest sphere from cube of side a	$(4/3)\pi(a/2)^3$
Largest cylinder from cube (axis through cube)	$\pi r^2 h$ with $r = a/2$, $h = a$
Largest cone from cube	$(1/3)\pi(a/2)^2(a)$
Largest cube inscribed in sphere of r	side = $(2r)/\sqrt{3}$
Diagonal of square inscribed in circle r	$2r$

PART J — 1-PAGE EXAM-DAY ARITHMETIC CHEATSHEET (revise day before)

Mental-math reflexes: - Squares 1-30 cold. Cubes 1-15 cold. - Tables 11-20 cold. - Fraction-percent table memorised ($1/2 = 50$, $1/8 = 12.5$, etc.). - $\pi \approx 22/7$; $\sqrt{2} \approx 1.414$, $\sqrt{3} \approx 1.732$.

Universal shortcuts: - 10% trick $\rightarrow$ decimal one place left. - 25% $\rightarrow \div 4$. 125 $\rightarrow \div 8$. 5% $\rightarrow$ half of 10%. - Percent symmetry: $x\%$ of $y = y\%$ of x . - Successive %: $a + b + ab/100$. - "r% more" $\rightarrow$ "r/(100+r)% less".

Speed multiplication: - $\times 11$: place sum of digits between. - $\times 9, 99, 999$: subtract from $N0, N00, N000$. - $\times 25$: $\times 100 \div 4$; $\times 125$: $\times 1000 \div 8$. - Squares of 11-19: 121, 144, 169, 196, 225, 256, 289, 324, 361. - Squares ending in 5: lead $\times$ (lead+1), append 25.

Time-Speed-Distance: - km/h to m/s: $\times 5/18$. - Average speed (equal distance): $2ab/(a+b)$. - Boat: down = $b+s$, up = $b-s$.

Profit & Loss: - Equal % gain & loss on same SP $\rightarrow$ always loss = $(r/10)^2$. - MP-CP-SP: Profit% = $(SP-CP)/CP \times 100$. Discount% = $(MP-SP)/MP \times 100$.

SI / CI: - Doubles in n yrs at SI $\rightarrow$ rate = $100/n$ %. - CI – SI for 2 yrs = $P(r/100)^2$.

Trigonometry standard angles: - $\sin(0, 30, 45, 60, 90) = 0, \frac{1}{2}, 1/\sqrt{2}, \sqrt{3}/2, 1$. - $\cos$ = reverse.

Geometry: - Sum interior angles polygon = $(n-2) \times 180^\circ$. - Equilateral triangle area = $(\sqrt{3}/4)a^2$. - Heron's: $\sqrt{s(s-a)(s-b)(s-c)}$.

Mensuration: - Sphere $V = (4/3)\pi r^3$, $SA = 4\pi r^2$. - Cone $V = (1/3)\pi r^2 h$, slant = $\sqrt{r^2 + h^2}$. - Hemisphere TSA = $3\pi r^2$ (NOT $2\pi r^2$).

Algebra: - $a + 1/a = k \rightarrow a^2 + 1/a^2 = k^2 - 2$; $a^3 + 1/a^3 = k^3 - 3k$. - $(a + b + c) = 0 \rightarrow a^3 + b^3 + c^3 = 3abc$.

Print this page and revise the morning before the exam. THIS is the 95%+ ammunition.